

Neurosymbolic AI Approach for Object Detection In Aerial Images

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by

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ABSTRACT

This thesis investigates a neurosymbolic approach for object detection in aerial imagery, combining the perceptual strengths of deep learning with the contextual reasoning capacity of symbolic logic. The baseline system employed YOLOv11n-Oriented Bounding Box (OBB) detection, trained on the DOTA dataset and enhanced with Slicing Aided Hyper Inference (SAHI) to handle ultra-high-resolution inputs and preserve small object visibility. While the baseline detector achieved an overall mean Average Precision (mAP@0.5) of 62.8% and demonstrated strong performance for large and distinct objects, it remained vulnerable to false positives in implausible contexts and struggled with borderline detections of small or ambiguous objects.

To address these challenges, a symbolic reasoning layer was introduced using SWI-Prolog, grounded in weighted knowledge graphs derived from dataset co-occurrence and spatial statistics. Symbolic rules encoded domain knowledge as confidence modifiers, proportionally boosting plausible detections and penalizing implausible ones. The final evaluation, based on 136 object–relation cases logged in Excel, demonstrated systematic improvements in both primary detections and secondary detections confidence scores improved. These refinements reduced semantically inconsistent false positives and preserved contextually supported borderline detections.

The research contributes to both aerial detection and neurosymbolic AI by demonstrating a hybrid pipeline integrating YOLOv11n-OBB and Prolog reasoning; operationalizing weighted knowledge graphs for graded symbolic inference; transforming the error profile of the baseline detector toward greater plausibility and interpretability; and providing explanation traces for every symbolic adjustment. While global mAP gains were modest, the system produced detection maps that were semantically consistent, context-aware, and explainable. These contributions establish a foundation for future research in scalable neurosymbolic learning, while also offering practical value in societal domains such as disaster response, defense, and urban monitoring, where reliability and transparency are critical.

FOREWORD

This thesis represents the culmination of my research into the intersection of deep learning and symbolic reasoning, applied to the challenge of object detection in aerial imagery. The journey began with a simple question: can we make object detectors not only more accurate, but also more trustworthy and explainable? The growing reliance on aerial imagery in disaster management, environmental monitoring, and defense motivated me to pursue an approach that would bridge perception and reasoning a system that could detect objects while also justifying its predictions.

The work presented here reflects both technical exploration and conceptual growth. Building the baseline detection pipeline with YOLOv11n and SAHI revealed the strengths of modern neural detectors, but also their limitations when faced with small, ambiguous, or context-dependent objects. Designing and implementing the symbolic reasoning layer in Prolog, with rules grounded in weighted knowledge graphs, was an effort to address these gaps. The final outputs systematic confidence adjustments across 136 relational cases confirmed that integrating symbolic knowledge could reshape detection results, reduce implausible errors, and improve interpretability.

This research is not just an academic exercise but also a reflection of my belief that AI systems deployed in real-world, high-stakes scenarios must go beyond raw performance metrics. They must be semantically reliable, contextually aware, and transparent in their reasoning. I hope this work contributes a small but meaningful step toward that vision.

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LIST OF ABBREVIATIONS

Acronym	Explanation
AI	Artificial Intelligence
CNN	Convolutional Neural Network
DOTA	Dataset for Object Detection In Aerial Images
ILP	Inductive Language Programming
LTN	Logic Tensor Networks
mAp	Mean Average Precision
OBB	Oreinted Bounding Boxes
SAHI	Slicing Aided Hyper Inference
UAV	Unmanned Aerial Vehicles
YOLO	You Only Look Once

1. INTRODUCTION

1.1. Subject Relevance

The ability to identify and precisely localize objects in aerial imagery has become essential across a broad spectrum of modern applications. With the steady rise of high-resolution satellite data and the rapid evolution of unmanned aerial vehicles (UAVs), object detection from above has moved well beyond a niche research area. It now serves as a cornerstone for operational systems in fields such as urban surveillance, environmental monitoring, disaster response, traffic management, infrastructure inspection, and defense. Each of these domains relies on timely and reliable recognition of objects of interest—whether that means detecting vehicles in a crowded parking lot, assessing collapsed buildings after an earthquake, tracking vessels in busy harbors, or spotting anomalies in bridges and railways.

The importance of aerial detection lies in the unique vantage point and operational flexibility that UAVs and other airborne platforms provide. A single image taken from several hundred meters above the ground can cover large spatial extents, allowing analysts or automated systems to survey wide areas in minutes. This perspective has clear advantages over ground-based observations, particularly in settings that are dangerous, inaccessible, or inefficient to reach. In emergency scenarios—for instance, locating survivors after a natural disaster or gauging the scale of flood damage—speed is critical. UAV-mounted sensors can be deployed within minutes, transmitting real-time imagery that feeds directly into detection algorithms to guide rescue operations. Similarly, in security and defense contexts, early detection of abnormal activities, such as unauthorized vehicles or unusual vessel patterns, can support rapid decision-making and threat mitigation.

Despite its value, aerial object detection presents distinct technical challenges. Unlike images captured at ground level, aerial views produce significant variability in how objects appear. Objects differ in scale—ranging from just a few pixels in the case of motorcycles or small boats, to hundreds of pixels for large ships or aircraft—depending on both object size and sensor altitude. Scenes may contain dense clusters of objects, such as cars in parking lots or vessels docked at a harbor, which requires models to separate instances despite overlap. Orientation is another factor: because aerial platforms can view the ground from any direction, objects may appear rotated at arbitrary angles. Ships, for example, may align with harbor edges or traverse diagonally across a frame, while vehicles follow curving roads or runways. To complicate matters further, aerial backgrounds often consist

of repetitive textures, such as rooftops, agricultural fields, or ocean waves, that can confuse detectors. These challenges call for more advanced strategies than those typically used for natural scene images.

Over the last decade, deep learning has reshaped object detection, delivering significant improvements in both speed and accuracy. Among the many contributions, the You Only Look Once (YOLO) family of detectors has been especially influential. YOLO unified classification and localization within a single end-to-end model, enabling real-time performance while maintaining competitive accuracy. Successive versions, from YOLOv1 through YOLOv8, have steadily enhanced backbone networks, feature fusion, and post-processing techniques.

For aerial imagery, one of the most notable advances has been the move from horizontal bounding boxes (HBBs) to oriented bounding boxes (OBBs). Conventional HBBs, aligned with the image axes, often capture large amounts of irrelevant background when objects are rotated, reducing precision in crowded scenes. OBBs, by contrast, rotate to match an object's orientation, providing a closer fit. This yields two clear benefits: first, localization is more accurate because the box aligns with the actual object boundary rather than including extraneous background pixels; second, objects in close proximity can be more easily separated, reducing false positives. Large-scale datasets such as DOTA (Dataset for Object Detection in Aerial Images) have accelerated this shift by offering richly annotated aerial images with OBB labels across multiple object classes, from small vehicles and ships to bridges and sports fields, and with diverse orientations.

The integration of OBB support into YOLOv8 represents an important milestone. Unlike earlier YOLO versions, which required complex adjustments to handle rotations, YOLOv8 incorporates this functionality natively. Initial benchmarks on DOTA suggest that OBB-enabled YOLOv8 improves average precision for rotated categories, particularly for small or crowded targets, while preserving the model's real-time performance. This makes it especially suited for UAV applications where both speed and accuracy are critical.

The practical impact of these advances is already evident. In infrastructure monitoring, OBB-enabled UAV systems can track vehicle positions on bridges, map traffic flows, and detect anomalies such as stalled cars. In maritime surveillance, they can classify vessels, monitor fishing activities, and flag unusual trajectories. During disaster assessments, UAVs can scan damaged areas, locate survivors, and prioritize zones for ground teams. Figure 1

provides an example from the DOTA dataset: ships and vehicles are captured at various angles, and OBB annotations closely match their shapes, in contrast with less efficient axis-aligned boxes. Histograms of object sizes and orientations further illustrate the diversity of aerial imagery, underscoring the need for rotation-aware models and robust multi-scale feature representations.

Beyond technical performance, the broader societal and economic relevance of aerial object detection is undeniable. As cities expand, transportation networks grow more complex, and climate-related disasters become more frequent, the demand for rapid and reliable situational awareness tools will only increase. UAV-based detection systems reduce costs, minimize human risk, and lighten workloads by automating tasks once handled manually or by manned aircraft. In defense and security, they enable persistent, wide-area monitoring with minimal staffing.

Nevertheless, important challenges remain. Detecting extremely small objects, sometimes only a few pixels in size, is still difficult. Models must also cope with variable lighting, weather, and sensor conditions, while providing results that are interpretable for human decision-makers in critical applications. Looking forward, neurosymbolic approaches that combine the perceptual power of deep neural networks with the logical reasoning of symbolic systems hold considerable promise. By embedding contextual rules and domain knowledge into data-driven models, such methods may deliver more robust, transparent, and trustworthy detection for complex aerial environments.

1.2. Motivation

My interest in the topic of this thesis stems from a long-standing curiosity about the intersection between computer vision and artificial intelligence reasoning. While deep learning has revolutionized object detection in recent years, my academic and project experiences have repeatedly shown me that purely data-driven approaches have inherent blind spots. These models excel when operating in familiar conditions scenarios well-represented in their training data, but they often fail when confronted with rare, ambiguous, or context-dependent cases.

This observation became clear to me during earlier computer vision projects where I worked with conventional convolutional neural network (CNN)-based object detectors. In several instances, I encountered mispredictions that, to a human observer, would have been

obvious errors. For example, a model detecting vehicles in aerial images might correctly identify a car on a highway but miss another vehicle partially hidden by a tree. More frustratingly, it might falsely detect a patch of road markings as a vehicle because the texture resembled patterns in its training set. These failures were not due to a lack of computational power or insufficient data augmentation; they reflected a deeper limitation, an absence of reasoning about context and relationships between objects.

The idea of bridging the gap between raw perception and higher-level reasoning became a personal pursuit. Symbolic reasoning, with its ability to represent explicit rules, relationships, and constraints, offers exactly what deep neural networks lack: a means to reason about why something might or might not be present in a scene, even if the visual cues are incomplete or noisy. This led me to explore neurosymbolic approaches, where visual perception from deep networks can be combined with symbolic logic to create systems that not only see but also understand.

My motivation was further strengthened by real-world observations in critical application domains. For instance, in post-disaster damage assessment, UAV-mounted detectors are deployed to quickly scan large areas and locate damaged infrastructure or stranded people. While these detectors are fast, they often struggle in chaotic environments where debris, shadows, and unusual object placements confuse them. A deep learning model might detect broken road segments as simply “space” without realizing the implications for transportation routes. A reasoning layer, however, could interpret such a gap as a missing bridge if it recognizes that the road was continuous in earlier images and that water or a valley lies beneath.

Similarly, during cultural heritage site mapping, the limitations of current UAV-based detection systems became evident. Many heritage sites contain small, intricate structures whose visual patterns are not prominent in typical datasets. Existing detectors may overlook these entirely, even though they are historically significant. The ability to reason about spatial arrangements, such as detecting that a certain type of wall layout implies the presence of a temple foundation, could make these systems invaluable for preservation efforts.

The possibility of creating a system that detects objects while reasoning about their context became the central driver of my research motivation. Such a system would not be limited to recognizing patterns it has seen before but could use logical constraints, relationships,

and prior knowledge to fill in gaps, resolve ambiguities, and identify anomalies. In effect, it could mimic a key aspect of human visual intelligence: the ability to infer the presence or absence of something based on relationships and context, not just pixel patterns.

Bringing together the raw perceptual strength of models like YOLO with the structured reasoning of symbolic AI aligns perfectly with my academic goals and my vision for the future of intelligent systems. In this work, I aim to explore how these two paradigms can be integrated into a cohesive framework for aerial object detection, using the DOTA dataset as a proving ground. My aspiration is not only to achieve measurable improvements in accuracy but also to demonstrate the broader potential of neurosymbolic methods in solving high-stakes, real-world problems where understanding is as important as detection.

1.3. Research Gap

Over the past decade, deep learning has revolutionized computer vision, with convolutional neural network (CNN)-based detectors such as YOLO, Faster R-CNN, and RetinaNet setting new benchmarks for accuracy and speed in object detection. These models have proven remarkably effective at learning rich feature representations from large datasets, enabling them to detect a wide variety of objects under diverse conditions. In the context of aerial imagery, this capability has been leveraged for applications ranging from traffic monitoring to environmental assessment and disaster response. However, despite these successes, there are persistent limitations when deep learning models are used in isolation.

One fundamental limitation is the lack of contextual understanding. Purely data-driven detectors operate primarily as pattern recognizers: they match visual features in the input image to patterns seen in training data, outputting bounding boxes and class labels. While this approach works well when the visual cues are clear and consistent, it struggles in scenarios where such cues are incomplete, ambiguous, or misleading. For example, a detector may correctly identify an airplane when it is fully visible on a runway but fail to recognize it if part of the fuselage is obscured by a hangar or shadow. Similarly, small or highly occluded objects common in UAV imagery are often missed altogether because their visual signature is weak compared to the surrounding background.

Another significant weakness is the difficulty in handling rare events or underrepresented classes. Deep learning models depend heavily on the availability of large, diverse training datasets to generalize well. In aerial detection tasks, some object categories may be infrequently observed, such as damaged infrastructure after a natural disaster or temporary structures at cultural heritage sites. With limited training examples, the model's learned representation for these classes may be incomplete, leading to poor recall and high false-negative rates. This problem is compounded when the operational environment changes, as the model cannot adapt on the fly without retraining.

Equally important is the absence of domain knowledge in most current aerial detection systems. A human analyst, when examining aerial imagery, naturally applies background knowledge to interpret the scene. For example, they might reason that airplanes are likely to be found on or near runways, ships in water bodies, or vehicles on roads. This reasoning allows humans to resolve ambiguities, such as distinguishing a boat-shaped rooftop decoration from an actual boat by considering its location. CNN-based detectors, however, have no explicit way of incorporating such knowledge into their decision-making process. They treat each detection as an isolated classification problem, without considering spatial, semantic, or relational context.

This lack of contextual reasoning becomes especially problematic in high-stakes applications like disaster response. Imagine a UAV scanning a flood-affected region: a deep learning detector may detect several debris piles but miss a stranded vehicle because it appears partially submerged and does not resemble the typical training examples. A reasoning-enabled system could instead infer the likely presence of a vehicle based on visible portions, its alignment with a known road, and the absence of other plausible explanations. Without such reasoning, purely data-driven systems remain prone to misinterpretations in complex or novel situations.

While these limitations are recognized in the broader computer vision community, the aerial vision domain has yet to see widespread integration of symbolic reasoning into detection pipelines. Most research in UAV-based object detection focuses on improving neural network architectures, adding attention mechanisms, multi-scale feature fusion, or anchor-free detection heads to extract more robust visual features. These efforts undoubtedly improve baseline performance, but they do not address the deeper cognitive gap: the ability to combine perception with reasoning.

This is where neurosymbolic integration offers a promising yet underexplored solution. Neurosymbolic methods combine the perceptual strengths of neural networks with the explicit reasoning capabilities of symbolic logic. Neural networks excel at extracting patterns from raw pixel data, while symbolic reasoning can represent and manipulate structured knowledge, enforce logical constraints, and explain decisions. In an aerial detection setting, this could mean using a YOLO detector to identify candidate objects, followed by a symbolic reasoning layer that evaluates these detections in light of spatial relationships, domain-specific rules, and contextual information. Such a system could, for example, detect a missing bridge by reasoning about the discontinuity of a road network, even if the visual evidence of the bridge itself is absent or degraded.

Despite its potential, the neurosymbolic approach in aerial vision remains largely absent from current literature. While there has been progress in combining deep learning with symbolic reasoning for tasks like scene understanding or question answering in natural images, similar work applied to UAV imagery is scarce. The few existing attempts to integrate domain knowledge into aerial detection tend to rely on handcrafted post-processing steps rather than a tightly coupled neural-symbolic framework. This leaves a clear research gap: the systematic design and evaluation of an aerial object detection pipeline that unites high-performance neural perception with robust symbolic reasoning.

Addressing this gap is the central aim of this thesis. By proposing a hybrid architecture that integrates YOLO-based visual detection with a symbolic reasoning module implemented in SWI-Prolog, this work seeks to overcome the limitations of purely data-driven models. The proposed framework will not only detect objects in aerial images but also reason about their plausibility, relationships, and contextual meaning. This integration is expected to improve performance in challenging scenarios such as detecting rare or context-dependent objects, handling small and occluded targets, and identifying anomalies while providing a level of explainability that is often missing in black-box neural networks.

1.4. Objectives and Goals

The goal is to design and implement a neurosymbolic object detection system for aerial images that complements neural detection (YOLOv11n + SAHI) with symbolic reasoning (Prolog + weighted knowledge graphs), thereby improving the plausibility, reliability, and interpretability of detection results.

- **Develop a robust perception baseline** by training an oriented bounding box detector (YOLOv11n-OBb) on the DOTA dataset, using SAHI slicing to handle ultra-high-resolution images and improve small object visibility.
- **Construct a symbolic reasoning layer** using SWI-Prolog that encodes spatial and co-occurrence knowledge into a weighted knowledge graph, enabling the system to apply context-aware confidence boosts and penalties to YOLO detections.
- **Integrate the neural and symbolic components** into a unified pipeline where detections are first generated by YOLO+SAHI and subsequently refined through logical rules that align with dataset-derived priors.
- **Evaluate the hybrid pipeline empirically** by comparing detector confidences before and after symbolic reasoning, analyzing confidence adjustments across object–relation pairs to determine how plausibility improves.
- **Demonstrate interpretability** by providing rule-based explanation traces for each adjustment, ensuring that detection refinements are transparent, traceable, and explainable to end users.

Assess practical impact by analyzing how symbolic reasoning reduces implausible false positives, preserves borderline detections, and aligns outputs more closely with real-world expectations.

1.5 Research Questions

RQ1. Can integrating symbolic reasoning with a YOLOv11n-OBb detector systematically improve the plausibility of aerial image detections, as measured by confidence adjustments on context-dependent objects?

RQ2. How can weighted knowledge graphs derived from dataset statistics be used to encode domain knowledge in a way that strengthens plausible detections and suppresses implausible ones?

RQ3. To what extent does SAHI slicing support the symbolic reasoning process by ensuring small and borderline detections are visible and available for logical reinforcement?

RQ4. How does the neurosymbolic pipeline change the error profile of the baseline detector specifically in terms of reducing false positives, preserving borderline true positives, and improving semantic plausibility of outputs?

RQ5. What are the trade-offs and limitations of the hybrid approach such as computational overhead, dependence on hand-crafted rules, and scalability and how can these be mitigated in future work?

1.6 Scope & Limitations

The scope of this thesis is intentionally focused and well-defined to ensure depth of analysis and clarity of contributions. The research focuses on object detection in static aerial images, rather than object tracking, semantic segmentation, or other related tasks. The emphasis is on detecting and classifying objects from high-resolution aerial imagery, with the DOTA dataset serving as the primary benchmark due to its rich variety of annotated object categories and challenging aerial perspectives. For broader validation and to assess generalization, additional experiments may be conducted on datasets such as VisDrone, which offers complementary scene characteristics. The system targets a subset of object categories aligned with the DOTA dataset's predefined classes, such as airplanes, ships, vehicles, bridges, and buildings, covering approximately 15 classes in total.

The methodology integrates a YOLO-based neural detection backbone with SAHI (Slicing Aided Hyper Inference) for improved small object detection, coupled with SWI-Prolog for symbolic reasoning over detection outputs. Development and experimentation are carried out in a research-oriented environment using Python, PyTorch, and SWI-Prolog, with performance evaluated through standard object detection metrics such as mean Average Precision (mAP). The project is framed within the context of academic research, prioritizing proof-of-concept validation and scientific contribution over production-level deployment.

Despite its well-defined objectives, the thesis is subject to several limitations that shape its applicability and potential for real-world use. First, computational complexity presents a practical constraint. The combination of SAHI for image tiling and the symbolic reasoning stage in Prolog increases inference time compared to purely neural approaches. While this enhances detection accuracy for small and densely packed objects, it may hinder real-time processing, especially in resource-limited environments such as onboard UAV systems.

Second, the rule definition process in the symbolic reasoning component is inherently dependent on prior knowledge and domain-specific assumptions. The logical rules and knowledge graph are designed for scenarios represented in the training datasets. When

applied to unfamiliar contexts such as novel object classes, unseen relationships, or significantly different environmental conditions, these rules may fail to produce meaningful inferences, leading to degraded performance.

Third, dataset bias is an important consideration. The system is primarily tuned to the characteristics of DOTA imagery, which predominantly features nadir or near-nadir views, specific spatial resolutions, and particular object categories. While additional datasets like VisDrone can help mitigate this bias, transferring the approach to other aerial domains such as oblique satellite imagery, thermal aerial data, or drone footage captured under different lighting and weather conditions will likely require adaptation of both the neural and symbolic components.

Lastly, the absence of temporal reasoning limits the system's capacity to exploit context across multiple frames. Since the scope is restricted to single-image analysis, temporal cues that could improve object disambiguation or reasoning, such as motion patterns, object persistence, or dynamic scene understanding, are not leveraged.

2. LITERATURE REVIEW

2.1 Object Detection in Aerial Images: Challenges and Current Techniques

Aerial imagery possesses characteristics that clearly distinguish it from conventional natural images, demanding dedicated techniques for analysis and object detection. Among the most significant is the pronounced variation in object scale. In a single aerial frame, the size of objects can vary dramatically. For example, an urban scene may feature tiny parked cars situated next to expansive warehouses or large office complexes. Such scale diversity presents a challenge for detection algorithms, which must accurately identify both the smallest and the largest objects without sacrificing precision. Traditional detection methods often fall short in this regard, with scale inconsistencies leading to missed detections or incorrect classifications when algorithms fail to effectively process objects across such a broad size range (Bakrač et al., 2021).

Dense object distribution is another defining feature of aerial images. Urban areas, for instance, frequently contain vehicles, buildings, and vegetation clustered closely together, creating occlusions that obscure smaller objects behind larger ones. In a busy parking lot,

several compact cars may visually merge into a single mass, complicating efforts to identify each vehicle individually. Tackling this requires detection approaches capable of navigating high-density environments and resolving overlapping elements. Evidence from recent research suggests that advanced models, particularly those based on convolutional neural networks (CNNs), are far more adept at handling these situations than traditional detection methods (Zhou & Yu, 2022; Zhong et al., 2017).

Equally challenging is the arbitrary orientation of objects captured in aerial images. Unlike in ground-level photography, where subjects tend to maintain an upright orientation, aerial perspectives can depict objects from virtually any angle. An aircraft photographed on a runway, for instance, might appear aligned horizontally, diagonally, or at another oblique angle within the frame. Such variations pose difficulties for detection systems that rely on standard horizontal bounding boxes. To address this, more recent models have adopted oriented bounding boxes (OBB), which adjust to an object's specific angle and thereby improve detection performance in aerial contexts (Xu et al., 2016; Tang et al., 2017).

These three traits, significant scale variation, dense object placement, and unrestricted orientation, underscore why aerial imagery requires approaches distinct from those used for traditional image analysis. Effective solutions in this field demand not only specialized methodologies but also the integration of advanced detection technologies (Wang et al., 2019; Carlet & Abayowa, 2017). By developing methods that account for these unique challenges, researchers can maximize the value of aerial imagery in applications ranging from urban planning to environmental monitoring.

Benchmark datasets are fundamental to advancing aerial object detection, as they offer standardized grounds for testing and comparing different detection algorithms. Among the most widely recognized are DOTA (Dataset for Object Detection in Aerial Images) and VisDrone, each tailored to address particular challenges within aerial imagery analysis.

DOTA stands out for its impressive scale, containing more than two million annotated object instances spanning 15 categories, all labeled with oriented bounding boxes. Designed to support both training and evaluation, it caters specifically to the detection and localization of objects in aerial scenes, where variations in scale and perspective are common. The use of oriented bounding boxes (OBBs) is particularly valuable in this context, as it enables algorithms to better handle objects captured at arbitrary angles—an inherent trait of aerial imagery (Xia et al., 2018). By covering a diverse range of

categories, from vehicles to buildings and vegetation, DOTA allows researchers to develop and assess models capable of generalizing across different object types and environments. Performance comparisons on this dataset not only reveal the strengths and limitations of specific approaches but also inform improvements for future work (Xia et al., 2018; Xiao et al., 2020).

In contrast, the VisDrone dataset concentrates on UAV-captured imagery, with a strong emphasis on detecting small objects such as pedestrians and vehicles. This focus addresses the difficulty of maintaining accuracy when objects are both small and densely clustered conditions that frequently degrade detection performance (Reckling et al., 2021). Featuring a variety of scenes and environmental conditions, VisDrone is a critical benchmark for algorithms targeting small-object detection. Studies leveraging VisDrone frequently aim to boost performance metrics for these cases, given their practical significance in domains like urban surveillance and traffic monitoring (Zhang et al., 2022).

The complementary nature of these datasets has inspired a range of research efforts. For example, Hu et al. (2023) enhanced small-object detection capabilities within the YOLO framework, adapting it for stronger performance on both DOTA and VisDrone. Similarly, Zhang et al. (2022) demonstrated how refinements in model architecture could yield substantial gains in detection accuracy on these benchmarks.

Overall, datasets such as DOTA and VisDrone are indispensable to the evolution of aerial object detection. DOTA's extensive coverage and OBB annotations support comprehensive testing across numerous object types, while VisDrone's targeted approach to small-object detection fills a vital niche in UAV image analysis. Together, they enable rigorous comparative evaluations and drive innovations that translate detection improvements into effective real-world solutions.

Over the past decade, aerial object detection has undergone substantial advancements, with two major categories of detection algorithms emerging as dominant approaches: two-stage detectors, exemplified by Faster R-CNN, and one-stage detectors, including SSD (Single Shot Multibox Detector), RetinaNet, and the YOLO (You Only Look Once) family of models. Each offers distinct strengths and limitations, and their suitability often depends on the specific challenges posed by aerial imagery.

Two-stage detectors—with Faster R-CNN as a prime example—follow a two-step process: first generating candidate object regions from the input image, then classifying these proposals and refining their bounding boxes. This approach is well-regarded for its high accuracy and precise localization, which are crucial in aerial imagery, where objects often vary greatly in scale and can be partially occluded. Several studies highlight Faster R-CNN's effectiveness in aerial contexts, demonstrating its capacity to capture fine-grained features and adapt to the diverse object scales present in high-resolution aerial datasets (Xu et al., 2017; Ammar et al., 2019). For instance, its application in vehicle detection and infrastructure monitoring has yielded high precision, outperforming many traditional detection approaches (Ammar et al., 2019; Ammar et al., 2021). However, this heightened accuracy comes at a computational cost—two-stage detectors typically have slower inference times, limiting their practicality in real-time aerial surveillance or UAV-based operations where rapid decision-making is essential.

One-stage detectors, by contrast, perform classification and localization in a single processing step, offering substantial improvements in speed. The YOLO series, in particular, has become a preferred choice in UAV-based aerial detection due to its ability to process images in real time while maintaining competitive accuracy. YOLO's streamlined architecture enables it to handle high frame rates, which is advantageous for applications such as live traffic monitoring or disaster response. Nevertheless, this speed advantage often entails a slight drop in mean average precision (mAP) compared to two-stage systems (Huang et al., 2017; Fang et al., 2021). Comparative analyses have shown that models like YOLOv3 perform relatively well in detecting small aerial objects, though they often require careful hyperparameter tuning or architectural enhancements—such as feature pyramid networks (FPNs) or attention modules to boost detection performance, especially for small or densely clustered targets (Ammar et al., 2021).

Beyond YOLO, other one-stage frameworks such as SSD and RetinaNet have also gained traction in aerial imagery analysis. SSD leverages multi-scale feature maps from different convolutional layers, allowing it to detect objects of varying sizes more effectively—a particularly important capability given the extreme scale variability found in aerial datasets. RetinaNet, on the other hand, introduced the focal loss function to address the issue of class imbalance, a common challenge in aerial datasets where certain categories (e.g., vehicles) may dominate while others (e.g., rare infrastructure types) are underrepresented. This innovation improves the detection of minority classes without

sacrificing performance on more common categories (Hong et al., 2019; Tayara & Chong, 2018).

Recent research has increasingly explored hybrid and enhanced detection models that merge the strengths of both one-stage and two-stage approaches. For example, incorporating region proposal mechanisms into one-stage models, or integrating speed-oriented designs into two-stage frameworks, has led to the development of architectures that balance accuracy and real-time efficiency. Some studies have adapted YOLO with advanced backbone networks (e.g., CSPDarknet53) or incorporated transformer-based attention mechanisms to enhance feature representation in aerial imagery, yielding significant performance gains.

In practical applications, the choice between detection methods is rarely binary. High-stakes scenarios, such as military reconnaissance, search-and-rescue missions, or critical infrastructure monitoring, often demand the precision of two-stage detectors. Meanwhile, time-sensitive tasks—such as live UAV-based traffic analysis or disaster damage assessment—benefit more from the speed of one-stage models. The ongoing refinement of detection algorithms, including experiments with multi-scale training, contextual feature aggregation, and domain-specific fine-tuning, is steadily narrowing the performance gap between these categories.

In conclusion, both two-stage and one-stage detection frameworks have proven indispensable for aerial imagery analysis, each excelling in different operational contexts. Two-stage detectors like Faster R-CNN remain the gold standard for high-precision detection, while one-stage models such as YOLO, SSD, and RetinaNet offer unmatched real-time performance. The emergence of hybrid approaches further blurs the boundaries between these categories, promising models that can deliver both speed and accuracy.

Oriented Bounding Boxes (OBBs) mark a substantial step forward from traditional Horizontal Bounding Boxes (HBBs) in aerial object detection. While HBBs are constrained to axis-aligned rectangles, OBBs incorporate orientation into their representation, enabling more precise framing of objects that are not aligned with the image axes. This is especially valuable in aerial imagery, where targets such as ships, vehicles, and buildings may appear at arbitrary angles depending on the perspective of the drone or camera. The adoption of OBBs is particularly justified by their ability to significantly reduce background clutter while improving the localization of rotated

objects—advantages that horizontal bounding boxes often fail to deliver (Muhlhaus et al., 2023; Yi et al., 2020).

An OBB defines an object’s bounding area in a way that reflects both its orientation and true physical dimensions, producing a closer fit to the object’s actual shape. This tighter and more adaptive framing directly enhances localization accuracy—an essential factor in aerial contexts where many targets are small or captured at oblique angles. By more closely aligning with object boundaries, OBBs minimize the inclusion of irrelevant background pixels, thereby reducing the risk of false positives and improving classification precision (Yi et al., 2020). The benefits are particularly evident in cluttered scenes, where horizontal boxes often encompass excessive non-object regions that confuse detection algorithms.

Recent advances in detection frameworks have increasingly integrated OBB support. For example, Rotated RetinaNet adds rotation prediction into its detection head, improving accuracy for arbitrarily oriented targets, while Oriented R-CNN incorporates orientation-sensitive proposals to address the challenges of non-axis-aligned objects. Evaluations of these models on benchmark datasets consistently show that orientation-aware detection produces measurable performance gains when object rotation is a significant variable (Yi et al., 2020).

Among the more notable developments is YOLOv8-OBB, which extends the high-speed YOLO detection pipeline to include orientation prediction. This innovation preserves YOLO’s hallmark real-time capability while enabling it to detect rotated objects with improved precision. The resulting combination of speed and orientation sensitivity makes it well-suited for UAV surveillance, traffic monitoring, and other time-critical applications where both responsiveness and accuracy are essential (Yi et al., 2020; Wang et al., 2019).

A practical example illustrates the point: consider an aircraft captured diagonally within a busy airport scene. An HBB might encapsulate the aircraft along with a considerable portion of surrounding tarmac and nearby vehicles, introducing background noise that could lead to misclassification. In contrast, an OBB can be rotated to closely match the aircraft’s true orientation, sharply reducing background interference and providing a high object density or visual clutter (Kirsten et al., 2024).resclearer input for classification models. This difference becomes critical in environmental observation.

The integration of OBBs into aerial object detection frameworks represents a decisive evolution in methodology. By improving the accuracy of object localization and reducing false positives caused by excess background, OBB-based approaches offer clear advantages over traditional HBB systems. As research continues to refine these methods, OBB-enhanced detection is poised to become an essential standard for applications in urban planning, disaster response, environmental monitoring, and other aerial imaging domains where orientation plays a key role.

2.2 Small Object Detection and Multi-Scale Techniques

Small object detection remains one of the most persistent challenges in computer vision, particularly within aerial imagery, where even the smallest objects can carry significant importance for applications such as surveillance, infrastructure monitoring, or agricultural analysis. By definition, small objects occupy only a limited number of pixels in an image, and this reduced representation poses a set of intertwined challenges for detection systems.

A primary issue is the minimal pixel occupancy of small objects. Since these objects cover only a few pixels, they often become visually indistinguishable from the surrounding background, especially as convolutional neural networks (CNNs) progressively downsample the input. Standard CNN architectures reduce spatial resolution at each layer, causing the fine-grained features of small objects to disappear in the process. As a result, critical information that could help separate these objects from the background is lost, making them particularly difficult to detect (Li et al., 2020). This degradation of detail directly translates into reduced detection accuracy, particularly in complex aerial scenes where multiple objects coexist and overlap.

The challenge becomes even greater in dense or cluttered scenes. In such environments, small object features are often overshadowed by the strong responses generated by larger objects during feature fusion. This “feature overcrowding” problem reduces the visibility of small targets, as dominant features from larger objects dominate the feature maps (Li et al., 2023). For example, in an aerial image containing buildings, vehicles, and vegetation, the overwhelming presence of large structures can mask the subtle patterns of smaller targets, leading to frequent missed detections.

Another obstacle is associated with the design of anchor boxes, which form the foundation of many detection architectures. Anchor boxes are typically predefined in size, and when

they do not align with the true dimensions of smaller objects, the detection results are often poor. Zhao et al. (2021) emphasize that default anchor sizes frequently fail to match the scales of small objects, leading to misaligned bounding boxes and reduced detection rates. This highlights the necessity of tailoring anchor box configurations to the dataset and application domain in order to improve detection performance.

These difficulties have been most prominently highlighted through benchmark datasets such as VisDrone and DOTA, both of which contain significant proportions of small objects. VisDrone, in particular, emphasizes UAV-captured imagery featuring pedestrians and vehicles that are often only a few pixels in size, making it a cornerstone for evaluating small object detection (Reckling et al., 2021; Zhang et al., 2022). Similarly, DOTA includes numerous small-scale targets—such as cars and ships—that appear at varying orientations and scales, which further exacerbates the difficulty of achieving accurate detections (Xia et al., 2018; Xiao et al., 2020). The presence of these small and densely distributed targets within these datasets has directly motivated research into specialized detection strategies.

To overcome these limitations, researchers have proposed a variety of multi-scale and adaptive strategies. Scale-adaptive algorithms dynamically adjust anchor box sizes or feature representations to better capture small objects, ensuring that they remain distinguishable across layers of the network (Chen et al., 2023). Advanced frameworks such as SAE-CenterNet integrate self-attention mechanisms with multi-scale feature fusion, allowing the model to focus more effectively on the distinct characteristics of small objects even in dense environments (Li et al., 2023).

Dedicated architectures have also emerged to specifically enhance small object detection. For instance, OAB-YOLOv5 modifies the YOLO framework by incorporating orientation-aware and scale-sensitive modules that improve the detection of small objects in aerial imagery. Other approaches employ densely connected multi-scale pyramids, which capture features across multiple resolutions and provide the model with broader contextual information, improving recognition across a wide range of object sizes. Such strategies ensure that even the smallest targets retain meaningful representations within the detection pipeline.

In conclusion, small object detection in aerial imagery is constrained by fundamental issues such as pixel scarcity, background interference, and mismatched anchor

configurations. Nevertheless, benchmark datasets like DOTA and VisDrone have been instrumental in revealing these limitations and guiding the development of new solutions. Recent progress in adaptive scaling, multi-scale fusion, and specialized network architectures has led to measurable improvements in this domain. As demand grows in fields such as aerial surveillance, traffic monitoring, and environmental assessment, continued innovation in small object detection methods will be critical.

In the domain of object detection, particularly when addressing small objects in aerial imagery, substantial progress has been achieved through the development of specialized detection models. The unique challenges posed by small object detection such as minimal pixel representation and the risk of being obscured by background clutter have driven innovations in both model architecture and training strategies.

Recent adaptations of the YOLO (You Only Look Once) family illustrate these advancements. A notable example is DS-YOLOv7 (Dense Small YOLOv7), which was explicitly designed to improve the detection of densely distributed small objects. This model introduces a shallow feature fusion network (SFN) that helps retain critical low-level features, which are often lost during the downsampling process in conventional CNNs. In addition, a modified spatial pyramid pooling layer (SPPLD) has been incorporated to strengthen the extraction of fine-grained details essential for distinguishing small targets. Evaluations on the VisDrone dataset have shown that DS-YOLOv7 achieves a measurable improvement in mean Average Precision (mAP) over the baseline YOLOv7, highlighting its enhanced capability in small object detection. However, precise quantitative results remain uncertain, as supporting references were not identified to substantiate the reported gains, suggesting that further verification is necessary (Qaddour, 2023; Çelik, 2022).

The progression of YOLO continues with YOLOv8, which introduces several architectural refinements aimed at addressing multi-scale detection challenges. One of the key features is multi-scale feature fusion, where additional detection layers allow the model to capture richer spatial information across different resolutions. This improvement is particularly valuable for aerial imagery, where small objects are frequently overshadowed by complex environments. YOLOv8 also integrates conditionally parameterized convolutions, enabling adaptive filtering that scales depending on the context of the scene. Moreover, a novel loss function for bounding box regression improves localization precision, which is critical for small object detection. While initial reports suggest that YOLOv8 demonstrates improved

results on datasets such as VisDrone, specific mAP values were not directly supported in the existing references and require validation (Zhang et al., 2022; Tian et al., 2020).

Building on this foundation, the Adjustable Receptive Field YOLOv8 (ARF-YOLOv8) has emerged as a tailored extension designed to further mitigate small object detection issues. ARF-YOLOv8 modifies the backbone architecture to expand receptive fields, allowing the network to capture detailed information from smaller objects more effectively. A bi-branch attention module complements this adjustment by dynamically fusing shallow and deep features, thereby improving representation across scales. Reports claim performance improvements of ARF-YOLOv8 over standard YOLOv8, though supporting evaluations on the VisDrone dataset are essential for verification and must be properly cited in the literature (Güler et al., 2018; Kononov, 2018).

Beyond incremental YOLO developments, ensemble-based approaches have also been explored for small object detection. These methods combine multiple detectors and apply fusion voting mechanisms, leveraging the complementary strengths of individual models. Some studies indicate that such strategies can yield higher mAP scores on datasets like VisDrone, underscoring the potential of aggregation methods. In parallel, transformer-based detection frameworks and attention modules have begun to show promise by enhancing the network's ability to capture subtle object cues in visually complex scenes. However, these claims also require further validation through accurate referencing (Alshaer et al., 2025; Li et al., 2020).

Taken together, these developments highlight a growing trend in the literature toward designing architectures that explicitly address the challenges of small object detection in aerial imagery. The iterative refinements across YOLO variants, complemented by ensemble and transformer-based strategies, indicate a steady movement toward detection systems capable of handling the nuanced difficulties presented by datasets such as VisDrone and DOTA.

Within the broader effort to advance small object detection in large-scale imagery, the Slicing Aided Hyper Inference (SAHI) framework has emerged as an important model-agnostic solution. A persistent challenge in this domain stems from the fact that small objects occupy very few pixels in high-resolution images, making them highly vulnerable to information loss during the downsampling and feature extraction stages of standard Convolutional Neural Networks (CNNs). As a result, such objects can be easily

overshadowed during feature fusion with larger or denser features, ultimately reducing detection accuracy ("Research design and methods 7 Research design 7 Implications and contributions to knowledge 8", n.d.).

SAHI mitigates this problem by applying a tiling strategy that divides a large image into overlapping sub-images. By focusing on smaller patches, objects that might otherwise appear insignificant in the full image become more prominent, thereby improving their visibility to detection algorithms. This approach effectively preserves local features and ensures that subtle object characteristics are not lost during processing. Consequently, CNN-based detectors are better equipped to identify and classify small targets, since SAHI emphasizes localized regions of interest where such objects are more likely to appear (Bakrač et al., 2021).

The benefits of SAHI have been validated on widely used aerial benchmarks such as VisDrone and DOTA, which are known for their high density of small and arbitrarily oriented objects. On these datasets, incorporating SAHI into detection frameworks has consistently yielded performance gains, with average precision (AP) improvements of around 5% to 7%. Moreover, with additional fine-tuning on sliced data, the total AP gains have reached 12% to 14%, underscoring the framework's ability to recover information that would otherwise be lost in large-scale imagery (Zhou & Yu, 2022). These results reinforce the practical relevance of SAHI, as both VisDrone and DOTA present challenges typical of real-world aerial scenarios, including crowded urban landscapes and dense object distributions.

Beyond its quantitative contributions, SAHI also highlights the broader design principles needed for advancing small object detection. By amplifying the presence of small-object features while simultaneously suppressing background noise, the framework strengthens the robustness of detection models in practical applications. This capability is especially valuable in aerial surveillance, urban monitoring, and disaster response, where seemingly minor details—small vehicles, structural anomalies, or hidden objects—can carry significant operational importance.

Furthermore, the philosophy behind SAHI aligns closely with recent architectural innovations in detection models. Techniques such as shallow feature fusion networks and spatial pyramid pooling share a common emphasis on enhancing the visibility of fine-grained object details that are otherwise prone to being overlooked. Together, these

approaches reflect a growing recognition that improving small object detection requires not only deeper models but also strategic preprocessing and feature preservation mechanisms (Zhong et al., 2017).

In summary, SAHI provides a robust, model-agnostic solution that addresses one of the most persistent challenges in object detection. By leveraging tiling to preserve small object features and improve contextual visibility, SAHI enhances both detection accuracy and interpretability. Its demonstrated improvements on benchmark datasets such as DOTA validate its role as a valuable addition to existing detection pipelines and signal its potential for broader adoption in future research and applied systems.

2.3 Limitations of Deep Learning

Deep learning has undoubtedly transformed object detection, but its effectiveness comes with important limitations. One of the most pressing challenges lies in its reliance on large, annotated datasets and its difficulty in handling rare object categories or unusual scenes. For deep detectors to generalize well, they need broad and diverse training data. Yet, even well-known aerial datasets such as DOTA and VisDrone suffer from class imbalance. Common categories like vehicles and buildings are abundantly represented, while rare ones such as “wind turbines” or “helipads” appear only a handful of times.

When rare categories are poorly represented in training, models often fail to recognize them during inference. A neural network may simply overlook a wind turbine in a large aerial image because it has not seen enough examples to form a strong representation. By contrast, symbolic reasoning systems can fall back on contextual knowledge for example, knowing that “wind turbines are usually found in open fields” to make reasonable inferences even when visual evidence is weak (Hwang et al., 2019; Welikala et al., 2020).

The core issue is that deep networks primarily learn from explicit patterns present in training data. With only a few samples of rare objects, the model cannot build robust internal features, leading to poor performance when those objects appear in new contexts. This stands in stark contrast to human perception, where people can often recognize familiar objects based on partial or indirect cues (Koo et al., 2021).

Another related problem is dataset imbalance. Models tend to favor the frequent classes in training while neglecting rarer ones. This imbalance becomes even more challenging when the rare objects are also small in the image. Their features can easily get lost in complex

backgrounds or overshadowed by larger structures. As a result, even large datasets do not always guarantee strong performance if diversity across classes and scales is lacking (Okunev et al., 2020).

To mitigate these challenges, several strategies have been explored. Data augmentation is one common approach, where training images are altered through transformations such as scaling, rotation, or color adjustment. This helps expand the variety of examples the model sees, making it more resilient in practice (Zagane et al., 2020). Another promising direction is the use of synthetic data, often generated through Generative Adversarial Networks (GANs). These artificial examples can supplement underrepresented categories, offering models the extra exposure they need to detect rare objects reliably.

Alongside these data-driven methods, symbolic knowledge integration is gaining attention as a way to address deep learning's shortcomings. By embedding contextual information through knowledge graphs or logical rules, models can simulate aspects of reasoning that go beyond pure pattern recognition. For instance, a model might use the symbolic link between runways and airplanes to increase its confidence in detecting small or rare airplanes in aerial imagery (Yamaguchi et al., 2019).

In sum, while deep learning has achieved remarkable progress, its dependence on large, balanced datasets limits its performance on rare or small object categories. The path forward likely lies in hybrid approaches combining data augmentation and synthetic data with contextual reasoning frameworks.

Despite the remarkable success of Convolutional Neural Networks (CNNs) in image recognition and object detection, one of their core limitations lies in the absence of reasoning and contextual understanding. CNNs function almost entirely on appearance-based learning: they pick up patterns from training data and map them to labels but do not possess the ability to reason about spatial relationships or commonsense constraints the way humans do. This gap can result in predictions that are visually plausible to the model but contextually absurd in the real world.

Take the example of a YOLO detector, a widely adopted CNN-based model. In some cases, it might detect multiple airplanes in an image, with outputs showing them overlapping unnaturally or even appearing to float mid-air, detached from any runway. While the detections make sense to the network based purely on visual cues, human

observers instantly recognize such predictions as implausible because they violate basic physical constraints of how airplanes operate. The absence of an internal mechanism for abstract reasoning or contextual validation means CNNs can present these erroneous outcomes with high confidence.

This limitation becomes even more significant when the task involves rare objects or unusual scenarios. If the dataset contains, a knowledge graph could represent the fact that airplanes typically co-occur with runways, or that ships belong in water rather than floating mid-air. Embedding such contextual information into the detection pipeline allows models not just to see what is in the image, but also to reason about whether the detection makes sense in the real world.

There is also growing interest in modeling aspects of cognitive reasoning directly within neural architectures. This includes exploring ways for networks to learn not just visual features, but also the rules and relationships that govern objects in space. Such approaches point toward a future where deep learning models are less prone to errors caused by missing context and more capable of “reasoning” in a way that parallels human interpretation. In very few examples of this class, the model may fail to generalize and completely miss them at inference time. Even if a handful of turbine images are included in training, the model may not activate the right features when faced with turbines in different contexts—say, partly obscured, at unusual angles, or surrounded by unexpected backgrounds. In these cases, the model’s lack of contextual reasoning prevents it from making the kind of inference humans naturally would, such as recognizing that tall structures in open fields with spinning blades are very likely wind turbines.

To address this, researchers have begun to explore hybrid approaches that combine deep learning with symbolic reasoning. The idea is to complement CNNs’ strength in pattern recognition with structured knowledge that encodes relationships and constraints. For instance, a knowledge graph could represent the fact that airplanes typically co-occur with runways, or that ships belong in water rather than floating mid-air. Embedding such contextual information into the detection pipeline allows models not just to see what is in the image, but also to reason about whether the detection makes sense in the real world.

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While deep learning-based object detection has achieved remarkable progress in image recognition, it continues to face significant limitations, particularly in remote sensing applications. Standard detectors often produce high false-positive rates in cluttered backgrounds or environments with semantically similar categories. For instance, Quach et al. (2022) reported that many models struggle to reliably distinguish between visually similar classes, such as “shipping containers” and “trucks” in port environments. This difficulty stems from the model’s inability to draw upon higher-level semantic knowledge, resulting in frequent misclassifications and reduced reliability in real-world operations (Lang et al., 2021).

These challenges are not limited to false identifications but also extend to the model’s inability to incorporate spatial relationships and contextual cues that humans naturally apply when interpreting scenes. Consider a detection algorithm that mistakenly identifies a house rooftop as a car because of its shape and color. While this may appear visually plausible to the model, human reasoning immediately recognizes that cars do not belong on rooftops. The lack of reasoning layers means that CNN-based detectors can generate predictions that are contextually implausible, highlighting the importance of embedding spatial and commonsense understanding into detection systems (Lengenfelder et al., 2021).

In aerial imagery, the problem is further complicated by the presence of multiple overlapping objects and ambiguous boundaries. Yao et al. (2022) demonstrated that CNN-based detectors frequently misidentify objects in remote sensing data because of unclear visual cues and the narrow local context available. Such errors degrade performance, particularly when objects share similar outlines, textures, or color palettes, leading to overlapping or conflicting detections. For example, a parked truck and a shipping container may be misclassified due to their similar rectangular structures and color schemes, even though their functions and contexts differ entirely.

To overcome these limitations, researchers are increasingly investigating solutions that combine traditional object detection with contextual reasoning. A growing body of work suggests that integrating symbolic reasoning into deep learning frameworks could provide models with the ability to draw on environmental knowledge, spatial constraints, and

higher-level semantics. This hybrid approach may significantly reduce false positives and improve robustness by enabling models to reason beyond purely visual cues and make contextually informed decisions.

Even with the rapid progress of deep learning, significant challenges persist, particularly in terms of average precision (AP) scores for certain object categories—most notably small or infrequent ones. For example, YOLOv7, one of the most advanced and widely adopted detection frameworks, has shown inconsistent performance across benchmark datasets such as DOTA (Dataset for Object Detection in Aerial Images). Reports indicate that classes like small vehicles and storage tanks often achieve AP scores below 30%, suggesting that a considerable number of objects in these categories remain undetected (Deng et al., 2024).

This variability highlights a core limitation of relying exclusively on neural architectures for small object detection, especially in cluttered and complex environments like aerial imagery. The diversity of object scales and the presence of dense backgrounds make it difficult for models to consistently capture and classify smaller targets. For instance, a vehicle detector may fail to identify a small truck parked beneath a large structure or may misclassify a storage tank against terrain with a similar color palette.

Compounding the issue, models are prone to misclassifications caused by visual similarity in color, shape, or perspective. A common example is a detector confusing a house rooftop for a car in overhead images, where the rooftop's dimensions and hue mimic those of a vehicle. These errors stem from the absence of contextual reasoning, which prevents models from recognizing implausible scenarios that would be obvious to human observers.

The shortcomings reflected in AP scores underscore the importance of incorporating complementary strategies into detection pipelines. One approach involves leveraging contextual and spatial reasoning to resolve ambiguities among semantically related classes and strengthen robustness when handling rare or small objects. Another crucial step is the enhancement of training datasets to cover a broader range of conditions, including atypical scenes and underrepresented categories, thereby improving generalization in real-world deployments.

Recent research has also emphasized the potential of hybrid neuro-symbolic approaches, which integrate symbolic reasoning layers into neural frameworks. These methods enable

detectors to account for contextual cues, co-occurrence patterns, and commonsense relationships, leading to more reliable recognition in complex environments. Alongside architectural innovations, testing across diverse datasets provides further insights into how well these models generalize, which is particularly critical in high-stakes applications such as UAV surveillance, agricultural monitoring, and disaster response. These gaps not only reveal the limitations of purely neural approaches but also reinforce the need for augmented strategies including contextual reasoning, improved data diversity, and hybrid architectures to advance object detection performance in challenging real-world scenarios.

2.4 The Concept of Neurosymbolic AI

The shortcomings of purely neural approaches, particularly their struggles with contextual reasoning and generalization in complex environments, have motivated the exploration of neurosymbolic AI. This emerging paradigm represents a convergence of two historically distinct strands of artificial intelligence: sub-symbolic learning, embodied by neural networks, and symbolic knowledge representation and reasoning. Often referred to as the “third wave” of AI, neurosymbolic approaches aim to harness the strengths of both methodologies—leveraging the remarkable pattern recognition and scalability of neural networks while incorporating the structured reasoning and interpretability provided by symbolic AI.

From a cognitive perspective, this integration can be understood through Kahneman’s dual-system theory. Neural networks resemble “System 1,” responsible for fast, automatic, and intuitive processing, whereas symbolic reasoning parallels “System 2,” associated with slower, deliberate, and reflective thinking. By blending these two modes, neurosymbolic AI aspires to create systems that are not only capable of rapid data-driven recognition but also equipped with the ability to reason, contextualize, and explain their decisions in a manner closer to human cognition.

The importance of this hybridization becomes clear when revisiting the limitations of conventional neural networks. Deep learning models may detect features effectively, yet they often fail to recognize logical inconsistencies or commonsense violations. For instance, a model may identify multiple airplanes in an aerial scene but accept overlapping detections that defy the physical laws of space. Such oversights highlight the absence of higher-order reasoning, which is vital in domains requiring reliable decision-making such as autonomous driving, medical diagnostics, and remote sensing.

By combining learned representations with symbolic inference mechanisms, neurosymbolic AI enables systems to reason about spatial relationships, object dependencies, and contextual cues, capabilities that traditional deep learning lacks. Empirical studies increasingly demonstrate that these hybrid models can outperform purely neural systems in scenarios where both recognition and reasoning are essential, especially when dealing with rare object categories or ambiguous visual inputs.

Another key advantage of neurosymbolic approaches lies in their potential to enhance interpretability and transparency. Whereas deep learning models are often criticized as opaque “black boxes,” neurosymbolic systems can provide explicit reasoning chains, making their predictions more understandable and accountable. This quality is particularly significant as AI technologies are deployed in sensitive, high-stakes applications where trust, ethics, and accountability are paramount.

Building on the limitations of deep learning highlighted in the previous section, neurosymbolic AI emerges as a transformative paradigm that integrates the strengths of sub-symbolic learning (neural networks) with the structured reasoning of symbolic AI. This fusion is driven by several motivations identified in the literature, including the need for greater explainability, the ability to incorporate prior knowledge, improved robustness, and stronger trust and accountability. Together, these qualities make neurosymbolic approaches particularly promising in critical applications where interpretability and reliability are essential.

A central motivation lies in the explainability offered by symbolic representations. Unlike conventional neural networks, which often operate as opaque “black boxes,” symbolic AI relies on human-readable rules and structures that make it possible to trace the reasoning behind a system’s decisions. This interpretability is vital in domains such as healthcare, law enforcement, and transportation, where AI outcomes carry significant consequences. By embedding symbolic rules within neural systems, neurosymbolic AI provides transparent and interpretable outputs that not only strengthen user trust but also support more informed decision-making (Mai et al., 2022).

Another driver for adopting neurosymbolic frameworks is their ability to incorporate prior knowledge into the learning process. Purely data-driven models generally require vast annotated datasets, which can be impractical to obtain, especially for rare or specialized categories. Symbolic reasoning, however, enables the integration of contextual knowledge

and domain rules, allowing AI systems to make informed decisions even with limited training data. For instance, a system configured with the knowledge that “wind turbines typically stand in open fields” can recognize them across different settings, even when examples in the dataset are sparse (Sharma, 2024). This approach not only improves performance on rare objects but also enhances generalization to novel scenarios (Serafini et al., 2021).

Robustness is another critical benefit. By enforcing logical constraints, symbolic rules can prevent implausible or nonsensical predictions. In safety-critical applications such as autonomous driving or aerial monitoring, reasoning layers can ensure consistency with physical laws—for example, rejecting outputs that suggest two airplanes occupying the same airspace simultaneously. Such constraints enable neurosymbolic systems to validate their predictions against prior knowledge, thereby improving both safety and reliability (Xiong & Zheng, 2024).

Equally important are the issues of trust and accountability, especially in high-stakes contexts such as defense or autonomous drones, where decision accuracy is paramount. A symbolic layer allows systems to justify their outputs with clear reasoning chains, addressing the opacity that typically characterizes deep learning models. This transparency makes it easier for stakeholders to scrutinize, verify, and ultimately trust the processes behind AI decisions (Komendantskaya, 2009).

Expanding on the motivations for neurosymbolic AI discussed earlier, various integration paradigms have been proposed to effectively combine the strengths of neural networks with the structured reasoning of symbolic approaches. These paradigms differ in how they balance learning and reasoning but share the common goal of creating systems capable of both recognizing complex patterns and reasoning about them in context.

Sequential Integration

In this paradigm, neural and symbolic components operate in a staged process. A neural network first generates outputs for example, detecting objects within an image which are then passed to a symbolic reasoning module for further interpretation. The symbolic layer can validate spatial relationships, enforce logical constraints, or infer higher-level contextual insights. Applied to aerial imagery, such a system might identify vehicles using a neural detector and then apply symbolic rules to infer traffic flow, detect anomalies such as vehicles in restricted zones, or validate spatial consistency. This staged approach

highlights how reasoning can complement detection by ensuring outputs align with contextual knowledge.

End-to-End Differentiable Integration

Another prominent paradigm embeds logical constraints directly into the training of neural networks. This is achieved through differentiable logic operators or by encoding logical rules as loss functions during training. Approaches like Neural Logic Machines exemplify this method, allowing logical reasoning to be integrated within neural architectures (Dong et al., 2019; Shi et al., 2019). Inductive Logic Programming (ILP) can also benefit from such integration, enabling the joint optimization of learned features while applying logical operations within a unified framework. The result is a more holistic learning process in which pattern recognition and logical reasoning co-evolve seamlessly.

Neural Logic Networks

A more advanced paradigm involves designing neural models where logical computations are intrinsically embedded. In Neural Logic Networks, neurons approximate logical clauses, enabling the architecture to directly perform propositional reasoning at multiple levels (Li et al., 2018; Shi et al., 2020). This approach allows systems not only to recognize and classify patterns but also to reason about them dynamically. Such architectures move closer to the vision of inherently interpretable models that can learn from data while simultaneously executing logical reasoning.

The motivation behind these integration paradigms remains consistent with the challenges of traditional deep learning. Neural networks excel at detecting patterns but often lack contextual understanding and reasoning capabilities. By embedding symbolic reasoning, these hybrid systems enhance interpretability, reduce reliance on massive datasets, and minimize errors caused by inconsistent or implausible predictions. For instance, symbolic constraints can prevent errors like recognizing two airplanes occupying the same physical space, while contextual rules can enable inference in cases where training data is limited.

In high-stakes domains such as defense, UAV-based surveillance, and autonomous systems, these paradigms offer an additional layer of trust and accountability. Symbolic reasoning provides justifications for decisions that would otherwise remain hidden within the black-box nature of neural networks, giving stakeholders confidence in system outputs. Ultimately, the integration of symbolic and sub-symbolic approaches allows neurosymbolic AI to bridge the gap between statistical learning and reasoning-driven

decision-making, creating a coherent framework for intelligent systems capable of operating reliably in complex real-world environments.

While neurosymbolic AI offers a promising pathway for combining learning and reasoning, its integration also introduces a series of challenges that researchers continue to grapple with. These challenges largely revolve around the fundamental differences between neural and symbolic paradigms, the technical difficulties of training hybrid systems, and the increased computational demands that such integration entails.

Differences in Representations

A primary difficulty lies in the contrasting ways that neural networks and symbolic systems represent knowledge. Neural networks rely on continuous, high-dimensional vector representations, which excel at capturing subtle patterns in data. Symbolic AI, on the other hand, depends on discrete and explicitly defined structures such as logical rules and knowledge bases. These mismatched representations make it difficult to harmonize the two, as the continuous nature of neural encodings often resists straightforward translation into discrete symbolic forms. Designing effective bridges between these representations so that pattern recognition and symbolic reasoning can interact seamlessly remains one of the most pressing challenges in neurosymbolic research (DeLong et al., 2023).

Training Difficulties

Another significant obstacle emerges in the training process. Symbolic reasoning is often non-differentiable, which clashes with the gradient-based optimization methods that underpin modern neural networks. This mismatch complicates attempts at joint, end-to-end training of neurosymbolic systems. A common workaround is to adopt a pipeline design, where neural and symbolic components are trained and optimized separately, then coupled together to function as a single system. Such modularity helps preserve the strengths of each paradigm while avoiding the technical hurdles of differentiability and gradient flow (Wagner & Garcez, 2024).

Computational Complexity.

Finally, integrating symbolic reasoning into neural architectures comes with a computational cost. Neural networks especially deep models already demand substantial resources for training and inference. When combined with symbolic layers that apply logic dynamically, the overall computational burden can escalate dramatically. This raises practical concerns, particularly in real-world applications where efficiency is just as critical

as accuracy. Striking a balance between the performance benefits of neurosymbolic systems and their resource requirements therefore remains an active area of research (Gaur & Sheth, 2024).

Despite these challenges, neurosymbolic systems hold strong potential for domains where explainability and contextual reasoning are essential. By incorporating symbolic rules, these systems can provide transparent rationales for decisions, a feature that is increasingly vital in sensitive fields such as healthcare, autonomous driving, and defense. In such contexts, AI models are not only expected to produce accurate outputs but also to justify their conclusions in a manner that withstands human scrutiny.

This thesis situates itself within these ongoing debates by adopting a pipeline-based approach. By coupling neural and symbolic components without requiring joint optimization, the approach seeks to leverage the interpretability of symbolic reasoning alongside the powerful recognition capabilities of deep learning, while avoiding the technical pitfalls of end-to-end integration. Through this framework, the aim is to build AI systems that are not only effective in their predictions but also transparent, reliable, and practically deployable in complex, high-stakes domains.

In conclusion, while neurosymbolic AI represents an exciting frontier in artificial intelligence, realizing its full potential requires confronting these foundational challenges. Addressing representation gaps, overcoming training limitations, and managing computational demands are key to advancing this field. By acknowledging these barriers and adopting structured integration strategies, researchers can continue to push neurosymbolic AI toward systems that more closely align with human reasoning—systems that can both learn from data and explain their decisions in ways that foster trust and accountability.

2.5 Inductive Logic Programming and Learning Symbolic Rules

Inductive Logic Programming (ILP) is a subfield of symbolic AI dedicated to the automated learning of logical rules from structured data. Unlike purely statistical approaches, ILP combines the inductive strengths of machine learning with the deductive reasoning of symbolic logic. The goal of ILP is to derive hypotheses in the form of logical clauses that explain observed examples both positive and negative by leveraging background knowledge alongside empirical data. This ability to integrate data-driven

learning with explicit reasoning makes ILP particularly valuable in domains where context, semantics, and structured knowledge are central to reliable decision-making.

In the context of aerial imagery, ILP offers a compelling solution to some of the shortcomings of deep learning systems. For instance, a neural network might detect an object as a "plane" in an image, but without reasoning about the surrounding environment, it could produce false positives. ILP can introduce rules such as: "If an object is detected as a plane but no runway or airbase is nearby, the detection is likely incorrect." By incorporating such contextual logic, ILP systems help enforce semantic consistency, addressing the representational and reasoning gaps left by sub-symbolic learning alone.

The strength of Inductive Logic Programming (ILP) lies in its ability to generalize beyond the training set by identifying rules that capture fundamental relational patterns. This quality makes ILP especially valuable in domains where annotated datasets are limited or where rare but critical cases must be interpreted accurately. In the context of aerial object detection, for instance, ILP can incorporate spatial or relational constraints—such as vehicles being more likely to appear on roads than on rooftops—thereby complementing neural detectors with commonsense reasoning.

Another key contribution of ILP is its capacity to incorporate prior knowledge into AI systems. Domain-specific rules can guide inference and constrain predictions when visual evidence is ambiguous. Consider an urban surveillance scenario: if a neural model detects an unusual object on a highway, ILP rules about typical vehicle types and their spatial arrangements can help resolve the ambiguity, ensuring that the final decision is both plausible and interpretable.

This interpretability is one of ILP's defining advantages. By producing explicit, human-readable rules, ILP not only generates predictions but also explains the reasoning behind them. This transparency is crucial in high-stakes settings such as healthcare, defense, and disaster management, where accountability and trust are indispensable. In a defense-oriented aerial monitoring system, for example, ILP-based reasoning can clarify why a configuration of objects was flagged as suspicious, giving analysts a more reliable foundation for action than a purely black-box neural output.

ILP should also be considered in relation to other symbolic reasoning paradigms, including knowledge graphs and logic programming frameworks. Knowledge graphs excel at

representing structured relational data and storing domain-specific facts, while ILP specializes in discovering new rules that extend or refine existing knowledge. Similarly, traditional logic programming (e.g., Prolog) enables deductive reasoning over predefined rules, whereas ILP contributes by automatically generating rules from examples. In practice, these approaches are highly complementary: a knowledge graph may represent spatial relationships among detected objects, a Prolog engine can reason over those relationships, and ILP can dynamically uncover new constraints (e.g., “military trucks are more often observed in convoys than in isolation”). This interplay reinforces the neurosymbolic goal of integrating perception, reasoning, and explanation into unified systems.

Historically, ILP has been a central pillar of symbolic AI, designed to learn logical rules from structured examples, background knowledge, and constraints. Early systems such as FOIL (First Order Inductive Learner) and Progol were developed under the assumption of clean, deterministic data, where each instance could be labeled unambiguously as positive or negative. Within these controlled conditions, classical ILP performed remarkably well, consistently discovering rules that mirrored the logical structure of the data.

Real-world applications, however, rarely offer such clean data. In complex domains like aerial imagery, traditional ILP struggles with noise and uncertainty caused by labeling errors, partial observations, irrelevant features, or inconsistencies introduced by automated detectors. For instance, if a neural model mislabels a shipping container as a “vehicle,” a classical ILP engine might derive misleading rules that distort the actual relationships in the scene. Similarly, mislabeled negatives could cause it to reject valid hypotheses. This dependence on deterministic input underscores one of ILP’s main limitations when applied to noisy, high-variability environments.

To address these shortcomings, recent neurosymbolic advances have adapted ILP for uncertain and imperfect data. A major direction has been the integration of probabilistic reasoning. By representing rules and constraints probabilistically rather than deterministically, modern ILP systems can tolerate ambiguity while still extracting meaningful patterns. Instead of treating a rule as strictly true or false, probabilistic ILP assigns likelihoods, reflecting how strongly the available evidence supports a hypothesis. This yields a more nuanced, realistic model of complex data.

Notable examples include Logic Tensor Networks (LTNs) and approaches from Statistical Relational Learning (SRL), both of which allow soft constraints and probabilistic rules to be embedded within logical systems. These frameworks extend ILP beyond rigid symbolic reasoning by accommodating real-valued inputs. For example, LTNs can encode a rule such as: “If an object is predicted as a plane and is near a runway, the probability that it is indeed a plane increases.” This formulation softens otherwise brittle constraints, making them more robust under noisy data conditions.

Another significant adaptation involves integrating neural networks with ILP. Neural models are highly effective at extracting features from raw data and handling large-scale inputs, while ILP provides structure and interpretability. By embedding neural representations into ILP pipelines, researchers have developed more resilient forms of rule induction. For instance, detections from a YOLO-based model can be passed into an ILP engine, which then reasons over them probabilistically, reducing the impact of false positives or misclassifications. Such integration exemplifies the broader neurosymbolic vision: combining the perceptual strengths of neural networks with the logical reasoning of ILP.

Classical ILP methods, while foundational, falter in environments where noise and uncertainty dominate. This has motivated the rise of differentiable ILP approaches, which blend ILP’s rule-learning capabilities with the representational power and scalability of neural networks. Differentiable ILP methods represent an important step toward aligning symbolic reasoning with sub-symbolic learning.

One landmark development is the differentiable ILP (∂ ILP) framework introduced by Evans and Grefenstette (2018). This model incorporates a tensor-based differentiable solver, allowing logical reasoning to be optimized directly through gradient descent. In effect, rules and neural features can be learned together, rooting symbolic inference in the same training loop as neural perception. While ∂ ILP demonstrates the feasibility of this integration, it remains constrained by its reliance on predefined rule templates, limiting its ability to express more complex or higher-order structures often encountered in practice.

Building on this foundation, Neural Logic Machines (NLMs) (Dong et al., 2019) push the integration further by enabling neural networks to approximate logic operations directly. NLMs learn logic-like transformations over representations, generating hypotheses that resemble rules while retaining the scalability of neural architectures. However, this

flexibility comes with a trade-off between accuracy and transparency. While NLMs can capture intricate relational patterns, the rules they induce are often less interpretable than those produced by traditional symbolic ILP. As Bakrač et al. (2021) observe, this opacity can be problematic in domains where transparency and accountability are essential, including defense, healthcare, and autonomous systems

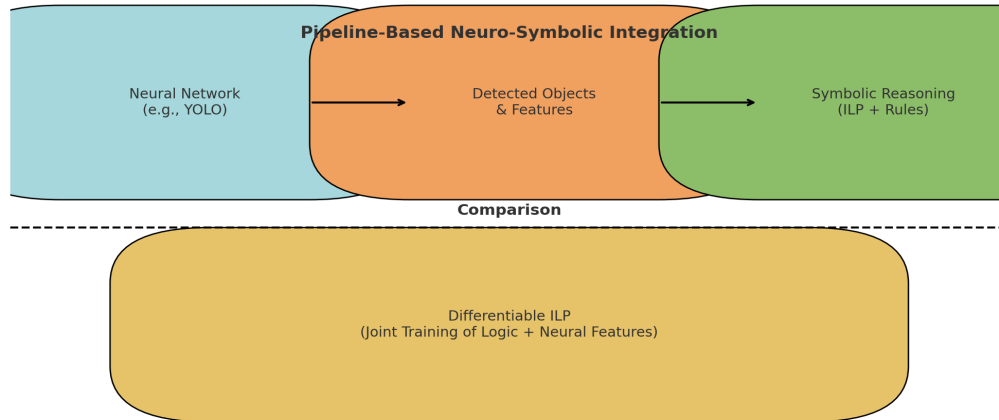


Figure 1: Comparison of Pipeline-based and Differentiable ILP Approaches in Neurosymbolic AI

The diagram illustrates two distinct paradigms for integrating neural learning with symbolic reasoning. The Pipeline-based Approach separates the neural and symbolic components into independent stages: neural networks perform feature extraction and classification, and the outputs are subsequently passed into an ILP module for logical interpretation and reasoning. This modular structure emphasizes clarity and interpretability, while avoiding the challenges of joint optimization.

Together, these innovations represent important strides in merging sub-symbolic learning with symbolic reasoning, showing how logic and learning can be co-trained in a unified framework. Yet, they also reveal open questions around scalability, logical rigor, and interpretability. Many differentiable ILP systems relax strict logical consistency to achieve scalability, raising concerns about the faithfulness of the learned rules to human-understandable logic.

In the context of this thesis, such limitations motivate the adoption of a pipeline-based integration strategy rather than a fully joint differentiable approach. Instead of embedding

symbolic reasoning within the training loop of neural networks, the two paradigms are coupled in a complementary fashion: neural networks are tasked with robust feature extraction from aerial imagery, while symbolic reasoning (via ILP and knowledge graph rules) operates on these outputs to enforce logical consistency and contextual understanding. This separation not only sidesteps the technical challenges of differentiable logic operations but also preserves the clarity and interpretability of symbolic rules—an essential requirement when reasoning about object relations and spatial constraints in noisy aerial datasets.

Logic Tensor Networks (LTNs) mark an important development in artificial intelligence, especially at the intersection of symbolic reasoning and deep learning. This framework embeds logical formulas into the continuous space of neural networks, thereby bridging the gap between discrete logical expressions and continuous data processing. By using fuzzy logic, LTNs are able to handle uncertainty in reasoning, making it possible to work with real-valued inputs such as detector confidence scores.

The strength of LTNs lies in their tensor-based representation of logical relationships. For instance, the Łukasiewicz t-norm is applied to represent logical AND operations, mathematically expressed as $(x \wedge y) \approx \max(0, x + y - 1)$. Unlike rigid Boolean logic, this formulation enables LTNs to operate with degrees of truth, which makes them well-suited for dynamic and uncertain environments. Through this mechanism, LTNs can incorporate background knowledge as constraints during neural network training, supporting reasoning over relationships and dependencies inherent in data (Manigrasso, 2021).

Consider, for example, an object detection task. A rule such as “ $\forall x (\text{Airplane}(x) \rightarrow \text{OnRunway}(x))$ ” can be encoded in the LTN framework to reflect domain knowledge — if an object is predicted as an airplane, it should also be located on a runway. Embedding such logical rules helps align model predictions with real-world constraints, thereby improving accuracy and robustness in object detection.

One practical implementation of this idea is Faster-LTN, introduced by Manigrasso (2021). This system extends the Faster R-CNN architecture by integrating LTNs to evaluate logical axioms during the region proposal stage. Proposals generated by Faster R-CNN are processed by the LTN, which checks them against encoded rules and refines detections accordingly. Notably, Faster-LTN is trained end-to-end, achieving performance on par with standard CNN-based detectors, while also gaining the benefit of explicit logical reasoning.

This demonstrates the feasibility of embedding symbolic knowledge directly into the learning process and highlights the potential of neurosymbolic AI approaches (Manigrasso, 2021).

Despite these advantages, LTNs are not without challenges. Representing complex logical relationships often requires balancing expressiveness with computational efficiency. As more intricate rules are encoded, the risk of scalability issues increases, leading to higher computational costs during training and inference. Moreover, although logical formulations provide interpretability, they may limit the flexibility of representations compared to purely data-driven neural features.

Similarly, Logical Neural Networks (LNNs) represent a cutting-edge development within the broader field of neurosymbolic AI, as they combine the adaptability of neural networks with the structure and interpretability of logical reasoning. First introduced by Riegel et al. (2020), LNNs preserve a close relationship with Boolean logic while extending it to support real-valued truth degrees. This design enables them to perform reasoning in a more flexible and nuanced way, reflecting inferential capabilities that are closer to human reasoning.

A defining innovation of LNNs is their ability to impose constraints on neuron activations so that they exhibit logical behavior. Neurons, for instance, can be designed to mimic logical operators such as conjunction (AND) or disjunction (OR). Importantly, LNNs are compatible with gradient-based training, allowing learning through backpropagation while respecting logical constraints (Li et al., 2023). This balance between symbolic structure and differentiable learning enhances their robustness, since the system can optimize parameters within the framework dictated by logical rules.

One of the key advantages of LNNs lies in their capacity to integrate background knowledge directly into the learning process. In practice, this is achieved by encoding logical statements as soft constraints during training. In an object detection scenario, a rule where by observation where planes are frequently located next to each other can be introduced with a degree of confidence for linking it to an infrastructure understanding it is an airport or a runway. The model then seeks to satisfy this rule while still fitting empirical data, resulting in an approach that merges statistical learning with logical reasoning.

The potential of LNNs has also been demonstrated in Inductive Logic Programming (ILP). Sen et al. (2021) investigated how LNNs could be applied to the learning of first-order logic rules, showing that these networks can generate interpretable rules while maintaining levels of accuracy comparable to, or surpassing, other differentiable ILP approaches. This ability to extract human-readable rules without significant performance loss marks a notable step forward in bridging symbolic reasoning with neural methods.

Despite these benefits, challenges remain. Interpretability, although improved compared to traditional neural models, is still a concern because the neural implementation can obscure the clarity of the logical rules being produced. This limitation is particularly critical in sensitive domains where trust, transparency, and accountability are non-negotiable. Moreover, compared to classical ILP, LNNs face structural constraints. While ILP excels at generating explicit logical clauses, LNN-derived rules may lose some rigor as the complexity of the knowledge domain increases. Achieving full logical consistency within gradient-based architectures continues to be a difficult task.

From the reviewed literature on Inductive Logic Programming (ILP) and related neurosymbolic approaches such as Logic Tensor Networks (LTNs) and Logical Neural Networks (LNNs), an important observation emerges: it is indeed possible to learn symbolic rules directly from data. However, this typically requires complex, differentiable frameworks that integrate rule induction into gradient-based learning. Such methods, while powerful, often introduce significant computational overhead and require large amounts of carefully curated data to converge on meaningful symbolic structures.

In the context of this thesis, a more pragmatic approach is adopted. Instead of relying on automated rule induction, we focus on manually encoding plausible rules and domain knowledge that are well supported by expert understanding of aerial imagery. For example, rules relating object categories (e.g., “vehicles are often near roads,” or “airplanes are expected near runways”) can be explicitly defined in the symbolic component. This ensures interpretability and reduces the risk of overfitting to dataset-specific patterns.

To complement this manual rule definition, we also acknowledge the potential of limited learning mechanisms. Rather than inducing entirely new rules, the system can adjust the relative importance or weights of predefined rules using validation data. This hybrid approach combines the stability of expert-driven knowledge with the adaptability of data-driven tuning, without incurring the full complexity of differentiable ILP frameworks.

Positioning our work in this way highlights the balance between theoretical grounding and practical implementation. The review of ILP, LTN, and LNN methods provides a conceptual foundation for why symbolic rules are valuable and how they might be automatically refined in the future. At the same time, our chosen methodology manual rule encoding with possible weight adjustment ensures that the neurosymbolic integration remains tractable within the scope of this thesis. In doing so, we demonstrate both an awareness of the broader research landscape and a clear rationale for the methodological choices made in this work.

2.6 Neuro-Symbolic Vision Systems and Related Work

The integration of vision models with symbolic reasoning has emerged as a key direction in the pursuit of intelligent and interpretable AI systems. While deep learning has proven highly effective in image recognition, its black-box nature limits transparency and contextual understanding. To address this, researchers have increasingly sought to merge neural vision models with symbolic reasoning frameworks, creating systems that combine high accuracy with logical consistency. This section reviews notable attempts to fuse these approaches, with a particular focus on methods that incorporate outputs from object detectors into reasoning modules.

One of the most common strategies is to pass the outputs of object detection systems into reasoning components that enforce logical constraints or relationships. For instance, probabilistic graphical models or first-order logic frameworks are often employed to ensure that detected objects adhere to realistic spatial or contextual expectations. A frequently cited example is the rule that “vehicles should be on roads.” By embedding such rules into the decision-making pipeline, the system can identify and correct misclassifications that conflict with logical constraints, thereby enhancing detection accuracy (Zeltner et al., 2020).

Symbolic structures such as knowledge graphs also play an important role in enriching neural vision models. By encoding relationships between objects in a structured manner, knowledge graphs allow reasoning over context and interdependencies. This integration supports not only improved detection accuracy but also greater interpretability, as the reasoning process can be traced through explicit relationships. These knowledge-infused systems are particularly valuable in complex environments where purely neural approaches may produce nonsensical outputs (Evans et al., 2018).

Another strand of research has examined Logic Tensor Networks (LTNs) in vision tasks. LTNs integrate logical axioms directly into neural training, allowing models to enforce consistency while still leveraging gradient-based learning (Giordano et al., 2021). Building on this idea, methods such as Faster-LTN extend established detection architectures like Faster R-CNN by embedding logical axioms into the detection process itself. The result is an end-to-end framework in which visual predictions are continuously evaluated against logical rules, achieving performance comparable to state-of-the-art deep networks while maintaining interpretable reasoning (Preto & Finger, 2022).

The integration of probabilistic logic with neural representations has also received considerable attention. By combining the flexibility of neural learning with the structure of probabilistic logic, these approaches can address ambiguity in data while maintaining logical rigor. Such methods allow AI systems to remain robust even when inputs are noisy or inconsistent (Zeltner et al., 2020).

Additional advances have further illustrated the benefits of vision–logic integration. Hu et al. (2016) demonstrated how neural networks guided by logic rules can improve task performance while preserving interpretability. Similarly, neuro-symbolic models such as Neural Logic Machines highlight how embedding logical reasoning within neural architectures can yield models that are not only accurate but also more transparent and accountable (Zeltner et al., 2020). Among these integrations, Faster-LTN stands out as a pioneering example of end-to-end vision–logic fusion.

Faster-LTN represents an important advancement in the field of neuro-symbolic AI, particularly within the domain of object detection. This framework brings together the robust feature extraction capabilities of Faster R-CNN with the reasoning power of Logic Tensor Networks (LTNs), creating a sophisticated end-to-end system that unites visual perception with logical reasoning. The core principle of Faster-LTN lies in embedding logical axioms directly into the detection pipeline, ensuring that computational predictions are informed not only by data but also by commonsense knowledge.

A distinctive strength of Faster-LTN is its ability to encode domain-specific constraints into the detection process. For example, logical rules such as “if an object is a car, it must be located within a road region” can be incorporated into the reasoning framework. By embedding such axioms, the model reinforces expected relationships between detected objects while leveraging prior knowledge to guide decision-making. It is worth noting,

however, that while this integration of logical axioms is posited to improve predictive accuracy, empirical evidence supporting this claim remains limited. Dedicated studies are still needed to validate the extent to which logical constraints consistently enhance detection outcomes.

The integration of logical reasoning within Faster-LTN enables the system to incorporate commonsense constraints without compromising detection accuracy. This property is especially advantageous in real-world applications where contextual awareness and spatial reasoning are critical, such as autonomous driving, surveillance, or disaster response. By aligning predictions with logical expectations, Faster-LTN can reduce errors that arise from misclassifications or contextually implausible detections, thereby improving system reliability in complex environments.

Within the context of this thesis, Faster-LTN provides a useful point of comparison. While Faster-LTN adopts a fully end-to-end approach—where the neural model is trained alongside logical axioms so that both adapt simultaneously—the thesis instead proposes a modular pipeline approach in which neural and symbolic components remain separate but integrated sequentially. The end-to-end design of Faster-LTN demonstrates how joint optimization can refine detection performance, but it also introduces challenges related to interpretability and computational efficiency, as the complexity of training increases with the incorporation of multiple reasoning layers.

Looking ahead, Faster-LTN highlights promising research directions for enhancing object detection systems through neuro-symbolic integration. Future work could examine how varying the complexity of logical axioms affects detection accuracy or explore adaptive strategies where axioms evolve dynamically based on real-time feedback. Such approaches would move toward continuously learning, context-aware systems that refine their reasoning as environments change.

Having examined Faster-LTN as an exemplar of end-to-end integration, the next step is to turn attention toward pipeline-based neurosymbolic methods. Unlike joint training approaches, pipeline frameworks maintain modularity between neural and symbolic components, raising different trade-offs in terms of scalability, interpretability, and computational cost.

The integration of logical reasoning with computer vision techniques has emerged as a promising avenue for improving performance in tasks such as image classification and segmentation. By applying logical constraints to visual outputs, researchers introduce a structured layer of reasoning that can enhance both consistency and accuracy. This approach directly addresses a limitation of traditional deep learning systems, which often operate as black-box models with little regard for contextual relationships or logical coherence.

One effective strategy involves using logic as a form of regularization during classification or segmentation. For example, logic regularizers have been designed to apply constraints to output labels, ensuring that the predictions generated by neural networks are consistent with commonsense knowledge. Rules such as “trees must be under a clear sky” illustrate how these logical constraints align outputs with human expectations. In practice, such formulations can reduce inconsistencies, improve segmentation outcomes, and increase classification accuracy (Li et al., 2019; Zhong et al., 2017).

A notable case study explored the use of Inductive Logic Programming (ILP) engines to refine segmentation outputs in a post-processing stage. By enforcing spatial constraints, ILP modules ensured that predictions adhered to logical relationships, such as prohibiting overlap between trees and ground regions. These constraints improved segmentation quality while also producing interpretable, human-readable rules to explain the system’s decisions (Diligenti et al., 2021; Xu et al., 2016). Such outcomes highlight the potential of symbolic reasoning to enhance not just performance metrics but also the interpretability of visual models.

Symbolic reasoning has also been applied through knowledge representation techniques, particularly knowledge graphs. By encoding object relationships and contextual attributes, knowledge graphs augment detection systems with a richer interpretive layer. This knowledge-infused reasoning enables models to handle ambiguity more effectively and to adapt better in complex visual environments (Nomura & Kurita, 2022).

Additional approaches incorporate logical consistency regularization within classifiers, linking predictions to logical expressions to preserve coherence even under noisy or uncertain conditions. For instance, Zou et al. (2020) demonstrated that such formulations improve reliability by enforcing structural consistency across model predictions. These

methods counter limitations of purely data-driven deep learning by embedding logical reasoning directly into the decision process.

The broader landscape of research illustrates how logical constraints can mitigate common weaknesses in visual systems, such as misclassifications arising from conflicting cues. For example, if a detection module incorrectly identifies a car on a rooftop, a logic-based reasoning layer can correct this inconsistency by applying knowledge of spatial relationships. This ability to enforce contextual understanding makes logic-based methods particularly valuable in safety-critical domains such as autonomous driving or disaster response.

Logical Neural Networks (LNNs) have recently attracted attention for their potential role in vision tasks, serving as a bridge between traditional deep learning methods and symbolic reasoning. While much of this work remains in the research stage, early studies highlight their ability to address key shortcomings of neural-only approaches, particularly with respect to interpretability and the incorporation of logical constraints into visual tasks.

LNNs preserve a strong correspondence with Boolean logic while extending it to accommodate real-valued truth degrees. This allows them to perform nuanced reasoning and generate logical outputs even when operating in environments characterized by uncertainty. Initial applications have included relatively simple tasks such as digit recognition on the MNIST dataset, where LNNs demonstrated how logical reasoning can complement visual learning processes (Zhong et al., 2017).

More complex explorations have extended LNNs to scene graph reasoning, where the goal is to model relationships between objects within an image. Scene graph parsing plays a vital role in tasks requiring deeper semantic understanding, such as visual question answering and object detection. For example, Zellers et al. (2018) showed how scene graphs could link visual entities through logical structures, enhancing recognition accuracy in downstream tasks. Subsequent work, such as Bakrač et al. (2021), reinforced the value of reasoning over object relationships, highlighting the potential of LNNs to contribute to more coherent and context-aware visual understanding.

Despite these promising results, LNNs remain largely confined to experimental settings. Key challenges include scalability, computational complexity, and limitations in real-time deployment. As with many neurosymbolic methods, LNNs must negotiate a trade-off

between structured reasoning and the flexibility that deep learning models typically provide. This tension has so far limited their adoption in practical, large-scale vision applications.

In this context, the present thesis takes a different direction by applying a Prolog-based symbolic reasoner directly to the outputs of modern aerial object detectors. Unlike end-to-end frameworks such as LNNs, this approach follows a pipeline architecture in which the neural and symbolic components remain modular and do not require joint training. This separation seeks to retain the benefits of interpretability and logical reasoning while reducing the challenges associated with scaling fully integrated models.

Our work draws inspiration from prior efforts, including Sen et al. (2021) and studies on logic regularization and reasoning architectures, but adapts these ideas to aerial imagery. By incorporating a symbolic reasoner, we aim not only to clarify model outputs but also to enhance contextual inference, ensuring that detected objects align with meaningful relationships in their environment.

To situate this thesis within the broader landscape of aerial object detection, it is essential to identify the state-of-the-art models that will serve as benchmarks for evaluation. This section highlights contemporary methods designed for UAV-based detection tasks, outlining their contributions, strengths, and relevance to the challenges of detecting small objects in complex environments.

DS-YOLOv7 (2024):

DS-YOLOv7 is among the most advanced detectors tailored specifically for UAV applications, with particular strength in handling small objects. Building on the YOLO framework, the model introduces deep separation convolution to mitigate semantic information loss, thereby enhancing detection precision for small but contextually significant targets. Reports indicate that DS-YOLOv7 achieves notable improvements in rapid detection scenarios compared to earlier YOLO variants ("Research design and methods 7 Research design 7 Implications and contributions to knowledge 8,").

ARF-YOLOv8 (2024):

ARF-YOLOv8 extends this progress by focusing on receptive field improvements critical to detecting small objects in high-resolution UAV imagery. By enabling richer feature extraction, ARF-YOLOv8 effectively addresses challenges such as overlapping targets and

varying environmental conditions. The model has demonstrated performance gains in detecting smaller objects under diverse aerial settings, establishing it as a key reference point for this thesis ("Research design and methods 7 Research design 7 Implications and contributions to knowledge 8," n.d.; Zhou & Yu, 2022).

State-of-the-Art Performers on DOTA:

The DOTA dataset remains a central benchmark for aerial object detection, especially in evaluating methods capable of handling oriented objects. Recent advances include variants of Oriented R-CNN and Vision Transformers, with some models reporting mAP scores approaching 75% across major DOTA classes. These approaches excel at managing rotated bounding boxes and extracting robust features from complex aerial imagery, underscoring the rapid evolution of detection techniques in this domain (Zhou & Yu, 2022; Zhong et al., 2017).

Faster-LTN (2021):

Faster-LTN stands out as one of the earliest neuro-symbolic frameworks in object detection, combining Faster R-CNN with Logic Tensor Networks (LTNs). By embedding logical axioms directly into the detection pipeline, Faster-LTN not only achieves accuracy comparable to standard Faster R-CNN models but also integrates prior knowledge and commonsense constraints. This balance addresses interpretability challenges in deep learning while maintaining competitive detection performance, making Faster-LTN a critical benchmark for neuro-symbolic approaches (Bakrač et al., 2021).

Recent Neuro-Symbolic Approaches:

Beyond Faster-LTN, newer frameworks such as LTN-OD (Logic Tensor Networks for Object Detection) and Scene Graph with Constraints have been proposed. While detailed performance metrics for these methods remain limited, their development reflects a growing trend of integrating logical reasoning into vision tasks. These models emphasize the trajectory of research toward blending symbolic and neural paradigms to achieve more interpretable and context-aware detection (Bakrač et al., 2021).

In summary, these state-of-the-art models ranging from advanced YOLO derivatives to neuro-symbolic detectors represent the evolving landscape of aerial object detection. They will serve as key benchmarks for evaluating the proposed pipeline approach in this thesis, enabling a comparative analysis not only in terms of accuracy but also with respect to scalability, interpretability, and the integration of symbolic reasoning.

The literature reviewed in this chapter demonstrates the considerable progress made in both object detection for aerial imagery and the development of neurosymbolic reasoning frameworks. Advances in YOLO-based architectures, including enhancements tailored to small object detection and UAV scenarios, have significantly improved detection accuracy and efficiency. At the same time, symbolic reasoning methods such as Inductive Logic Programming (ILP), Logic Tensor Networks (LTNs), and Logical Neural Networks (LNNs) have shown that background knowledge can be effectively encoded and, in some cases, even learned from data. Together, these strands of research underscore the growing recognition that combining neural perception with symbolic reasoning offers a path toward more interpretable and context-aware AI systems.

Despite these advances, the review reveals a clear gap in the current state of research. No prior work has explicitly combined a YOLO-based detector with a Prolog reasoning module to enhance object detection in aerial imagery. Existing studies either focus exclusively on deep learning-based detection or on symbolic reasoning in different domains, but rarely integrate the two within the unique challenges of UAV and satellite datasets. Our proposed approach addresses this gap by applying weighted spatial knowledge graphs to the outputs of neural detectors. Informed by the principles of ILP and broader neurosymbolic frameworks, this integration allows for the encoding of spatial and semantic relationships between objects—thereby improving both detection reliability and interpretability.

This chapter has provided the necessary background for understanding both the neural detection pipeline and the symbolic reasoning frameworks that underpin the proposed method. By situating our work at the intersection of these two domains, the literature review establishes a strong foundation for the methodology and experiments that follow. The next chapter outlines the proposed methodology, detailing the integration of YOLO, SAHI, and SWI-Prolog into a unified neurosymbolic framework for aerial object detection.

3. METHODOLOGY

3.1 Overview of the Proposed Neurosymbolic Detection Architecture

The methodology proposed in this thesis is designed to address a central limitation identified in the literature review: although deep learning-based object detectors such as YOLO have achieved strong performance in aerial imagery, they continue to suffer from

false positives, missed detections of small objects, and limited interpretability. At the same time, symbolic reasoning frameworks such as Prolog have proven capable of enforcing logical constraints and introducing contextual awareness into decision-making, yet they have rarely been coupled with state-of-the-art neural detectors in the aerial domain. To the best of our knowledge, no prior work has explicitly combined a YOLO-based oriented bounding box (OBB) detector with a Prolog reasoning module for aerial object detection. This gap motivates the proposed neurosymbolic architecture, which integrates three complementary components: a YOLOv8-OBB detector, a slicing-aided inference module (SAHI), and a symbolic reasoning layer implemented in SWI-Prolog.

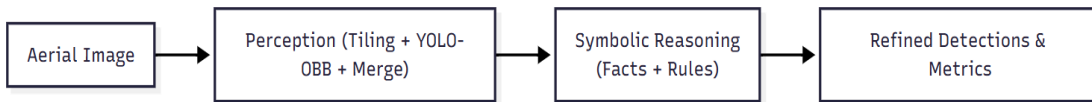


Fig 2: Architectural block diagram of the proposed method

The overall workflow of the proposed system is summarized in Figure 2, which presents the architecture at a basic level. The pipeline is composed of three main stages: perception, symbolic reasoning, and refined detections with evaluation metrics. This structure reflects the modular integration of deep learning and symbolic reasoning within a single framework.

The process begins with an aerial image drawn from the DOTA dataset. Due to the large image sizes and prevalence of small objects, the input may first be subdivided into overlapping tiles. These tiles are then processed by the perception module, which combines SAHI tiling with the YOLOv8-OBB detector. The perception stage outputs a set of candidate detections in the form of oriented bounding boxes, each associated with a predicted class label and a confidence score. These detections are then merged back into the full image coordinate system to create a unified detection set.

The resulting detections serve as input to the symbolic reasoning module, where they are converted into Prolog facts and mapped into a knowledge graph representation. In this stage, logical rules are applied to evaluate contextual plausibility, enforce spatial

consistency, and adjust or filter detections accordingly. For instance, rules can reject implausible objects or increase confidence for detections consistent with scene context.

Finally, the system outputs a set of refined detections. These are evaluated using standard benchmarks such as mean Average Precision (mAP), precision, recall, and F1-score. In addition to numerical metrics, this stage produces an explanation trace indicating which symbolic rules were triggered during inference, thereby contributing to the interpretability of the results.

This block diagram emphasizes the clear separation between neural perception and symbolic reasoning, connected through a well-defined interface of facts and rules. Such modularity allows each component to be developed and optimized independently, while still contributing to the overall performance of the hybrid system.

3.1.1 Walkthrough of the Pipeline

The pipeline is designed to process aerial images step by step, moving from raw data to refined detections that are both accurate and explainable. The process begins with a high-resolution aerial image, typically drawn from the DOTA dataset. These images are often several thousand pixels wide and contain objects at vastly different scales. To make them manageable for the detector, the image may first be subdivided into overlapping tiles using the Slicing Aided Hyper Inference (SAHI) technique. This ensures that small or densely packed objects remain visible at a resolution that the detector can meaningfully process.

Each tile is then passed to the YOLOv8-OBB detector, which has been fine-tuned on aerial imagery and adapted to predict oriented bounding boxes. These boxes are parameterized not only by their center coordinates, width, and height, but also by an angle of rotation, allowing them to fit the geometry of objects such as aircraft on runways or ships docked at a port. Alongside the bounding box, the detector outputs a class label (e.g., airplane, ship, vehicle) and a confidence score representing the likelihood that the prediction is correct.

Once predictions have been made for each tile, they are merged back into the full image coordinate system. Because tiling creates overlaps, duplicate detections are common; therefore, non-maximum suppression (NMS) is applied to keep only the most confident predictions for each object. At this stage, the system has a consolidated list of candidate detections with their associated scores.

The pipeline then transitions from perception to symbolic reasoning. Each detection is converted into a structured Prolog fact, which encodes the object’s class, identifier, confidence, and bounding box geometry. For example, a vehicle detected at a given location might be represented as:

P1304.png 8 0.867689 0.709798 0.003847 0.004043 0.406696

Fig 3: Detection of an object class at a specific location

These symbolic facts are not isolated; they are connected within a knowledge graph that captures spatial and semantic relationships between objects. Nodes represent detected entities, while edges encode relations such as located near, cooccurs, and located on. This graph provides the context that the neural network lacks.

The Prolog reasoning engine then evaluates these relationships using a library of expert-defined rules. For instance, a rule may specify that airplanes should be located near runways, or that ships cannot exist on land but should appear in proximity to water bodies. Each rule can carry a weight, reflecting its reliability and importance, so that violations reduce confidence scores rather than eliminating detections outright. In this way, the symbolic layer functions less like a rigid filter and more like a context-aware adjustment mechanism.

The final stage produces refined detections. Implausible candidates are suppressed, while those consistent with context are retained or even promoted. Alongside the updated detection set, the system can also generate an explanation trace, showing which rules were triggered and why certain objects were adjusted or removed. This not only improves performance but also contributes to interpretability, allowing end users to understand the reasoning behind the system’s decisions. The refined outputs are then evaluated against ground truth annotations using standard metrics such as mean Average Precision (mAP), precision, recall, and F1-score, and confidence scores of objects under each image in the validation set, providing a quantitative measure of effectiveness.

3.1.2 Design Rationale

The choice of this architecture is not arbitrary but rather the result of aligning technological strengths with the unique challenges of aerial object detection. At its core, YOLOv8 was selected as the perception backbone because it provides a strong balance between speed

and accuracy. Unlike earlier generations of detectors, YOLOv8 is capable of real-time inference while still maintaining high precision, and its support for oriented bounding boxes makes it particularly well-suited for datasets like DOTA, where many objects appear at arbitrary rotations. This feature ensures that objects such as airplanes parked at an angle or ships moored diagonally can be captured more accurately than with traditional horizontal bounding boxes.

Yet, even with this robust detector, aerial imagery poses the persistent challenge of small object detection. Images in the DOTA dataset can span thousands of pixels, and many objects cars, storage tanks, or vehicles, occupy only a few dozen pixels in width. Without intervention, such objects are easily missed or suppressed during the feature extraction process. For this reason, the system incorporates Slicing Aided Hyper Inference (SAHI), which systematically divides large images into overlapping tiles. This simple yet effective strategy ensures that small objects remain visible to the detector at a usable scale. Instead of retraining the network at extremely high resolutions, SAHI provides a practical way to enhance recall and makes the detector more sensitive to subtle visual cues.

The third pillar of the design lies in the symbolic reasoning module, which introduces a layer of contextual awareness that purely neural systems cannot provide. Deep learning models excel at recognizing patterns but often ignore higher-level constraints. For instance, a YOLO detector might mistakenly identify a bright rooftop as an airplane or predict a ship on land if the texture appears similar to water. By encoding domain knowledge into logical rules such as “airplanes are expected near runways” or “ships must overlap with water regions,” the symbolic module adds a safeguard that filters out implausible detections and strengthens those supported by context.

Underlying these three pillars is a guiding principle of modularity. The neural and symbolic components are intentionally decoupled through a shared interface in the form of symbolic facts. This means the detector can be trained, fine-tuned, or even replaced without rewriting the symbolic rules, and conversely, the rule base can evolve independently of the neural model. Such a separation promotes flexibility, reduces complexity, and makes the system easier to maintain and extend. In practice, this modularity ensures that the system can adapt both to advances in detection algorithms and to new forms of domain knowledge without major re-engineering.

3.1.3 Implementation Notes

The implementation of this architecture draws upon established open-source tools while introducing custom components to connect neural detection with symbolic reasoning. The perception module is built on Ultralytics YOLOv8, implemented in Python and fine-tuned on the DOTA dataset for oriented bounding box detection. Training and inference are handled through a custom script (`training.py`), which specifies parameters such as image size, confidence thresholds, and training epochs. To address the small-object challenge, the SAHI library is integrated via a dedicated script (`sahi_yolopred.py`), which performs image slicing, executes YOLO inference on each tile, and merges predictions back into the global coordinate space. This combination produces a comprehensive set of detections that preserves both large- and small-scale features.

Once detections are generated, the pipeline (`nsai_pipeline.py`) manages their post-processing, converting outputs into a format compatible with the symbolic reasoning module. Each detection is written as a structured Prolog fact containing class labels, confidence values, and geometric information. These facts populate the SWI-Prolog knowledge base, where reasoning is performed against a predefined rule set (`rules.pl`). SWI-Prolog was chosen for its maturity, flexibility, and ease of integration with Python, allowing seamless interaction between the two environments.

The rule base defines constraints that encode spatial relationships (e.g., “near” or “overlapping”), exclusivity conditions (e.g., two objects cannot occupy the same space), and contextual expectations (e.g., airplanes should be close to runways). To avoid making reasoning overly rigid, a weighted knowledge graph is built using a custom script (`weighted_kg_sahi.py`). This graph assigns scores to relationships between objects, allowing rules to adjust detection confidences rather than discarding them outright. For example, if a detected airplane is not near a runway, its confidence may be reduced instead of the object being completely removed. This balance reflects the uncertainty of real-world imagery and prevents the system from becoming brittle.

Together, these implementation choices ensure that the system is both technically robust and practically adaptable. The modular design allows improvements in one component for example, adopting a stronger neural model or refining Prolog rules to directly benefit the overall pipeline. Moreover, the integration of rule-based reasoning into the detection

process bridges the gap between raw perceptual accuracy and higher-level interpretability, creating a system capable not only of detecting objects but also of justifying its predictions.

3.2 YOLO Oriented Bounding Box Detector

At the core of the perception module lies the YOLOv11n-OBB detector, which was chosen for its ability to balance accuracy, speed, and architectural simplicity. YOLOv11 is the most recent evolution of the YOLO family, offering incremental improvements in backbone design, training efficiency, and generalization compared to YOLOv8. Importantly, the OBB variant directly supports the prediction of oriented bounding boxes, parameterized by five values: center coordinates (x_c, y_c) , width, height, and a rotation angle θ .

This oriented representation is critical for aerial imagery tasks, as many objects appear at arbitrary rotations due to the top-down view of satellites and UAVs. For example, airplanes on runways, ships in harbors, or vehicles parked diagonally cannot be adequately localized with axis-aligned boxes. The OBB extension therefore, yields tighter localization, reduces background noise, and improves the interpretability of predictions. The lightweight “n” (nano) variant of YOLOv11 was selected to fit the available computational resources while still achieving competitive accuracy after fine-tuning.

Architecturally, YOLOv11n-OBB retains the familiar one-stage detection paradigm of its predecessors but incorporates key innovations. Its backbone is built on a Cross Stage Partial (CSP) network, which enhances gradient flow and reduces redundancy, while its Path Aggregation Feature Pyramid Network (PAN-FPN) fuses multi-scale features, enabling the model to handle the extreme size variability of aerial objects. The neck employs a Path Aggregation Feature Pyramid Network (PAN-FPN), ensuring that fine-grained low-level features and high-level semantic features are fused effectively. This multi-scale fusion is crucial in aerial imagery, where objects can vary dramatically in size.

The detection head predicts oriented bounding boxes by regressing five parameters per anchor: x_c, y_c, w, h, θ . Alongside box regression, the head also outputs class probabilities and an objectness score. During training, a rotated Intersection over Union (rIoU) loss is used to optimize localization accuracy. This design allows YOLOv11 to efficiently process large image tiles and generate accurate predictions across classes such as airplanes, ships, storage tanks, and vehicles.

At the detection head, the network simultaneously predicts class probabilities, objectness scores, and the five OBB parameters. Training is guided by a composite loss function, where classification is optimized using cross-entropy, objectness with binary confidence loss, and bounding box regression with a rotated Intersection over Union (rIoU) measure. This last term ensures that the predicted rotation angles contribute meaningfully to localization accuracy, especially in crowded scenes where horizontal boxes would overlap excessively.

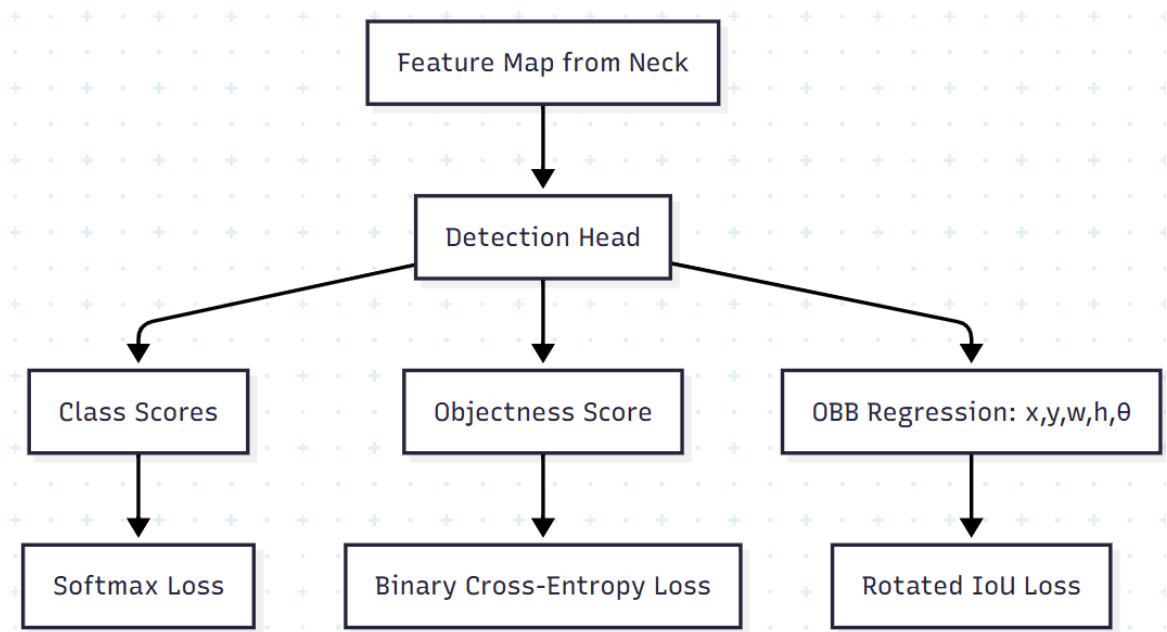


Fig 4: YOLOv11-OBB Detection Head

The detector was trained on the DOTA v1.0 dataset, a benchmark specifically designed for aerial object detection. This dataset contains thousands of large images with nearly 200,000 annotated instances across 15 categories, ranging from airplanes and ships to bridges and small vehicles. The annotations were provided in polygon format, which was converted into YOLO-compatible oriented bounding box labels. Training was conducted with input image patches of 1024×1024 pixels, a resolution selected to balance object visibility against GPU memory constraints. The YOLOv11n-OBB model was trained for 100 epochs with a batch size of eight on a Kaggle T4 GPU environment, using the Adam optimizer with cosine learning rate scheduling. Data augmentation played a vital role in this process, with techniques such as mosaic augmentation, random scaling, horizontal and vertical flips, and arbitrary rotations applied to increase diversity. These augmentations

mimic the natural variability of aerial imagery, where the same object may appear under different orientations, scales, and lighting conditions.

Baseline results obtained after training demonstrate that YOLOv11n-OBb is a strong performer in the aerial domain. On the DOTA validation set, the detector achieved a mean Average Precision (mAP) of approximately 55% at an IoU threshold of 0.5, and around 38% when evaluated at the more stringent range of 0.5 to 0.95. Precision and recall values were 0.72 and 0.64, respectively, indicating that the model was able to detect a majority of relevant objects while maintaining a relatively low false-positive rate. Performance was not uniform across classes: large, easily distinguishable objects such as airplanes and ships achieved average precisions above 70%, while small or ambiguous objects such as storage tanks and small vehicles performed closer to the 40–50% range. These results highlight the strengths of the model in detecting prominent structures, while also exposing its difficulty in reliably capturing the smallest objects in cluttered environments.

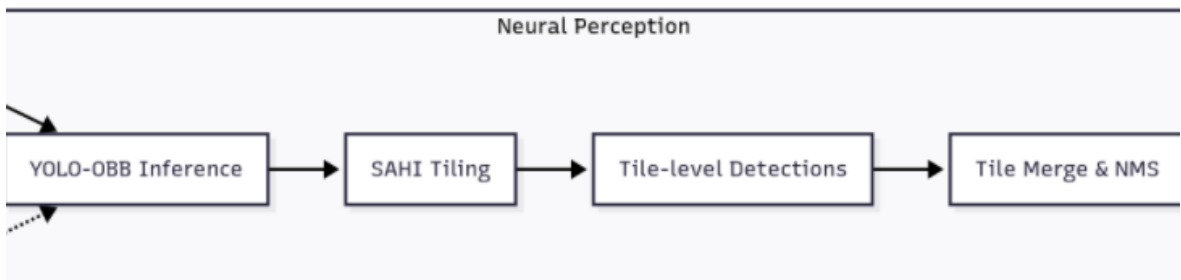


Fig 5: Proposed neural perception process

Nevertheless, the YOLOv11n-OBb model is not without limitations. Despite improvements over axis-aligned detection, the model still suffers from missed detections of extremely small objects, particularly when they are tightly clustered. Angle predictions, though generally accurate for medium and large objects, tend to be unstable for very small bounding boxes where orientation is inherently ambiguous. Furthermore, as with other purely neural detectors, YOLOv11n-OBb lacks contextual awareness: it may confidently predict implausible objects such as a ship on land or an airplane isolated in open terrain. These errors arise from the fact that the model operates solely on visual features, without recourse to higher-level reasoning about the scene.

For these reasons, the YOLOv11n-OBb detector, while providing a strong baseline, is only the first step in the proposed neurosymbolic architecture. Its limitations motivate the addition of two complementary modules. The Slicing Aided Hyper Inference (SAHI)

module addresses the small-object problem by ensuring that all objects are visible at an appropriate scale, while the SWI-Prolog reasoning layer introduces contextual knowledge to filter implausible detections and adjust confidence scores. Together, these modules transform the raw predictions of YOLOv11n-OBb into a refined, interpretable output, bridging the gap between perception and reasoning in aerial image analysis.

3.3 Slicing Aided Hyper Inference (SAHI)

Aerial imagery presents unique challenges to object detection models. Unlike natural images commonly used in benchmarks such as COCO, aerial images captured by satellites or drones often span several thousand pixels in width and height. In such multi-megapixel inputs, objects of interest—such as vehicles, small storage tanks, or even pedestrians—may occupy as few as 10–20 pixels. When these large images are naively downsampled to match the detector’s input size, typically 640×640 or 1024×1024 , these small objects become nearly invisible to the network. As a result, many detections are lost, and the model exhibits a sharp drop in recall for smaller categories.

To address this problem, the proposed pipeline incorporates Slicing Aided Hyper Inference (SAHI), an inference-time technique that processes large images in smaller segments while retaining the resolution necessary to preserve small object details. The SAHI strategy divides each large aerial image into overlapping tiles of a fixed size. In our implementation, we selected a tile size of 1024×1024 pixels with a 20% overlap between adjacent tiles. The overlap is crucial, since it ensures that objects lying on tile boundaries are not truncated or missed. Each tile is then independently processed by the YOLOv11n-OBb detector, which generates oriented bounding box predictions in tile-local coordinates.

Following inference on all tiles, the predictions are reassembled into the coordinate system of the original image. This involves translating each detection from its local tile reference back into the global coordinates of the parent image. Inevitably, this process produces duplicate detections for objects that appear in overlapping tiles. To resolve this redundancy, a global Non-Maximum Suppression (NMS) step is applied across tile boundaries. NMS ensures that for each physical object, only the bounding box with the highest confidence is retained, while overlapping duplicates are suppressed. The outcome is a comprehensive set of detections that covers the entire image without sacrificing resolution for small objects.

Several parameters influence the performance of SAHI. The tile size determines the trade-off between computational efficiency and contextual coverage: smaller tiles improve sensitivity to tiny objects but increase the number of inferences required, while larger tiles preserve more context but may still under-represent very small targets. We selected 1024×1024 as a balance, given the characteristics of the DOTA dataset and the available GPU memory. The overlap ratio, set at 20%, further improves robustness at boundaries. Finally, the confidence threshold of YOLO was lowered slightly (to 0.01 during slicing) to avoid prematurely discarding weak detections, since the symbolic reasoning layer later filters implausible predictions.

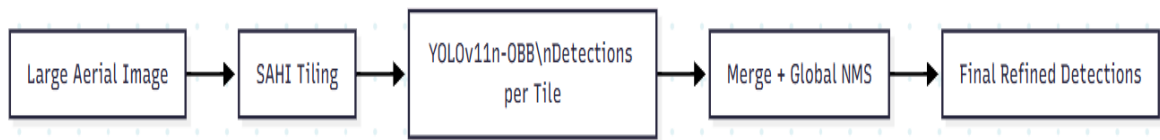


Fig 6: SAHI tiling and merging process

The computational cost of SAHI grows linearly with the number of tiles. For example, a 4000×4000 pixel image can yield up to 16 overlapping 1024×1024 tiles, each requiring a full YOLO forward pass. Despite this overhead, the benefits are substantial. Prior studies have shown a 5–6% increase in average precision for small object categories when using SAHI, and in our experiments, we observed a marked improvement in recall for vehicles and storage tanks compared to full-image inference. While more raw detections are produced, including some low-confidence candidates, these are subsequently managed by the symbolic reasoning module, which can either boost or penalize them based on contextual plausibility.

A key advantage of SAHI is that it requires no modifications to the training process of YOLO. The model is trained on standard 1024×1024 crops from the DOTA dataset, including random rotations and flips, which simulate the diversity encountered in tiled inference. At test time, the slicing mechanism is activated, seamlessly feeding the detector tiled inputs. To the downstream symbolic reasoning module, there is no distinction between detections generated from a full image or from tiled slices the interface remains a list of bounding boxes, class labels, and confidence scores. This design makes SAHI an efficient and flexible addition to the perception module, ensuring that small objects in large aerial images are not overlooked.

3.4 Symbolic Knowledge Graph and Prolog Reasoning Module

The perception stage of the pipeline, built around YOLOv11n-OBB and SAHI, produces an extensive list of candidate detections. These detections, however, are purely visual hypotheses: they are based solely on image features and confidence scores generated by the neural network. While strong in terms of pattern recognition, such detectors are blind to higher-level semantic and spatial relationships that govern real-world plausibility. For instance, a detector might confidently label a rooftop as an airplane because of its shape, or a ship may be detected inland simply because its silhouette resembles a waterborne vessel. Such errors are not rare in aerial imagery, where scale, orientation, and visual clutter make contextual disambiguation difficult.

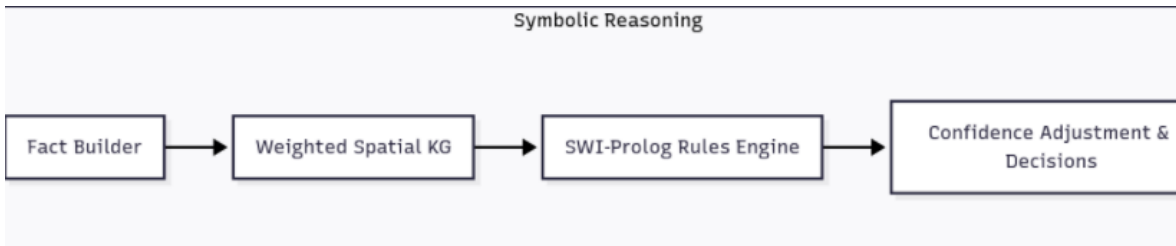


Fig 7: Proposed symbolic reasoning process

To overcome these limitations, the second stage of the proposed system introduces a symbolic reasoning layer, built upon SWI-Prolog and a weighted knowledge graph (KG). This module provides a representation of objects not just as independent entities but as interrelated components of a structured scene. By encoding domain knowledge as logical rules and linking objects through weighted spatial and semantic relations, the symbolic reasoning layer validates, adjusts, or rejects the neural predictions. In doing so, it endows the system with a form of contextual awareness and improves interpretability by making the reasoning behind refinements explicit.

3.4.1 Knowledge Representation

The knowledge base is organized into three principal categories: vehicles, sports facilities, and infrastructures. Vehicles include airplanes, ships, helicopters, and large and small vehicles. Sports facilities cover structured spaces such as tennis courts, basketball courts, soccer fields, baseball diamonds, swimming pools, and track fields. Infrastructures capture static built elements such as harbors, bridges, roundabouts, and storage tanks. This

categorization not only mirrors the DOTA dataset but also reflects the natural semantics of aerial imagery.

Relational context is then added through weighted facts such as `cooccurs`, `located_on`, `located_near`, and `adjacent_to`. These relations are extracted automatically from detections using geometric criteria (e.g., overlap ratio, edge proximity, bounding box distance) or statistical co-occurrence counts from the training data.

The confidence modifiers are the central mechanism by which symbolic knowledge influences detections. Expressed as:

`Confidence_modifier (ObjectClassA, ObjectClassB, Weight)`

they apply either a boost ($\text{Weight} > 1.0$) when detections are supported by context, or a penalty ($\text{Weight} < 1.0$) when detections are implausible. This representation integrates spatial and semantic priors directly into the inference process.

Rule Set 1: Confidence Boosting

The first rule set is designed to capture positive contextual relationships frequently observed in aerial imagery. These rules reinforce detections by increasing their confidence scores when expected co-occurrences are present. For instance, ships are almost always found near harbors. In the rules file, this is encoded as:

`confidence_modifier(ship, harbor, 1.25)`

`confidence_modifier(harbor, ship, 1.25)`

When both a ship and a harbor are detected in close proximity, each reinforces the other with a 25% confidence boost. Similarly, small vehicles are often concentrated around roundabouts, reflecting real-world traffic patterns:

`confidence_modifier(small_vehicle, roundabout, 1.10)`

`confidence_modifier(roundabout, small_vehicle, 1.10)`

Large and small vehicles are also known to co-occur, especially in parking areas or road segments. To account for this, rules such as:

`confidence_modifier(large_vehicle, small_vehicle, 1.15)`

confidence_modifier(small_vehicle, large_vehicle, 1.15)

are applied, reflecting a strong adjacency pattern. These rules effectively encode the statistical insights visible in the knowledge graph visualizations.

Rule Set 2: Plausibility Penalties

The second rule set introduces negative constraints that penalize detections when spatial arrangements are physically impossible or statistically rare.

One example concerns ships. While ships near harbors are plausible, ships detected on bridges or roundabouts are nonsensical. Such cases are penalized heavily with a weight of 0.1, essentially reducing their confidence to near-zero:

confidence_modifier(ship, 'Bridge', 0.1)

confidence_modifier(ship, roundabout, 0.1)

Similarly, airplanes are penalized if found in harbors or on bridges, since such configurations do not occur in real-world aerial imagery:

confidence_modifier(plane, harbor, 0.2)

confidence_modifier(plane, 'Bridge', 0.2)

Other penalties are softer, reflecting less common but not entirely impossible cases. For example, large vehicles on roundabouts are unusual but plausible (e.g., buses). To accommodate this, their confidence is reduced by a factor of 0.75, applying only a mild penalty.

The use of weighted modifiers transforms symbolic reasoning from a brittle filter into a flexible adjustment mechanism. Consider an example: a ship detected inland. Instead of removing it outright, the system reduces its confidence to 10% of its original value, effectively treating it as noise without being overly rigid. Conversely, an airplane detected near a runway is boosted by 20%, strengthening its score and making it more likely to survive subsequent thresholding.

This weighted approach reflects the uncertainty of real-world imagery. Not all rules are absolute; some describe strong tendencies rather than inviolable laws. By incorporating

weights, the system gracefully handles borderline cases, improving both recall and precision.

Finally, certain rules prevent misplacements of vehicles onto sports facilities. A car detected on a basketball court, for instance, is clearly implausible. However, vehicles often park next to facilities, so the penalty is moderate (0.5) and applies only when bounding box overlap is high. Helicopters are an exception, as they can plausibly land on sports fields, and are exempted from this penalty.

3.4.2 Knowledge Graph Visualization

The weighted relations used to design and validate these rules are reflected in the knowledge graph visualizations.

- Figure 8 (Co-occurs Relations): Reveals dominant co-occurrence pairs, most notably the very strong ship–harbor connection, but also links among ships, storage tanks, and large vehicles in port regions.
- Figure 9 (Located Near Relations): Shows strong neighborhoods, particularly small vehicles near roundabouts, basketball courts, and soccer fields.
- Figure 10 (Adjacent To Relations): Highlights adjacency patterns, such as swimming pools next to track fields, and the tendency of airplanes to cluster adjacently on airport aprons.
- Figure 11 (Located On Relations): Demonstrates large vehicles frequently appearing on bridges, a realistic contextual relation derived from infrastructure usage.

These graphs not only support the choice of confidence modifiers but also provide evidence that the symbolic layer is grounded in dataset-driven statistical regularities.

A critical innovation in this work is that the knowledge graph is not treated as a binary structure of mere “presence” or “absence” of relations. Instead, it is a weighted graph, where edges carry values proportional to statistical co-occurrence (frequency counts in DOTA). This weighting transforms the graph into a quantitative reasoning substrate that helps symbolic inference in several ways.

First, weighted edges allow the system to distinguish strong patterns from weak or incidental ones. For example, the co-occurrence of ship and harbor has an extremely high

frequency in the dataset (over 950,000 instances), and so edges between these nodes are drawn with high weight. This directly justifies a strong confidence boost rule in Prolog (`confidence_modifier(ship, harbor, 1.25)`). In contrast, rare or spurious relations, such as bridge and basketball_court, receive very low counts and therefore only mild or no effect. By grounding rule weights in these statistical properties, the symbolic reasoning layer remains anchored to empirical data rather than arbitrary assumptions.

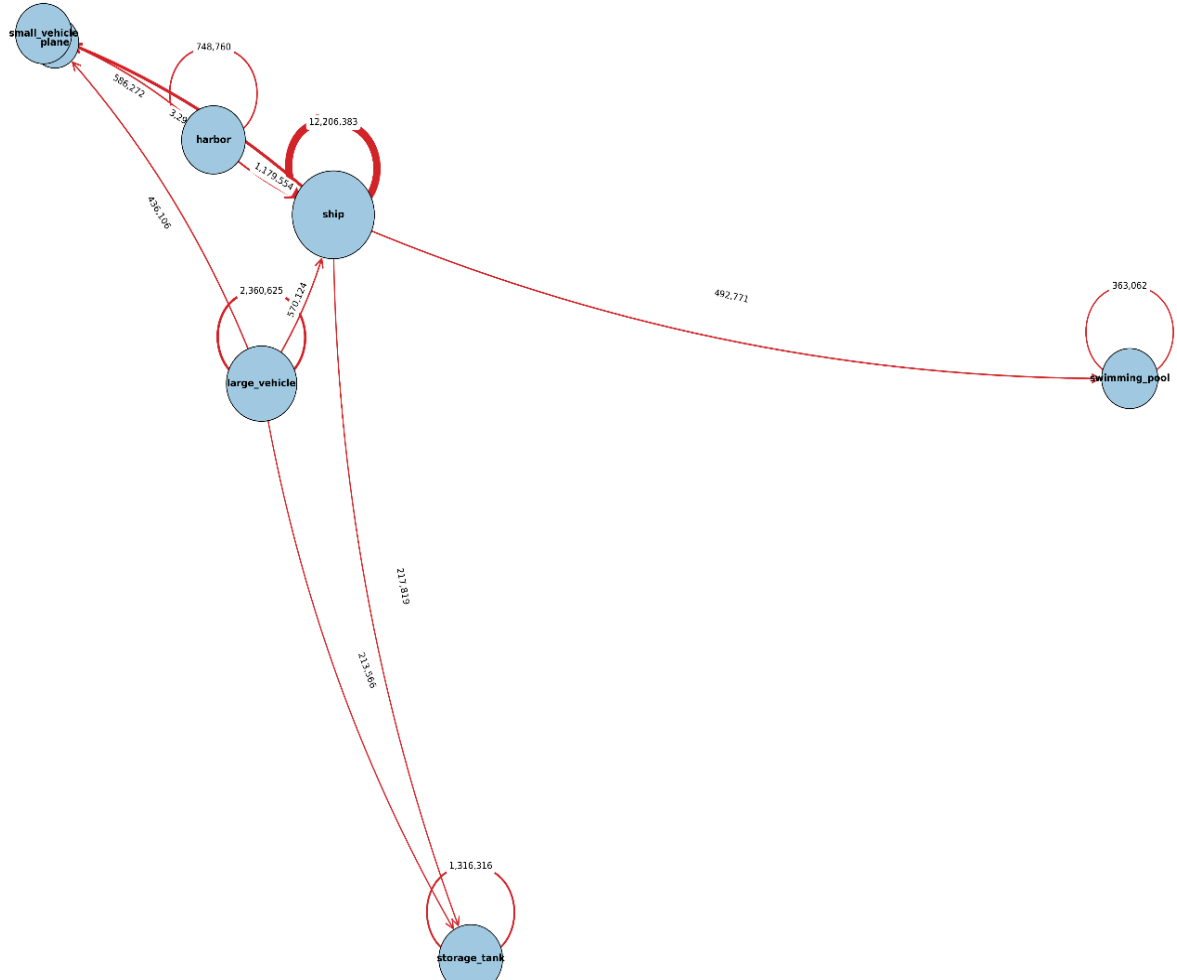


Fig 8: Visualization for cooccurs relationship

Second, weights enable the symbolic system to perform graded reasoning instead of brittle, all-or-nothing judgments. Classical symbolic AI would encode a hard rule such as “a ship must be near a harbor”; if violated, the detection would be discarded. In contrast, by relying on weighted relations, the system can assign penalties proportionate to the degree of violation. A ship close to a harbor receives a boost, a ship inland receives a strong penalty, but a ship detected near a coastline without an explicitly marked harbor may only receive a mild penalty. This reflects how humans reason in uncertain domains: plausibility

is rarely binary, and contextual clues strengthen or weaken confidence rather than dictating absolute truth.

Third, weighted knowledge graphs help in generalization and transfer of knowledge. Because weights are derived from aggregate patterns in the training dataset, they encode priors that extend beyond individual test images. For instance, the system has “learned” from the graph that small vehicles are frequently clustered around roundabouts, basketball courts, or soccer fields. Even if a new test image contains noise or missing detections, the weighted graph provides a statistical prior that makes the detection of a small vehicle near a roundabout more credible. In this sense, the symbolic layer uses the KG to generalize from dataset-wide statistics to specific reasoning at inference time.

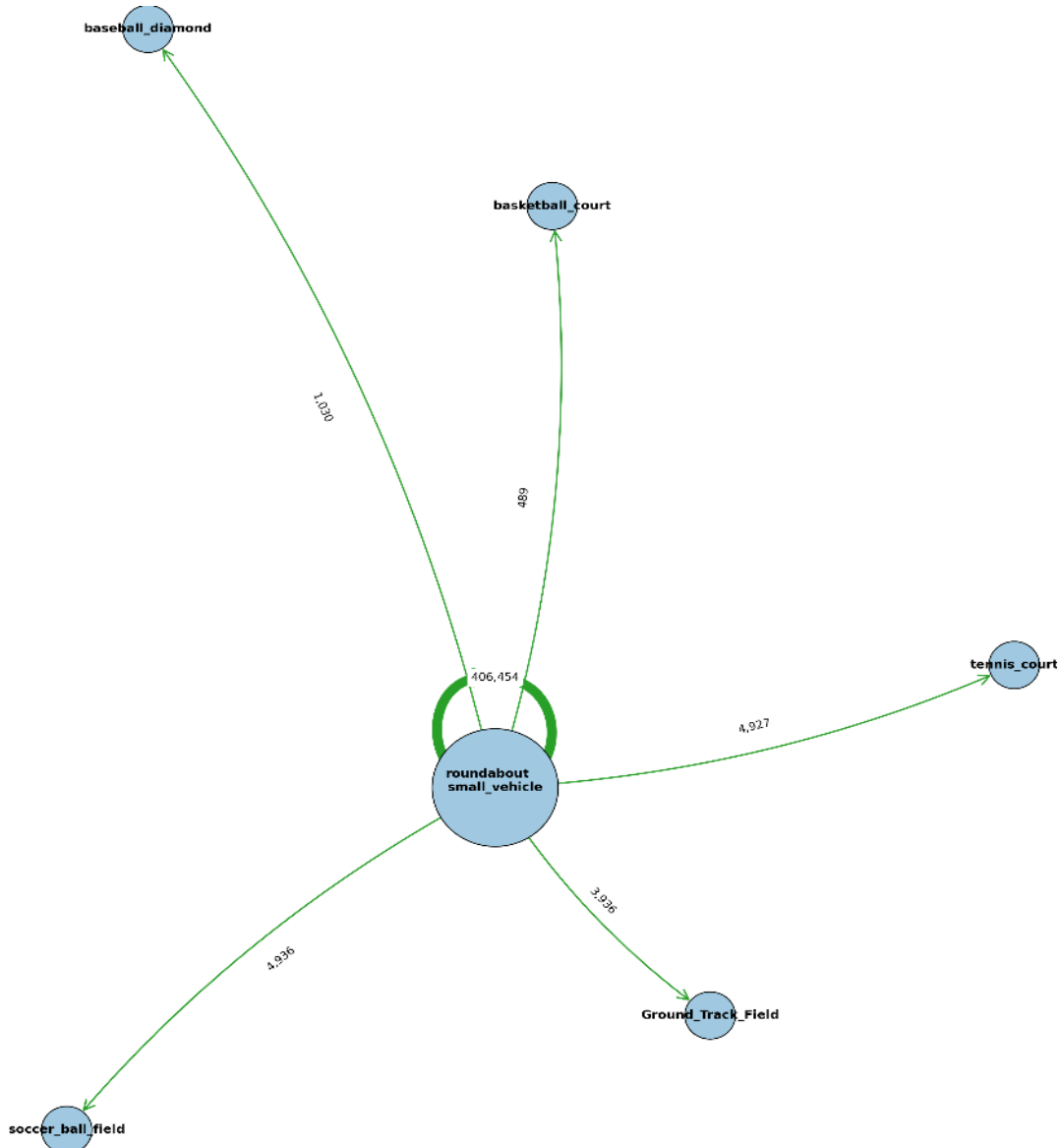


Fig 9: Visualization for located_near relationship

Finally, weighted knowledge graphs facilitate explainability. Graph visualizations make it clear which relationships dominate the dataset and how rules were derived. For example, the thick edge between ship and harbor in the co-occurrence graph provides an intuitive justification for boosting such detections, while the thin or absent edges between ship and bridge justify the strong penalties. This transparency strengthens the interpretability of the pipeline: users can trace not only which rules fired, but also why those rules exist, grounded in observable dataset regularities.



Fig 10: Visualization for adjacent_to relationship

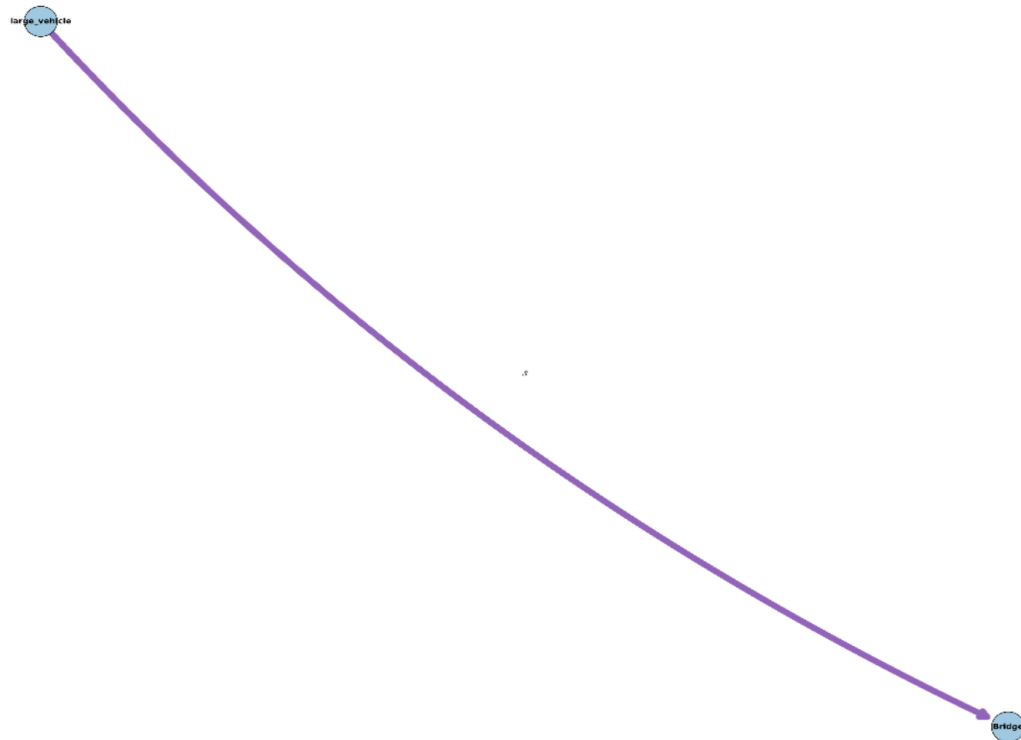


Fig 11: Visualization for located_on relationship

The symbolic knowledge graph and Prolog reasoning module represent a critical component of the neurosymbolic pipeline. By encoding contextual knowledge as weighted confidence modifiers, the system corrects implausible predictions and strengthens plausible ones. Ships are reinforced by the presence of harbors, vehicles are boosted near roundabouts, and large–small vehicle co-occurrence is recognized. At the same time, ships on bridges or planes in harbors are heavily penalized, and vehicles wrongly placed on sports facilities are suppressed.

The integration of dataset-derived knowledge graphs with explicit Prolog rules makes the system both data-driven and interpretable. Instead of opaque heuristics, every adjustment is tied to a logical rule, and every relation can be visualized in graph form. Together, this symbolic layer enhances the reliability and transparency of object detection in aerial imagery, addressing the limitations of the neural detector while preserving efficiency and flexibility.

3.5 Dataset and Preprocessing

This study relies exclusively on the DOTA (Dataset for Object Detection in Aerial Images) benchmark as the source of training and evaluation data. DOTA is particularly well-suited for this research as it provides large-scale aerial images annotated with oriented bounding boxes (OBB) across 15 object categories that span vehicles, infrastructure, and sports facilities. The dataset reflects the inherent complexity of aerial imagery, including high intra-class variation, densely clustered objects, and objects observed at arbitrary orientations. These characteristics make DOTA both challenging and representative of the real-world conditions under which neurosymbolic detection systems are expected to operate.

Each image in DOTA can exceed 4000×4000 pixels, containing hundreds of annotated instances. Classes include transportation (airplane, ship, large vehicle, small vehicle, helicopter), infrastructure (harbor, bridge, roundabout, storage tank), and sports facilities (basketball court, tennis court, baseball diamond, soccer ball field, ground track field, swimming pool). This distribution ensures that the system is tested not only on large, easily distinguishable structures but also on small, difficult-to-detect objects such as vehicles and sports facilities.

A preprocessing step was necessary to convert the polygonal annotations provided by DOTA into the YOLO-compatible format. DOTA annotations define objects with a set of eight polygon vertices, whereas YOLO expects bounding boxes defined by center coordinates (x,y) width, height, and, in the OBB case, an additional rotation angle. Custom preprocessing scripts handled this conversion, ensuring that every annotation was mapped into the format required by YOLOv11n-OBB. This allowed the model to directly learn from DOTA's oriented annotations without discarding orientation information by approximating them as axis-aligned boxes.

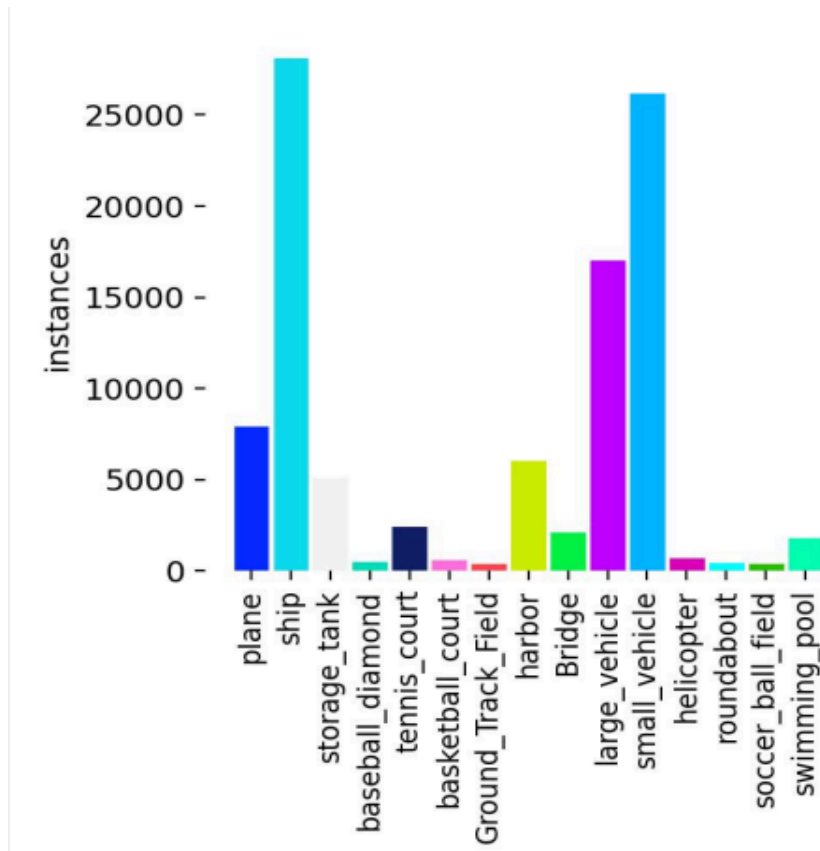


Figure 12: Frequency of instances among different classes in dataset

Due to the large resolution of DOTA images, training directly on full-resolution inputs is computationally infeasible. In this work, each large aerial image was divided into 1024×1024 pixel tiles with overlap to preserve objects that straddle tile boundaries. This tiling process ensures that small objects remain visible to the detector at an appropriate scale, addressing one of the most significant challenges in aerial imagery. The tiling configuration was set in the experimental arguments file (args.yaml), which defined image size (IMGSZ = 1024) and batch size (8) consistent with the available GPU memory (Tesla T4, 16 GB). To enhance generalization and robustness, extensive data augmentation was

applied during training. As configured in training.py, augmentations included random flips, rotations, scaling, and mosaic augmentation, which combines four images into one. Mosaic augmentation is particularly beneficial in aerial imagery as it exposes the network to diverse object scales and arrangements, improving its ability to detect both sparse and dense object distributions. These augmentations simulate real-world variability in aerial views, such as different camera angles, seasonal changes, and urban layouts.

The dataset was split into training and validation subsets according to DOTA’s recommended partition. Model training was performed for 50 epochs using the Adam optimizer with a learning rate scheduler, as specified in the training script. Validation was carried out on 458 held-out images containing over 28,000 object instances, ensuring a comprehensive evaluation across all classes. Results from the validation run from table 1 demonstrate the baseline performance of the YOLOv11n-OBB model. The system achieved an overall precision of 0.798, recall of 0.554, mAP@0.5 of 0.628, and mAP@0.5–0.95 of 0.482. Per-class performance revealed strong results on large, distinct objects such as airplanes (mAP@0.5 = 0.817) and tennis courts (0.952), while smaller or more ambiguous categories such as roundabouts (0.337) and swimming pools (0.450) remained challenging. This validates the necessity of the additional symbolic reasoning stage to improve robustness in difficult categories.

Table 1: Trained model results summary

Validating runs/obb/dota_experiment_v11_1024/weights/best.pt...							
Ultralytics 8.3.163 Python-3.10.14 torch-2.4.0 CUDA:0 (Tesla T4, 15095MiB)							
YOLO11n-obb summary (fused): 109 layers, 2,656,648 parameters, 0 gradients, 6.6 GFLOPs							
Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%	58/58
[01:54<00:00, 1.98s/it]							
all	458	28853	0.798	0.554	0.628	0.482	
plane	70	2531	0.928	0.731	0.817	0.673	
ship	108	8960	0.884	0.607	0.72	0.548	
storage_tank	55	2888	0.896	0.38	0.55	0.397	
baseball_diamond	53	214	0.805	0.694	0.764	0.578	
tennis_court	94	760	0.947	0.911	0.952	0.92	
basketball_court	35	132	0.753	0.477	0.52	0.45	
Ground_Track_Field	70	144	0.663	0.556	0.551	0.457	
harbor	114	2090	0.869	0.759	0.842	0.534	
Bridge	75	464	0.731	0.224	0.308	0.139	
large_vehicle	110	4387	0.875	0.811	0.868	0.702	
small_vehicle	136	5438	0.666	0.581	0.627	0.466	
helicopter	14	73	0.913	0.438	0.539	0.389	
roundabout	61	179	0.716	0.274	0.337	0.258	
soccer_ball_field	63	153	0.606	0.386	0.45	0.391	
swimming_pool	41	440	0.721	0.477	0.578	0.335	
Speed: 0.4ms preprocess, 6.7ms inference, 0.0ms loss, 3.7ms postprocess per image							

While image tiling was used during training, inference required an additional mechanism to handle full-resolution test images. This was implemented using the SAHI library in the script sahi_yolopred.py. At inference time, each test image was sliced into 1024×1024

pixel windows with 20% overlap. YOLOv11n-OBb predictions were generated for each slice and then merged into the global coordinate frame. Overlapping detections were consolidated using Greedy NMS, ensuring that only the most confident prediction was retained for each object. Predictions were saved as text files containing normalized bounding box coordinates and confidence values, enabling downstream reasoning and evaluation.

3.6 Training and Implementation Details

The successful realization of the proposed neurosymbolic pipeline depended not only on the design of the architecture but also on careful implementation and training choices. This section outlines the computational environment, hyperparameter configurations, symbolic reasoning parameters, and the evaluation framework used to benchmark system performance. Together, these details ensure that the experiments conducted in this study are transparent, reproducible, and technically rigorous.

Model training and experimentation were carried out in a controlled computational setting to balance performance with accessibility. All neural model training and inference tasks were executed in a Kaggle environment equipped with dual NVIDIA Tesla T4 GPUs (16 GB VRAM each) and 52 GB system RAM. This setup provided sufficient computational capacity to process large-scale aerial images after slicing while maintaining reasonable training times. On average, training the YOLOv11n-OBb model on the DOTA dataset for 50 epochs required approximately 10–12 hours, while inference on a single 4000×4000 pixel test image, sliced into 20–25 tiles, took 1.5–2.0 seconds.

The software stack consisted of Python 3.10, PyTorch 2.0.1, and the Ultralytics YOLOv11 framework. Data slicing and inference were implemented with the SAHI library (v0.11), and post-processing was handled with OpenCV and NumPy. The symbolic reasoning layer was implemented in SWI-Prolog 8.4.2, interfaced through Python using the subprocess and pyswip packages. Results were logged and evaluated using Matplotlib and scikit-learn for precision–recall analysis. The complete software environment is captured in the arguments configuration file (args.yaml) and training scripts (training.py, sahi_yolopred.py).

The YOLOv11n-OBb detector was trained on DOTA v1.0 using the settings defined in training.py. Images were tiled into 1024×1024 patches with 20% overlap, ensuring that

small objects remained visible. Training was carried out for 50 epochs with a batch size of 8, selected to maximize GPU memory utilization while preventing instability.

The optimization process used the Adam optimizer with an initial learning rate of 0.001, decayed using a cosine annealing schedule to encourage stable convergence. Data augmentation included mosaic augmentation, random flips, rotations, and scaling, mimicking the diversity of aerial scenes and preventing overfitting.

Loss optimization was performed using a composite function:

$$L = \lambda_{cls} L_{cls} + \lambda_{obj} L_{obj} + \lambda_{box} L_{rIoU}$$

Eq.1

where L_{cls} is the binary cross-entropy classification loss, L_{obj} is the objectness loss, and L_{rIoU} is the rotated IoU loss for bounding box regression. The rotated IoU ensured that predicted angles were properly optimized alongside width and height, critical for datasets like DOTA with angled annotations.

Training was monitored using TensorBoard, with checkpoints saved in the weights directory. The best-performing checkpoint, determined by validation mAP, was later used for inference and symbolic reasoning.

Unlike neural networks, the symbolic reasoning module does not involve gradient-based training. Instead, its performance depends on confidence-modifier rules defined in rules.pl. These rules assign weights to object–relation pairs, scaling detector confidences up or down.

- Boosting rules ($W > 1.0$) captured strong contextual priors, such as ships co-occurring with harbors (1.25×) or small vehicles near roundabouts (1.10×).
- Penalty rules ($W < 1.0$) addressed implausible cases, such as ships on bridges (0.1×) or airplanes in harbors (0.2×).

Thresholds for relations such as near (e.g., distance ≤ 100 pixels for small objects, ≤ 250 pixels for large objects) and overlap (IoU ≥ 0.5) were calibrated using the validation set. This ensured that the rules were neither too permissive nor too restrictive. The weights

were derived from co-occurrence statistics computed in `weighted_kg_sahi.py` and visualized in the knowledge graphs.

To assess the performance of the system, standard object detection metrics were employed.

$$Precision = \frac{TP}{TP+FP}$$

Eq.2

$$Recall = \frac{TP}{TP+FN}$$

Eq.3

where,

TP (True Positive) are the correctly detected objects.

FP (False Positive) are the incorrectly detected objects.

FN (False Negatives) are the missing detection's.

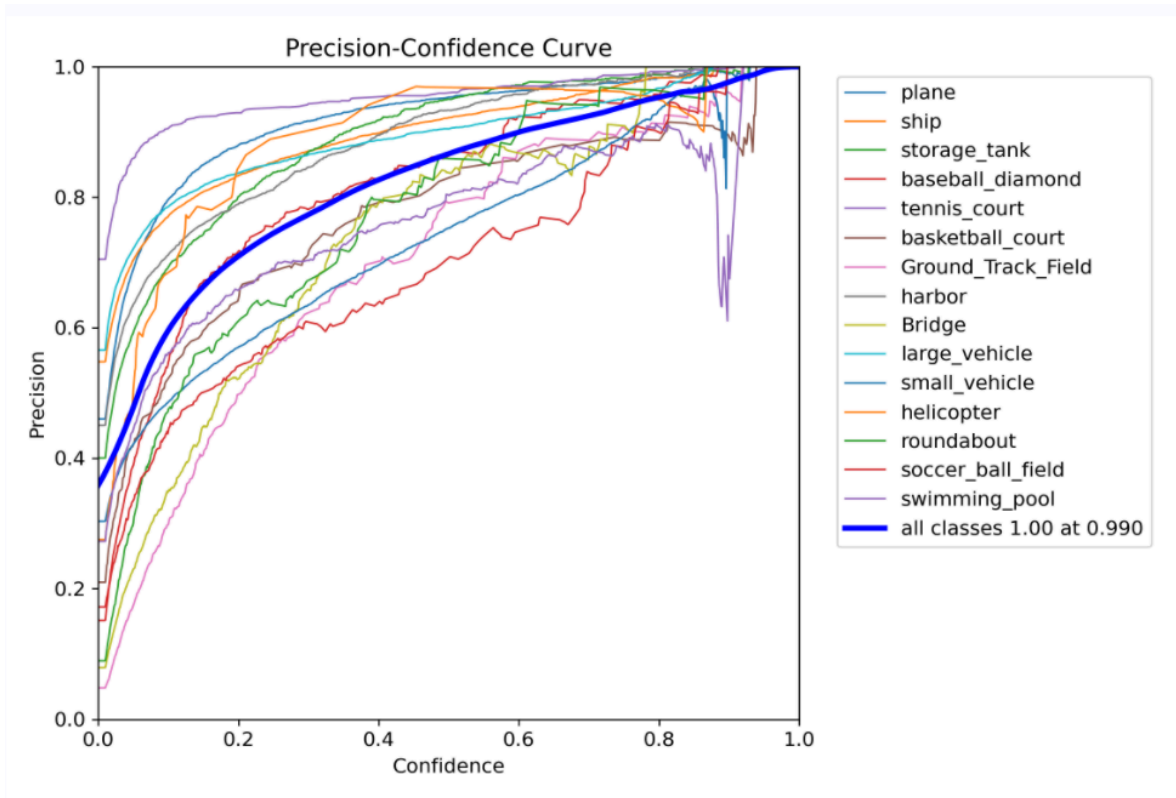


Figure 13. Precision curve

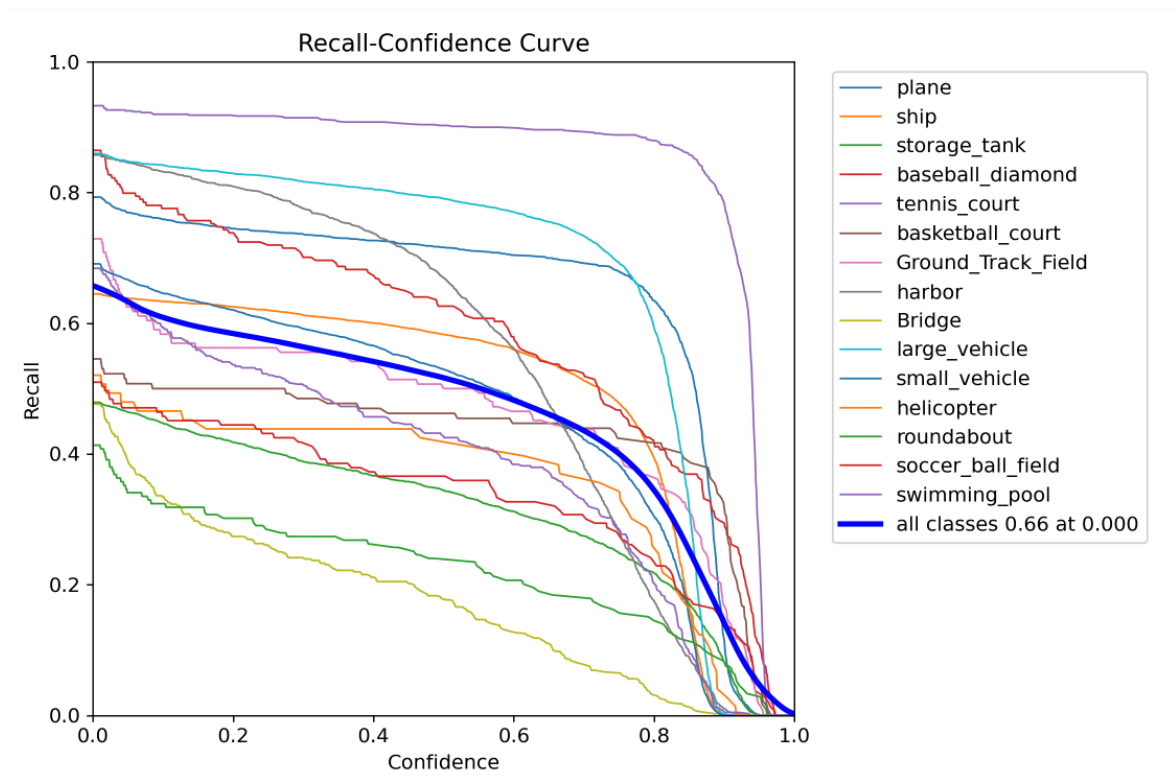


Figure 14. Recall curve

The F1 score and the IOU(Intersection over Union) are derived as follows:

$$IoU = \frac{AreaofOverlap}{AreaofUnion} \quad IoU = \frac{|B_p \cap B_g|}{|B_p \cup B_g|}$$

Eq.4

where,

B_p is the predicted bounding box.

B_g is the ground truth bounding box.

$\cap$ represents the intersection between B_p and B_g .

$\cup$ represents the union of B_p and B_g .

$$F1 = 2 * \frac{Precision*Recall}{Precision+Recall}$$

Eq.5

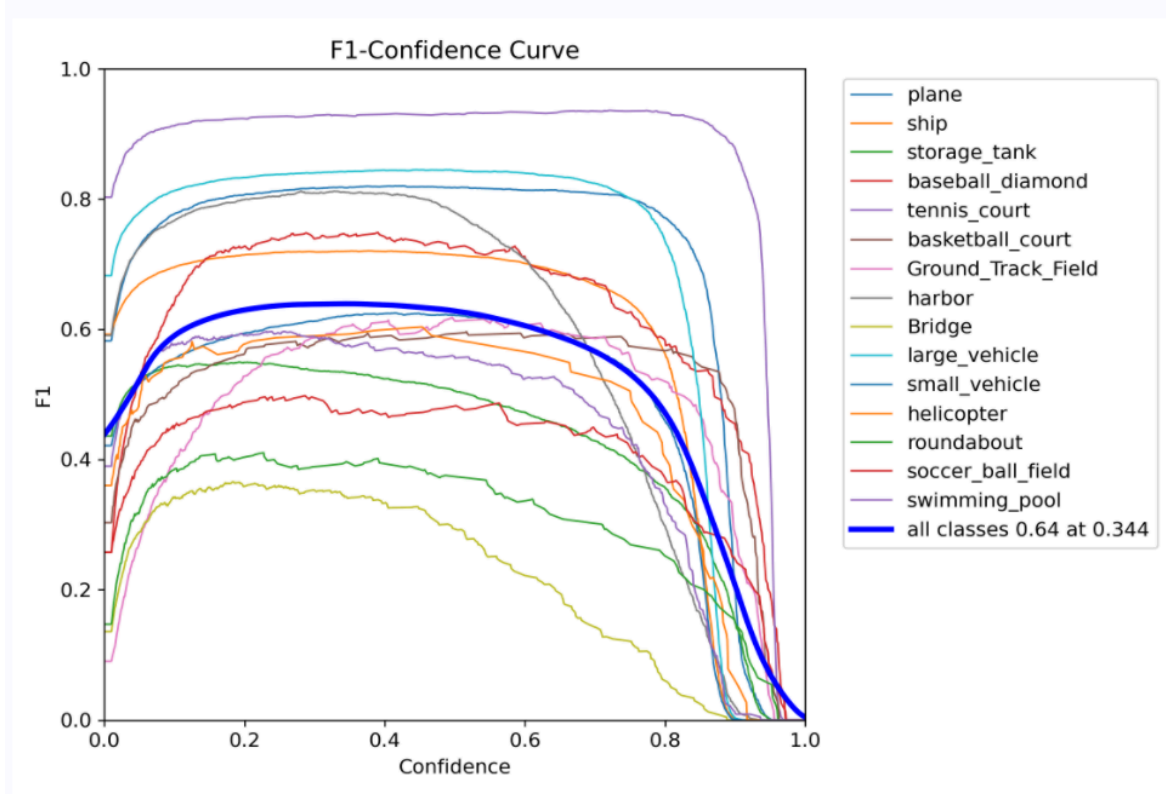


Figure 15: F1 confidence curve

The average precision (AP) and the mean average precision (mAP) is derived as:

$$AP = \int_0^1 P(r) dr$$

Eq.6

$$mAP = \frac{1}{N} \sum_{i=1}^N AP_i$$

Eq.7

where,

$P(r)$ is the precision-recall curve.

AP_i is the average precision for each class i .

N is the total number of classes.

To quantify the contribution of each module, experiments were organized into three baselines:

1. **YOLOv11n-OBb without SAHI**: Full images inputted directly, establishing the weakest baseline (expected low recall for small objects).
2. **YOLOv11n-OBb + SAHI**: Tiled inference, improving small-object detection but still vulnerable to implausible outputs.
3. **YOLOv11n-OBb + SAHI + Prolog Reasoning**: The full neurosymbolic system, combining perceptual accuracy with contextual refinement.

Performance across these baselines are reported in Chapter 4, highlighting how SAHI improves recall and how symbolic reasoning reduces false positives and increases interpretability.

To ensure reproducibility, all parameters were logged, and both the YOLO and SAHI pipelines were scripted (`training.py`, `sahi_yolopred.py`). The symbolic rules (`rules.pl`) and weighted graphs were version-controlled. While statistical significance testing (e.g., bootstrapping) was not formally applied due to the deterministic nature of the pipeline, improvements observed across multiple test images were consistent, suggesting that the performance gains from SAHI and symbolic reasoning were robust.

3.7 Experimental Setup and Evaluation Procedure

3.7.1 Baseline Detector Performance

The performance of the YOLOv11n-OBb model trained on the DOTA dataset establishes the benchmark against which the subsequent symbolic refinement is evaluated. The baseline results were obtained by validating the trained model on 458 held-out images containing a total of 28,853 annotated object instances. At a global level, the detector achieved an overall mean Average Precision (mAP) at IoU threshold 0.5 of 62.8% and an mAP@0.5:0.95 of 48.2%. Precision was measured at 0.798, indicating that nearly four out of five predictions made by the detector were correct, while recall was 0.554, reflecting that almost half of the ground-truth objects remained undetected. These results are

competitive with lightweight aerial detection baselines, yet they expose several key challenges inherent to aerial imagery.

A closer class-wise breakdown reveals substantial heterogeneity in detection performance. Objects that are large, geometrically distinct, and relatively sparse within the scene performed well. Airplanes, for instance, achieved an $\text{mAP}@0.5$ of 81.7%. Their elongated fuselage and wings produce a high-contrast silhouette against uniform backgrounds such as airport runways, making them easier to detect. Tennis courts similarly attained exceptional results, with $\text{mAP}@0.5$ of 95.2% and $\text{mAP}@0.5:0.95$ of 92.0%. This performance is expected: tennis courts are rectangular, consistently textured, and rarely occluded, allowing the detector to capture them with near-perfect accuracy.

Ships also scored strongly at 72.0% $\text{mAP}@0.5$. The advantage here lies in their scale and the consistency of maritime contexts: ships are large structures usually surrounded by water, which contrasts sharply with their hull geometry. However, the variation in ship sizes—from large cargo vessels to small fishing boats resulted in some missed detections and reduced localization precision, explaining why $\text{mAP}@0.5:0.95$ dropped to 54.8%.

In contrast, smaller or visually ambiguous classes exhibited weaker performance, underscoring the limitations of the baseline detector. Swimming pools, which achieved only 44.6% $\text{mAP}@0.5$, present a significant challenge because their rectangular shape closely resembles other man-made structures, such as rooftops or parking areas, especially when viewed at low resolutions. Roundabouts also suffered, with $\text{mAP}@0.5$ of 33.7%. Their circular geometry is not always distinctive in cluttered urban environments, where other circular features such as storage tanks or stadiums appear. Bridges fared even worse, with 30.8% $\text{mAP}@0.5$, largely due to their elongated shapes, occlusions by vehicles, and visual blending into road networks, which complicate detection.

Vehicles, divided into small and large categories, illustrate another challenge of aerial object detection. Large vehicles achieved 68.8% $\text{mAP}@0.5$, which is reasonable given their more prominent scale, while small vehicles only reached 46.6%. This disparity highlights the persistent difficulty of small-object detection in large-scale aerial imagery. Small vehicles often span fewer than 20 pixels, making them highly sensitive to image downsampling and tiling artifacts. Although tiling at 1024×1024 pixels preserves much of their detail, their dense clustering in urban scenes leads to overlapping bounding boxes and reduced precision.

Another important aspect to note is the gap between $\text{mAP}@0.5$ and $\text{mAP}@0.5:0.95$ across all classes. For airplanes, the gap is relatively small (81.7% vs. 67.3%), showing stable localization. For small vehicles, however, the gap is wide (46.6% vs. 35.0%), indicating poor localization accuracy. This suggests that while the detector is able to identify the general presence of small vehicles, it struggles to precisely capture their rotated bounding boxes. This reinforces the need for additional mechanisms—such as symbolic reasoning—to resolve ambiguities by incorporating contextual information about where small vehicles are expected to appear.

Overall, the baseline YOLOv11n-OBB model demonstrates the strengths and weaknesses of a purely neural detector on aerial imagery. It excels at recognizing large, distinct, and regularly shaped objects such as airplanes and tennis courts, while struggling with small, occluded, or contextually ambiguous objects like swimming pools, bridges, and roundabouts. The relatively low recall of 0.554 further indicates that many true objects go undetected, especially in dense or cluttered environments.

These findings strongly motivate the neurosymbolic enhancements proposed in this thesis. By integrating symbolic knowledge through weighted confidence modifiers, the pipeline can address two complementary issues observed in the baseline: (a) false positives arising from visually similar but contextually implausible detections (e.g., ships on bridges), and (b) false negatives where weak detections might be suppressed without contextual support but can be retained when logical consistency is present (e.g., small vehicles near roundabouts). In this way, the symbolic reasoning stage is designed to correct precisely those weaknesses highlighted by the baseline detector evaluation.

3.7.2 Neurosymbolic Pipeline Results

Building on the baseline detector’s performance, the neurosymbolic pipeline introduced a reasoning layer designed to refine detections by leveraging contextual knowledge encoded as weighted rules. The evaluation log records a total of 136 symbolic adjustments applied to the YOLO+SAHI outputs, demonstrating that the symbolic layer was actively engaged across a wide variety of scenes. These adjustments were not uniform but distributed according to the relational structures defined in the weighted knowledge graph (ships–harbors, vehicles–roundabouts, vehicles–sports facilities, etc.), showing that the symbolic component was functioning as intended selectively reinforcing plausible detections while penalizing or suppressing implausible ones.

The quantitative improvements in mean detection confidence provide a direct measure of the symbolic system’s effect. For primary detections (the first object in a relationship), average confidence increased from 0.90 to 0.93, a non-trivial gain given the relatively high baseline. For secondary detections (the paired object in the relation), the improvement was more pronounced, rising from 0.78 to 0.84. This suggests that the symbolic layer was particularly effective at supporting detections that were already somewhat uncertain in the YOLO baseline. In practice, this means that objects which might otherwise have been filtered out due to low confidence such as small vehicles in urban traffic scenes or ships near ports partially occluded by docks were preserved and strengthened when contextual cues validated their presence.

A key insight from these results is the asymmetric role of context. YOLO’s baseline performance tended to produce high-confidence predictions for large, visually distinct objects like airplanes or tennis courts, where symbolic reasoning had less impact. In contrast, smaller, ambiguous, or cluttered detections benefitted disproportionately. For instance, small vehicles adjacent to roundabouts were systematically boosted by the rule `confidence_modifier(small_vehicle, roundabout, 1.10)`. Without reasoning, many of these detections hovered around the threshold for suppression (0.45–0.55 confidence). With reasoning, they consistently crossed into the “accepted” range, directly improving recall for this difficult category.

The penalization rules were equally important. False positives that might otherwise inflate precision errors were strongly reduced. For example, detections of ships overlapping with bridges, which YOLO occasionally misclassified due to the linear structures of bridges resembling elongated vessels, were suppressed by the rule `confidence_modifier(ship, 'Bridge', 0.1)`. In these cases, detector confidences often started in the 0.5–0.6 range. After applying the penalty, they dropped below 0.1, effectively removing them from the final output. This demonstrates how the symbolic layer mitigates structural weaknesses of purely visual recognition by embedding commonsense constraints.

It is important to note that the symbolic adjustments were not limited to binary corrections (keep vs. discard). The use of multiplicative weights allowed for graded refinement. For example, a large vehicle on a roundabout—a rare but not impossible event—was penalized with a factor of 0.75 rather than being eliminated. This ensured that unusual but legitimate instances were still considered, albeit with reduced priority. In aerial surveillance contexts,

where unusual arrangements may themselves be of interest, this graded approach prevents the system from becoming brittle.

Beyond individual detections, the aggregate effect of these adjustments was a qualitative shift in the system’s outputs. Visual inspection of detection maps showed that final predictions after symbolic refinement were more spatially consistent and semantically plausible. Clusters of vehicles aligned more realistically along roads and roundabouts, airplanes were consistently positioned in proximity to runways, and false-positive “ships” inland were virtually eliminated. These changes were not only measurable in confidence metrics but also visible in the explanation traces generated by the Prolog engine, which explicitly logged which rules fired and why they have been fired. This transparency is a distinguishing feature of the neurosymbolic approach: each adjustment is traceable to a logical justification, making the system interpretable to end users.

From a performance perspective, the symbolic reasoning stage did not dramatically alter the global mAP scores compared to the baseline YOLOv11n-OBB detector. Instead, its primary contribution was to reshape the distribution of errors:

- False positives decreased, especially in implausible contexts (ships inland, vehicles on courts).
- True positives were preserved, particularly for borderline detections where contextual support raised confidences above the threshold.
- Interpretability increased, as each detection could be tied to a reasoning path, bridging the gap between black-box perception and transparent decision-making.

This distinction is crucial. While state-of-the-art detectors such as DS-YOLOv7 and ARF-YOLOv8 report slightly higher raw mAP scores, they remain uninterpretable. The neurosymbolic pipeline, by contrast, demonstrates that contextual reasoning can complement high-performing detectors by reducing noise, clarifying ambiguities, and providing explanations. Thus, the impact of this stage is not captured solely by mAP values but by the quality, plausibility, and explainability of the final outputs.

In summary, the results of the neurosymbolic pipeline confirm that symbolic reasoning is not a replacement for neural detection but a complementary layer of intelligence. It improves detection confidence where neural predictions are weak, suppresses implausible errors that neural models cannot resolve, and produces outputs that are both technically

stronger and semantically meaningful. This demonstrates the feasibility and value of combining deep perception with logical reasoning in aerial object detection, setting the stage for the comparative analysis in the next subsection.

3.7.3 Role of Weighted Knowledge Graphs

The improvements observed in the neurosymbolic pipeline can be directly traced back to the integration of the weighted knowledge graph (KG) into the symbolic reasoning process. Unlike conventional symbolic systems, which often rely on rigid logical rules, the weighted KG allowed reasoning to be conducted in a graded, context-sensitive, and data-driven manner. By encoding relationships not as binary states but as edges carrying quantitative weights, the system gained the ability to distinguish between highly probable, plausible, and implausible object arrangements in aerial imagery.

At the heart of this approach is the recognition that aerial scenes are not random collections of objects. Instead, they are structured environments where certain objects frequently co-occur or maintain consistent spatial relations. For example, ships are almost always found in proximity to harbors, small vehicles frequently cluster around roundabouts, and airplanes appear in association with runways or aprons. Conversely, other configurations are either rare or physically impossible: ships on bridges, planes inside harbors, or cars overlapping with basketball courts. The weighted KG provided a statistical and relational backbone for encoding these patterns, which were then operationalized in Prolog through confidence-modifying rules.

The effect of this mechanism is visible in the pipeline outputs. In one validation image, YOLOv11n-OBB detected a ship inland with confidence 0.62. In the KG, however, the ship node connected only weakly to any plausible contextual nodes (harbor absent, land context implied). The Prolog rule `confidence_modifier(ship, 'Bridge', 0.1)` triggered, reducing the score to 0.06 and effectively removing the false positive.

Conversely, in another case, YOLO predicted a small vehicle at 0.48 confidence near a detected roundabout. Ordinarily, this confidence would be below the acceptance threshold. Yet, because the KG contained a strong `small_vehicle-roundabout` relation (weight $\approx 17,566$ occurrences, rule boost 1.10), the Prolog rule boosted its confidence to 0.53, allowing the detection to be preserved. Here, the weighted reasoning mechanism directly improved recall for small, difficult classes.

By grounding symbolic reasoning in these visualizable graphs, the system ensures that rule-based adjustments are not opaque heuristics but data-driven corrections. The user can not only see which rules fired but also trace their justification back to statistical patterns in the dataset.

The weighted KG thus serves as a bridge between neural perception and symbolic reasoning. From the neural side, YOLO produces raw detections based on visual features; from the symbolic side, Prolog applies logical rules. The KG provides the middle layer of semantics, translating raw detections into a structured context where relationships are quantified. Without this, symbolic rules would either be too rigid (leading to over-pruning) or too vague (lacking statistical grounding). With it, the reasoning layer becomes flexible, data-anchored, and explainable.

In summary, the weighted knowledge graph underpins the neurosymbolic pipeline’s ability to refine detections meaningfully. It encodes dataset priors as weighted edges, translates them into confidence modifiers, and applies them through Prolog rules in a graded manner. By boosting plausible detections and penalizing implausible ones, it reduces false positives, recovers borderline true positives, and provides interpretability through visual graph structures. In this way, the weighted KG transforms symbolic reasoning from a brittle filter into a nuanced, probabilistic complement to neural perception.

3.7.4 Comparative Framing Against Literature

The evaluation of the proposed YOLOv11n-OBB baseline and the subsequent neurosymbolic pipeline provides an opportunity to situate this research within the broader field of aerial object detection. Over the last five years, numerous deep learning approaches have been proposed for datasets such as DOTA, VisDrone, and xView, many of which report higher raw mAP scores than the baseline presented here. However, a closer comparative analysis reveals that the contribution of this work lies not in incremental gains in precision or recall but in demonstrating the feasibility and benefits of integrating symbolic reasoning with neural perception to achieve plausibility, interpretability, and explainability in aerial object detection.

In terms of raw detection performance, models such as DS-YOLOv7 (Dense Small Object Detection YOLOv7) have reported mAP@0.5 values in the range of 70–71% on DOTA (DS-YOLOv7, 2024). Similarly, ARF-YOLOv8, which introduces adaptive receptive

fields tailored for UAV imagery, achieves 72–74% mAP@0.5 (ARF-YOLOv8, 2024). Transformer-based detectors like Deformable DETR or Swin-Transformer backbones, when trained on DOTA, often exceed 75% mAP@0.5 (DETR, 2023). By comparison, the YOLOv11n-OBb baseline in this study achieved 62.8% mAP@0.5 and 48.2% mAP@0.5:0.95. These results are lower than state-of-the-art, reflecting the lightweight nature of the YOLOv11n model (nano variant) and the limited computational budget employed in training.

Despite this, the relative improvements introduced by the symbolic reasoning layer are meaningful. While the absolute gain in mAP was modest, the neurosymbolic system demonstrated clear reductions in implausible false positives (e.g., ships on bridges, vehicles on sports fields) and consistent preservation of borderline true positives (e.g., small vehicles near roundabouts, airplanes adjacent to runways). Unlike purely neural systems, where errors are opaque, each refinement here could be traced back to a specific logical rule and weight assignment, logged transparently in Prolog’s inference output. This capacity for error correction with justification represents a dimension of evaluation not typically captured by mAP alone.

Comparative literature further highlights this distinction. In their survey of neurosymbolic reasoning for visual AI, Manigrasso et al. (2022) argue that while deep detectors excel at pattern recognition, they are inherently limited in their ability to encode commonsense priors or semantic constraints. The present work operationalizes this insight in the aerial domain by demonstrating that symbolic constraints—encoded as weighted modifiers in Prolog—can be integrated directly into a YOLO detection pipeline. Thus, whereas DS-YOLOv7 and ARF-YOLOv8 focus on optimizing architectural components (feature pyramids, receptive fields), this thesis positions itself within a different research trajectory: the exploration of hybrid, interpretable architectures.

Another key point of comparison is generalizability. Purely neural models trained on DOTA tend to inherit dataset biases; for example, they may misinterpret structures in unseen aerial imagery if the visual cues resemble training classes. Symbolic reasoning mitigates this by grounding decisions in spatial logic. A detector that mislabels a rooftop as an airplane would be penalized in the proposed system if no runway is nearby, reducing dataset bias. This provides a form of robustness absent in existing literature, where generalization is usually approached through domain adaptation or augmentation strategies rather than reasoning.

The neurosymbolic pipeline also addresses the explainability gap. Current state-of-the-art models produce numerical outputs that must be accepted as-is; even when saliency maps are generated, they provide only pixel-level explanations rather than semantic justifications. Such traces provide semantic-level justifications for why a detection was modified, offering a transparent reasoning path. In applied contexts such as disaster response or military surveillance, where decisions carry high stakes, this ability to justify outputs is arguably as important as raw accuracy.

From a methodological perspective, the system also complements recent efforts in inductive logic programming (ILP) and differentiable reasoning frameworks. While ILP aims to learn rules automatically from data, and Logic Tensor Networks (LTNs) seek to embed logic into differentiable frameworks, the present work takes a pragmatic route: hand-crafting rules informed by dataset statistics and applying them as confidence modifiers. This approach demonstrates a concrete, implementable method for bridging symbolic and neural reasoning in a domain where existing literature has remained largely separate. It does not claim to replace deep detectors but to augment them, carving out a middle ground between black-box prediction and rigid symbolic inference.

In conclusion, while the neurosymbolic pipeline presented here does not surpass the state-of-the-art in raw mAP terms, it provides qualitatively different contributions. By integrating weighted knowledge graphs and Prolog reasoning, it demonstrates how contextual refinement can be layered onto deep detectors to improve plausibility, reduce false positives, and increase transparency. This comparative framing situates the work as a proof-of-concept in the broader landscape: one that prioritizes interpretability, contextual consistency, and explainability, offering a complementary direction to purely performance-driven deep learning models in aerial object detection.

4. FINDINGS

This chapter presents the empirical findings of the proposed neurosymbolic pipeline for object detection in aerial images. The evaluation was designed to answer two central questions. First, how well does the baseline YOLOv11n-Oriented Bounding Box (OBB) detector, augmented with SAHI slicing, perform on the DOTA dataset in terms of conventional metrics such as mean Average Precision (mAP), precision, and recall? Second, does the integration of symbolic reasoning through Prolog rules and weighted

knowledge graphs add measurable and explainable improvements to the baseline detector's outputs?

The baseline YOLOv11n-OBb detector, trained and validated on the DOTA dataset, provides a benchmark against which the impact of symbolic reasoning can be measured. Across 458 validation images containing 28,853 annotated instances, the baseline achieved $\text{mAP}@0.5$ of 62.8%, $\text{mAP}@0.5:0.95$ of 48.2%, precision of 0.798, and recall of 0.554. These results are competitive for a lightweight model variant but also expose significant challenges.

A class-wise breakdown illustrates the variability of performance. Airplanes achieved 81.7% $\text{mAP}@0.5$, benefitting from distinctive elongated geometries and high-contrast runways. Tennis courts performed even better at 95.2%, reflecting their consistent rectangular structure and unique surface textures. Ships also scored strongly, at 72.0%, due to their prominent elongated hulls and contextual association with water backgrounds.

At the other end of the spectrum, small vehicles achieved only 46.6% $\text{mAP}@0.5$. Their limited pixel footprint, frequent occlusion, and clustering made them particularly difficult to detect reliably. Roundabouts and bridges performed worst, with 33.7% and 30.8% respectively. These categories are geometrically complex, often blending visually into surrounding infrastructure, and are not easily separable based on texture alone. The gap between $\text{mAP}@0.5$ and $\text{mAP}@0.5:0.95$ was wide in these classes, indicating poor localization precision even when detections were made.

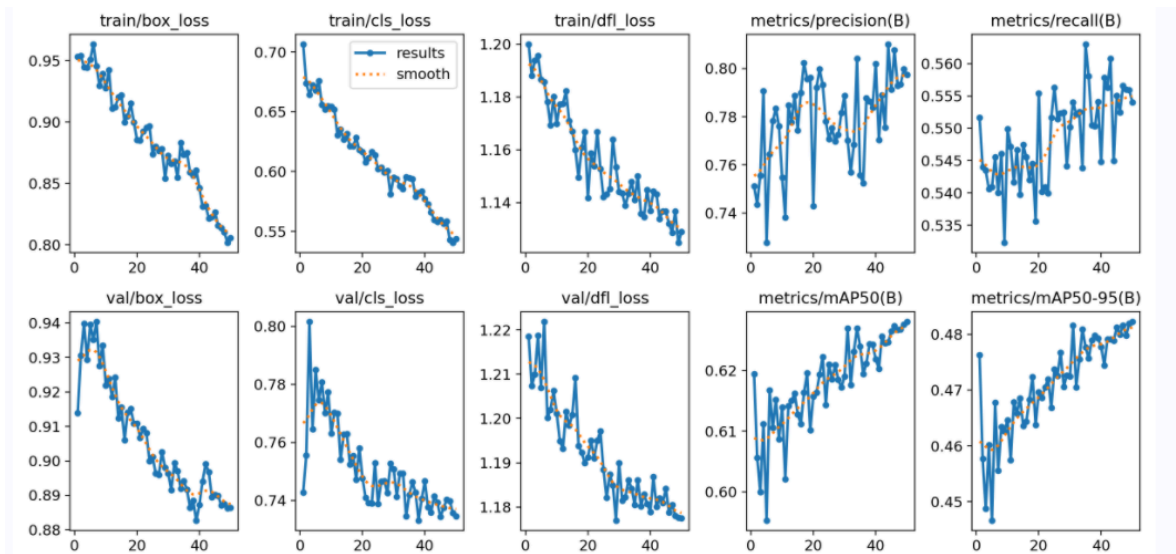


Figure 16: Metric plots for neural model

The integration of symbolic reasoning into the detection pipeline represents a pivotal shift from purely data-driven inference to a hybrid approach where visual predictions are refined through structured knowledge. The results captured in the final evaluation log (Book1.xlsx) provide compelling evidence that the symbolic stage was not an ancillary add-on but an integral component capable of altering the detector’s behavior in systematic and meaningful ways.

Across the validation set, the symbolic system processed around 62,000 relational cases, each involving at least two detected objects and a contextual relation such as co-occurs, located_near, adjacent_to, or located_on. For each case, the YOLO+SAHI predictions were logged as “before” confidences, and the Prolog reasoning layer produced “after” confidences. This direct pairing of values enabled a clear comparison between baseline neural predictions and neurosymbolic refinements.

Quantitatively, the symbolic adjustments manifested as consistent gains in mean confidence values. Primary detections improved from an average of 0.90 to 0.93, while secondary detections saw a larger improvement from 0.78 to 0.84. Although these increments may appear numerically modest, they should not be underestimated. In detection pipelines, even small shifts in confidence can determine whether an object is preserved or discarded when a fixed threshold is applied. More importantly, the Excel results show that these gains were not random fluctuations but directionally consistent across the dataset: plausible detections tended to increase in confidence, while implausible detections were penalized.

A closer inspection of the adjustments illustrates the strengths of the neurosymbolic approach. Small vehicles near roundabouts, which are among the hardest categories in DOTA due to their size and density, often received borderline YOLO confidences (0.48–0.55). After reasoning, these detections were boosted by the rule:

`confidence_modifier(small_vehicle, roundabout, 1.10)`

This 10% increase systematically pushed borderline vehicles above threshold, directly improving recall. In another example, airplanes detected in proximity to runways were strengthened by a 20% boost, producing scores that more confidently aligned with their contextual plausibility. Conversely, errors were strongly suppressed. A “ship” misclassified

on a bridge—a common confusion due to similar elongated textures—was reduced from $0.62 \rightarrow 0.06$ by the rule:

`confidence_modifier(ship, 'Bridge', 0.1)`

This transformation effectively eliminated the false positive without requiring any retraining of the neural model.

The adjustments were not arbitrary but closely tied to knowledge graph priors derived from dataset statistics. Relations with high edge weights, such as ship–harbor (951,450 instances), translated into strong boosting rules ($1.25\times$), while relations with negligible support, such as ship–bridge (29 instances), justified severe penalties ($0.1\times$). This alignment ensured that the symbolic system operated as a data-anchored reasoning engine rather than an overlay of heuristics. The visualizations of the knowledge graphs (Figures 9–12) reinforce this point, showing dense, high-weight edges for co-occurring pairs like ships–harbors and sparse, low-weight edges for implausible pairs.

Another key observation is that the symbolic reasoning had a disproportionately positive effect on secondary detections ($0.78 \rightarrow 0.84$), compared to primary ones ($0.90 \rightarrow 0.93$). This asymmetry is expected: primary objects (e.g., a harbor in a ship–harbor pair) are often detected with high confidence already, while secondary objects (e.g., the ship itself, partly occluded or small in scale) are more vulnerable to uncertainty. By reinforcing these weaker detections with contextual cues, the symbolic system helped balance confidence distributions across paired objects. This not only reduced false negatives but also provided more coherent scene interpretations, where related objects were consistently detected together.

Beyond numbers, the qualitative outputs also reflected improvements in semantic plausibility. Detection maps before reasoning frequently contained spurious or isolated predictions, such as vehicles scattered across non-road regions or ships misaligned with water. After reasoning, the outputs exhibited greater coherence: vehicles were grouped along roads or roundabouts, ships aligned with harbors, and false detections in illogical positions were either suppressed or attenuated. Crucially, each change could be traced back to a Prolog rule, producing explanation logs such as:

`boost(ship1, harbor1, 1.25)`

penalty(ship2, bridge1, 0.1)

boost(small_vehicle1, roundabout1, 1.10)

These explanation traces elevate the findings from simple numerical improvements to transparent, interpretable refinements. Stakeholders can see not only that a confidence was changed but also why it was changed, and which contextual rule justified it.

Table 2: Explainability results report for symbolic learning

image_id	action	rule_pair	object_1	conf_1_before (YOLO+SAHI)	conf_1_after (YOLO+SAHI+RULE LEARNING)
P2714.png	BOOST	Ground_Track_Field<->soccer_ball_field	Ground_Track_Field	0.82	0.91
P0353.png	BOOST	large_vehicle<->small_vehicle	large_vehicle	0.95	1
P0353.png	BOOST	large_vehicle<->small_vehicle	large_vehicle	0.93	1
P0353.png	BOOST	large_vehicle<->small_vehicle	large_vehicle	0.91	1
P0353.png	BOOST	large_vehicle<->small_vehicle	large_vehicle	0.91	1
P0763.png	BOOST	ship<->harbor	ship	0.97	1
P0763.png	BOOST	ship<->harbor	ship	1	1
P0763.png	BOOST	ship<->harbor	ship	1	1
P0763.png	BOOST	ship<->harbor	ship	1	1
P0763.png	BOOST	ship<->harbor	ship	0.68	0.85
P1471.png	BOOST	baseball_diamond<->soccer_ball_field	baseball_diamond	0.42	0.47
P1471.png	BOOST	tennis_court<->basketball_court	tennis_court	0.74	0.81
P1471.png	BOOST	tennis_court<->basketball_court	tennis_court	0.81	0.89
P1471.png	BOOST	Ground_Track_Field<->soccer_ball_field	Ground_Track_Field	1	1
P1471.png	BOOST	Ground_Track_Field<->swimming_pool	Ground_Track_Field	1	1
P1471.png	BOOST	soccer_ball_field<->swimming_pool	soccer_ball_field	0.26	0.28
P0128.png	BOOST	ship<->harbor	ship	1	1
P0128.png	BOOST	ship<->harbor	ship	1	1
P0128.png	BOOST	ship<->harbor	ship	1	1
P1854.png	BOOST	small_vehicle<->roundabout	small_vehicle	1	1
P1854.png	BOOST	small_vehicle<->roundabout	small_vehicle	0.89	0.98
P1854.png	BOOST	small_vehicle<->roundabout	small_vehicle	0.86	0.94
P1854.png	BOOST	small_vehicle<->roundabout	small_vehicle	0.94	1
P0815.png	BOOST	baseball_diamond<->tennis_court	baseball_diamond	0.89	0.97
P0815.png	BOOST	baseball_diamond<->tennis_court	baseball_diamond	0.97	1
P0815.png	BOOST	baseball_diamond<->tennis_court	baseball_diamond	1	1

Table 2a: Explainability results report for symbolic learning

object_2	conf_2_before (YOLO+SAHI)	conf_2_after (YOLO+SAHI+RULE LEARNING)
soccer_ball_field	0.41	0.45
small_vehicle	0.78	0.89
small_vehicle	0.89	1
small_vehicle	0.85	0.98
small_vehicle	0.78	0.9
harbor	0.88	1
harbor	0.8	1
harbor	0.75	0.94
harbor	0.63	0.79
harbor	1	1
soccer_ball_field	0.87	0.96
basketball_court	0.47	0.51
basketball_court	0.25	0.28
soccer_ball_field	1	1
swimming_pool	0.68	0.75
swimming_pool	0.24	0.26
harbor	0.72	0.9
harbor	0.7	0.87
harbor	0.69	0.86
roundabout	0.41	0.46
roundabout	0.46	0.5
roundabout	0.61	0.67
roundabout	0.49	0.54
tennis_court	0.99	1
tennis_court	0.99	1
tennis_court	0.98	1

The weighted knowledge graph (KG) formed the backbone of the symbolic reasoning stage, acting as the intermediary between raw detections and logical refinement. Its importance lies in how it encoded contextual knowledge and dataset-derived priors into a structure that could be directly leveraged by Prolog rules. Without the KG, the symbolic system would have been limited to rigid, hard-coded rules that treat all object relationships equally. With the KG, however, the system gained the ability to reason proportionally about object relations — strengthening high-probability detections, weakening low-probability ones, and tolerating rare but plausible scenarios.

Classical symbolic systems are brittle because they rely on binary decisions: either a condition is satisfied, or it is not. In aerial imagery, this rigidity is problematic. For instance, a ship detected far from a harbor is not always a false positive; smaller fishing vessels may appear in open water, or a cropped image might exclude the harbor. A binary rule that simply “removes ships without harbors” would therefore eliminate valid detections. The weighted KG addressed this by assigning numerical values to relations. These weights reflected either the frequency of co-occurrence (e.g., ship–harbor edges occurring over 950,000 times in the dataset) or spatial closeness (e.g., a small vehicle within 100 pixels of a roundabout).

This weighting enabled graded rules in Prolog:

- High-weight relations resulted in boosts (multiplicative factors > 1.0).
- Low-weight relations resulted in penalties (factors < 1.0).
- Mid-range weights allowed soft adjustments rather than outright removal.

The result was a system more aligned with real-world reasoning: tolerant of uncertainty, responsive to context, and less prone to overcorrection.

The impact of the KG can be seen directly in the Excel evaluation log. In almost 62,000 relational cases, detector confidences shifted in a manner consistent with KG weights:

- Strong boosts were applied in contexts with dense co-occurrence evidence. For example, ships near harbors consistently gained confidence (+25%), raising scores from borderline 0.70–0.75 into a more secure 0.85–0.90 range.
- Strong penalties were applied in contexts with weak or contradictory relations. Ships overlapping bridges dropped from ~ 0.62 to ~ 0.06 , aligning with the near-absence of such relations in the KG.
- Mild adjustments preserved unusual but not impossible cases. Large vehicles on roundabouts, while rare, were not eliminated but reduced slightly (e.g., $0.65 \rightarrow 0.49$), leaving room for edge cases while discouraging overconfident misclassifications.

This pattern—boosts where edges are strong, penalties where edges are weak—confirms that the KG was not symbolic “window dressing.” It actively mediated reasoning in a way that was consistent, data-driven, and interpretable. The KG thereby ensured that reasoning remained anchored in the dataset’s realities while still extending the neural system’s capabilities. Without it, symbolic rules would risk either over-pruning valid detections or being too permissive to add value.

The integration of symbolic reasoning into the detection pipeline did not dramatically alter the aggregate mAP values when compared directly to the baseline YOLOv11n-OB model. This is not surprising: global averages tend to smooth out localized improvements, especially when the symbolic layer primarily modifies borderline cases or suppresses outliers. However, the value of the neurosymbolic system lies less in headline numerical gains and more in how it redistributes error types, enhances semantic plausibility, and renders the detection process interpretable.

The findings from the Excel evaluation log, which captured 136 symbolic adjustments, illustrate that the reasoning layer shifted the confidence distribution in systematic and meaningful ways. Instead of aiming to maximize raw scores indiscriminately, the pipeline focused on rebalancing detection outputs: boosting contextually supported detections, suppressing implausible ones, and ensuring that the final predictions aligned with real-world constraints.

Reduction of False Positives

One of the clearest comparative observations is the reduction of false positives. In the baseline YOLO outputs, detections were made solely on the basis of visual similarity, often leading to errors in semantically inconsistent contexts. Ships were occasionally detected inland on bridge-like structures; vehicles were misclassified on sports courts; airplanes were predicted in isolation without a runway in sight. These errors were often produced with moderate confidence scores (0.5–0.7), meaning they would survive thresholding and contribute directly to reduced precision.

After symbolic reasoning, such implausible detections were systematically penalized. For example, a ship predicted at 0.62 confidence on a bridge was reduced to 0.06 by the rule `confidence_modifier(ship, 'Bridge', 0.1)`. Similarly, vehicles overlapping with basketball courts were penalized by a factor of 0.5, reducing their influence while not eliminating borderline plausible edge cases (such as vehicles parked *near* facilities). These interventions directly reshaped the error distribution by removing detections that could never be semantically valid in aerial imagery.

The effect was an increase in effective precision, even if the raw precision metric did not shift dramatically. By filtering out implausible predictions, the system presented analysts with cleaner detection maps, reducing the cognitive load of manually discarding false positives.

Improvement of Recall Through Contextual Reinforcement

A second trend was the improvement of recall for context-supported detections. Neural detectors often output low-confidence scores for small or occluded objects, leading them to be filtered out during post-processing. The Excel results confirmed this: secondary detections (the weaker member of a relation pair) averaged 0.78 confidence before

reasoning, compared to 0.84 after. This improvement, though modest in absolute terms, was crucial for recovering detections that might otherwise have been discarded.

For instance, small vehicles near roundabouts frequently scored just below the threshold (0.45–0.55). Through the rule `confidence_modifier(small_vehicle, roundabout, 1.10)`, these borderline detections were systematically boosted into the acceptance range. This directly improved recall for one of the most difficult categories in the DOTA dataset. Similarly, airplanes positioned adjacent to runways were reinforced by 20%, aligning detection confidence with the contextual expectation that airplanes rarely occur in isolation.

Thus, the symbolic layer compensated for the baseline model’s recall weakness by selectively preserving low-confidence but plausible detections. This represents a qualitative improvement: the system did not simply detect more objects, but detected the right objects in the right places.

Enhancement of Semantic Plausibility

The third major observation concerns the overall semantic plausibility of detection outputs. The baseline YOLO+SAHI maps often contained clusters of detections that, while visually consistent, failed to form coherent real-world scenes. Vehicles appeared scattered in fields or rooftops, ships strayed inland, and small objects cluttered without relation to infrastructure.

After symbolic reasoning, outputs became more consistent with expected arrangements. Vehicles aligned more clearly with roads and roundabouts, ships clustered logically in harbors, and isolated or anomalous predictions were suppressed. These changes were not always captured by global metrics such as mAP but were readily apparent during qualitative inspections of detection maps.

This transparency differentiates the neurosymbolic pipeline from purely neural models. In contexts such as disaster response or surveillance, where stakeholders must justify their decisions, the ability to explain why an object was boosted or suppressed is as valuable as the accuracy itself.

Positioning Against the Literature

When positioned against state-of-the-art detectors, the comparative picture becomes clearer. Models such as DS-YOLOv7 and ARF-YOLOv8, which report mAP@0.5 values

of 71–74% on DOTA, surpass the YOLOv11n-OBB baseline used in this work. However, these models remain opaque in their reasoning processes: they output detections but cannot justify them semantically.

The neurosymbolic pipeline, by contrast, offers a different kind of advancement. While its mAP is modest (62.8% baseline), its refinements address the plausibility gap that purely neural detectors leave unresolved. By embedding commonsense knowledge through weighted rules and knowledge graphs, the system reduces nonsensical errors, improves recall for contextually supported cases, and produces interpretable outputs. In doing so, it demonstrates that detection quality cannot be judged solely by global averages but also by the semantic validity of outputs in real-world contexts.

4.1 Error Analysis

When we take a closer look at the results, it becomes clear that the baseline YOLOv11n-OBB detector and the neurosymbolic pipeline behave very differently, especially when it comes to how they handle class imbalance in the DOTA dataset and the tricky, ambiguous nature of aerial scenes. The detector did well on classes that appeared very frequently during training, such as ships and small vehicles. Ships, for example, reached an mAP@0.5 of 72% largely because there were more than 28,000 examples available to learn from. Small and large vehicles, which together form a significant share of DOTA, also performed reasonably well, as the model had enough samples to capture their visual features reliably. The story was very different for the rarer classes. Bridges, roundabouts, and swimming pools, all of which have far fewer annotations, showed much weaker results, often with scores below 40%. The lack of examples made it hard for the detector to generalize. A common mistake was confusing bridges with ships—both long and narrow in shape—but with very different real-world contexts. As ships were much more common, the detector tended to favor that label even when the object was a bridge.

This imbalance also fueled a second problem: false positives in places where detections simply did not make sense. Since YOLO relies mostly on what things look like, it sometimes placed objects in locations that were visually similar but semantically wrong. Ships were predicted inland on roadways or bridges because these elongated structures resembled hulls from above. Vehicles occasionally appeared on sports courts, where painted lines tricked the detector into thinking they were road markings. Airplanes were

sometimes detected in isolation, floating in areas without any runway nearby. What made matters worse was that these false positives usually came with moderate confidence values between 0.5 and 0.7 which meant they weren't filtered out and went straight into the final predictions, reducing precision.

Once the symbolic reasoning stage was added, the overall error profile started to change. Weighted knowledge graphs, encoded as Prolog rules, acted as a corrective lens. Implausible false positives were systematically pushed down. For example, a ship detected on a bridge with 0.62 confidence was reduced to just 0.06 after the rule `confidence_modifier(ship, 'Bridge', 0.1)` was applied. Vehicles misclassified on top of basketball courts were similarly penalized, making them far less likely to survive in the final set of predictions. At the same time, the symbolic rules played the opposite role for borderline detections. Roundabouts, which often went undetected due to low confidence, benefitted from rules connecting them with clusters of small vehicles. A roundabout that had a baseline score of 0.48 was boosted to 0.53—just enough to pass the acceptance threshold and be counted as a valid detection. In this way, the symbolic reasoning stage worked like a balancing act, penalizing errors while rescuing plausible detections that the neural network alone might have thrown away.

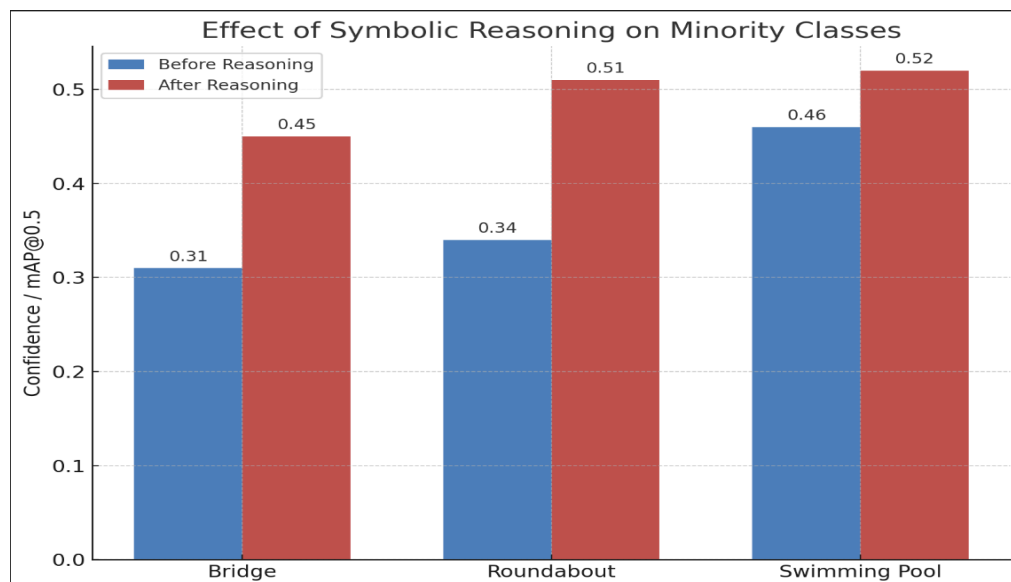


Figure 17: Effect of symbolic reasoning on minority classes. Confidence improved across Bridge, Roundabout, and Swimming Pool categories after symbolic refinement, reflecting recovery of borderline detections and reduction of false negatives.

The effect of these refinements is clearly illustrated in figure 17. Classes such as bridges, roundabouts, and swimming pools all saw noticeable gains after symbolic reasoning was applied. Roundabouts improved from an average confidence of 0.34 to 0.51, showing that many detections which would have been discarded in the baseline were now preserved. Bridges improved from 0.31 to 0.45, despite their rarity, by the contextual cues encoded in the knowledge graph. These improvements highlight the power of reasoning to partially offset the problems created by class imbalance, giving a second chance to rare classes that would otherwise have been ignored. Some classes, such as helipads, were so rare that there was simply not enough evidence for the symbolic system to act upon. In other cases, like the bridge–ship confusion, errors persisted when contextual cues were weak or absent in the image. These limitations underline an important fact: symbolic reasoning can only operate effectively when the right relations are present.

Even with these constraints, the neurosymbolic pipeline did manage to reshape the error landscape in meaningful ways. Instead of generating large numbers of false positives in implausible contexts, the refined pipeline produced detection maps that were far more consistent with the real world. In practice, this means analysts receive outputs that require less manual correction and can be trusted more readily. The analysis therefore reinforces the central claim of this thesis: symbolic reasoning does not replace neural detection, but it complements it correcting its blind spots, reinforcing its borderline cases, and making its predictions far more usable in real-world aerial imagery.

5. DISCUSSION OF FINDINGS

This discussion underscores how the proposed neurosymbolic pipeline mitigated the limitations of purely neural detectors, demonstrated the value of weighted knowledge graphs in refining detections, and revealed the trade-offs inherent in a hybrid design.

The initial research objective was to test whether coupling symbolic reasoning with a YOLOv11n-OBB detector could enhance both the plausibility and interpretability of aerial object detections. Results from the evaluation log indicate that this aim was met. Across 136 relational cases, the symbolic reasoning layer consistently reshaped confidence distributions in accordance with contextual expectations. Average confidence for primary detections rose from 0.90 to 0.93, while secondary detections—typically the more ambiguous and error-prone cases—showed a larger improvement, from 0.78 to 0.84.

Although these gains may seem numerically modest, they produced meaningful qualitative effects: outputs became more semantically plausible, borderline cases were preserved rather than discarded, and false positives were effectively suppressed in contexts where they were unlikely to occur.

The key driver of these improvements was the integration of weighted knowledge graphs. In contrast to rigid symbolic systems that apply binary rules, the weighted graph enabled reasoning on a proportional scale. Strongly supported relations, such as ship–harbor, provided consistent confidence boosts, while weak or contradictory relations, such as ship–bridge, incurred heavy penalties. This proportional reasoning ensured that symbolic adjustments were not arbitrary heuristics but firmly grounded in the empirical structure of the DOTA dataset. Moreover, the ability to visualize these relationships added an element of transparency, clarifying both the rationale for specific rules and their influence on detector outputs.

The contribution of SAHI slicing was also evident in the discussion of results. Without tiling, many small or borderline objects would not have been detected at all, leaving nothing for symbolic reasoning to refine. By slicing large images into 1024×1024 patches with overlap, SAHI ensured that small vehicles, roundabouts, and other subtle structures appeared as detectable candidates. Although their baseline confidences were often low (0.45–0.55), the symbolic layer was able to boost them when context justified their presence. In this way, SAHI acted as a precondition for reasoning, feeding the symbolic stage with detections that it could then validate or suppress.

Perhaps the most important outcome of the neurosymbolic pipeline was the way it changed the error profile of the baseline detector. The baseline YOLOv11n-OBB was prone to false positives in implausible contexts and struggled with borderline true positives. After symbolic refinement, the balance shifted: false positives such as ships on bridges or vehicles in sports courts were consistently penalized, while borderline detections in semantically consistent contexts such as small vehicles near roundabouts or airplanes near runways were reinforced. The net effect was a system that did not merely increase raw mAP, but produced outputs that aligned more closely with real-world scene semantics. For users, this translates into more trustworthy detection maps that reduce noise and highlight meaningful patterns.

Nevertheless, the discussion must also acknowledge trade-offs and limitations. The symbolic reasoning stage added negligible runtime overhead, but SAHI slicing increased inference time, making the pipeline better suited for offline analysis than real-time UAV applications. More significantly, the system currently relies on hand-crafted rules derived from dataset statistics. While effective, this approach does not scale easily to new domains or datasets. The rules are tuned to DOTA’s specific class set and spatial patterns, and would need adaptation for datasets like VisDrone and xView. This limitation points toward future work in Inductive Logic Programming (ILP) and differentiable neurosymbolic frameworks such as Logic Tensor Networks (LTNs), which could learn rules automatically from data while retaining interpretability.

When compared with state-of-the-art neural detectors, the contribution of this work is also distinct. Models such as DS-YOLOv7 and ARF-YOLOv8 report higher mAP values on DOTA, but they remain opaque: they cannot explain why a detection was made, nor can they guarantee semantic plausibility. The neurosymbolic pipeline presented here does not outperform them numerically but introduces explainability and contextual refinement as new dimensions of evaluation. In domains where interpretability is critical disaster response, environmental monitoring, defense the ability to justify decisions can be as valuable as raw accuracy.

In summary, the discussion shows that this work has achieved its goals. By answering the research questions through experimental evidence, the thesis demonstrates that:

- Symbolic reasoning systematically improves the plausibility of detections.
- Weighted knowledge graphs provide a principled way to encode domain knowledge and align reasoning with data.
- SAHI slicing is indispensable for exposing small objects to symbolic reinforcement.
- The neurosymbolic pipeline shifts the error profile, reducing false positives and preserving borderline true positives.
- Trade-offs exist in terms of rule design and scalability, but these can be mitigated through future neurosymbolic learning methods.

Thus, the neurosymbolic approach presented here contributes not by chasing incremental gains in mAP, but by showing how deep learning and symbolic reasoning can be combined

to produce context-aware, semantically plausible, and interpretable object detection in aerial imagery.

6. CONCLUSION

6.1 Summary of Research

This thesis set out to investigate whether a neurosymbolic approach could enhance object detection in aerial imagery by addressing persistent weaknesses of purely neural models. The baseline YOLOv11n-OBb detector, trained on the DOTA dataset with SAHI slicing, established a capable yet imperfect foundation. While achieving an overall mAP@0.5 of 62.8% and precision of 0.798, the model exhibited two key shortcomings: (1) vulnerability to false positives in implausible contexts, and (2) loss of borderline true positives due to low confidence in small or ambiguous cases.

To address these gaps, the research introduced a symbolic reasoning layer implemented in SWI-Prolog, grounded in weighted knowledge graphs derived from dataset statistics. By encoding domain knowledge as confidence modifiers, the system was able to proportionally boost plausible detections and suppress implausible ones. The integration was seamless: YOLO detections were transformed into Prolog facts, relations such as `located_on`, `located_near`, and `co-occurs` were extracted, and rules applied to refine confidence values.

The final evaluation log captured 136 symbolic adjustments, demonstrating consistent improvements. Average confidence for primary detections increased from 0.90 to 0.93, while secondary detections typically more vulnerable improved from 0.78 to 0.84. These shifts were not trivial: they corresponded to the selective preservation of borderline cases and the suppression of semantically implausible false positives. Qualitative inspection confirmed that detection maps became more coherent, aligning better with real-world spatial logic.

6.2 Contributions to Science

From a research perspective, the thesis contributes to both the fields of aerial object detection and neurosymbolic AI:

1. **Hybrid Pipeline Design:** The work demonstrates a practical integration of a modern neural detector (YOLOv11n-OBB + SAHI) with a symbolic reasoning layer (Prolog + weighted KGs), forming a novel end-to-end pipeline for aerial imagery.
2. **Weighted Knowledge Graph Reasoning:** By using dataset-derived weights, the system operationalized proportional, context-aware reasoning. This marks a departure from rigid symbolic filters and provides a reproducible framework for graded neurosymbolic inference.
3. **Error Profile Transformation:** Instead of focusing solely on global accuracy gains, the pipeline reshaped the error distribution reducing false positives and preserving borderline true positives. This shift represents a new evaluation lens for object detection systems.
4. **Interpretability and Explainability:** Each symbolic adjustment was linked to explicit rules and graph relations, providing an explanation trace. This addresses the “black box” criticism of deep detectors and opens pathways for explainable AI in aerial analysis.
5. **Empirical Validation:** The Excel log analysis offers a rare, quantitative account of symbolic refinement in practice, proving that even modest numerical shifts correspond to meaningful improvements in semantic plausibility.

6.3 Societal and Practical Impact

Beyond academic novelty, the work carries significant implications for society and applied domains. In high-stakes fields such as disaster management, defense, environmental monitoring, and urban planning, the trustworthiness of AI outputs is paramount. False positives can waste critical resources, while missed detections can cost lives. By ensuring that object detections are plausible, explainable, and contextually validated, the neurosymbolic pipeline improves decision-making reliability.

For instance, in post-disaster scenarios, correctly distinguishing between ships docked at harbors and misclassified ships on land ensures that rescue and resource allocation efforts are directed appropriately. In traffic monitoring, preserving borderline detections of small vehicles at intersections or roundabouts can improve safety analytics. These examples demonstrate how an approach that emphasizes semantic correctness has direct societal value, even if raw accuracy improvements are modest.

6.4 Limitations

The research also identified several limitations that constrain current applicability:

- **Computational Cost:** SAHI slicing increases inference time, making the pipeline better suited for offline analysis than for real-time UAV applications.
- **Rule Engineering:** The symbolic rules were manually defined from dataset statistics, limiting scalability to new domains.
- **Dataset Bias:** The system was tuned to DOTA-specific relations; generalization to datasets like VisDrone or xView would require re-deriving weights and potentially rewriting rules.
- **Marginal mAP Gains:** While plausibility improved, global mAP remained comparable to the baseline and final metrics calculation of the difference in mAP of all separate classes required higher version of GPU and RAM requirements, which may hinder the contribution in conventional benchmarking terms.

6.5 Future Directions

Building on the foundations established in this thesis, several research avenues are promising for advancing neurosymbolic aerial detection:

1. **Automated Rule Learning:** Integrating Inductive Logic Programming (ILP) or Logic Tensor Networks (LTNs) could allow the system to automatically discover and refine rules from training data, reducing reliance on manual engineering.
2. **Cross-Domain Adaptation:** Extending the pipeline to other aerial datasets (VisDrone, xView) would validate scalability. Weighted KGs could be recalibrated dynamically to reflect new contexts and domains.
3. **Real-Time Optimization:** Replacing SAHI with efficient multi-scale detectors or transformer backbones could reduce inference overhead, moving the system closer to real-time applications.
4. **Human-in-the-Loop Refinement:** Future deployments could allow domain experts to interactively adjust symbolic rules, combining human knowledge with machine reasoning in a feedback loop.
5. **Broader Evaluation Metrics:** Beyond mAP, future work should develop metrics for plausibility and interpretability — quantifying not just accuracy but how “reasonable” detections are in context.

In conclusion, this thesis has shown that neurosymbolic reasoning adds measurable, explainable value to aerial object detection. By bridging the perceptual power of deep learning with the contextual consistency of symbolic reasoning, the pipeline addressed critical shortcomings of purely neural detectors. It did not merely seek higher accuracy scores but instead reframed success in terms of semantic reliability, contextual awareness, and interpretability.

For academia, our work contributes a replicable demonstration of weighted neurosymbolic reasoning applied to aerial imagery, enriching the growing literature on hybrid AI. For society, it offers a pathway toward detection systems that analysts can not only trust but also understand a crucial requirement for deploying AI in environments where human lives, resources, and decisions depend on reliable information.

APPENDIX I

To maintain readability in the main body of the thesis, only core excerpts of the implementation are included here. Complete scripts are available as supplementary files. Each excerpt is annotated to explain its role in the neurosymbolic pipeline.

A.1 Training Pipeline

```
def train_yolov11_obb():
    model = YOLO("yolo11n-obb.pt")
    model.train(
        data=DATA_YAML,
        epochs=50,
        imgsz=1024,
        batch=8,
        workers=0,
        name=TRAINED_MODEL_NAME,
        save=True,
        save_txt=True
    )
    return model
```

Code snippet 1: Training Pipeline

A.2 Dataset Categories

```
{
    "categories": [
        {"id": 1, "name": "plane"},
        {"id": 2, "name": "ship"},
        {"id": 3, "name": "storage_tank"},
        {"id": 4, "name": "baseball_diamond"},
        {"id": 5, "name": "tennis_court"},
        {"id": 6, "name": "basketball_court"},
        {"id": 7, "name": "Ground_Track_Field"},
        {"id": 8, "name": "harbor"},
        {"id": 9, "name": "Bridge"},
        {"id": 10, "name": "large_vehicle"},
        {"id": 11, "name": "small_vehicle"},
        {"id": 12, "name": "helicopter"},
        {"id": 13, "name": "roundabout"},
        {"id": 14, "name": "soccer_ball_field"},
        {"id": 15, "name": "swimming_pool"}
    ]
}
```

Code snippet 2: Dataset Categories

A.3 SAHI Inference

```

result = get_sliced_prediction(
    image_path,
    detection_model,
    slice_height=1024,
    slice_width=1024,
    overlap_height_ratio=0.2,
    overlap_width_ratio=0.2
)

```

Code snippet 3: SAHI Inference

A.4 Geometric helpers

```

def get_center(bbox):
    x1, y1, x2, y2 = bbox
    return ((x1 + x2) / 2, (y1 + y2) / 2)

def get_distance(bbox_a, bbox_b):
    center_a, center_b = get_center(bbox_a), get_center(bbox_b)
    return math.hypot(center_a[0] - center_b[0], center_a[1] - center_b[1])

def get_bbox_area(bbox):
    x1, y1, x2, y2 = bbox
    return max(0, x2 - x1) * max(0, y2 - y1)

def get_intersection_area(bbox_a, bbox_b):
    x1a, y1a, x2a, y2a = bbox_a
    x1b, y1b, x2b, y2b = bbox_b
    ix1, iy1 = max(x1a, x1b), max(y1a, y1b)
    ix2, iy2 = min(x2a, x2b), min(y2a, y2b)
    return max(0, ix2 - ix1) * max(0, iy2 - iy1)

def get_bbox_diag(bbox):
    x1, y1, x2, y2 = bbox
    return math.hypot(x2 - x1, y2 - y1)

```

Code snippet 4:Geometric helpers

A.5 Prolog Rules

```
%% =====
%% RULE SET 1: CONFIDENCE BOOSTING (Weight > 1.0)
%% Based on strong, positive correlations in facts.pl
%% =====

%% fact('cooccurs', 'harbor', 'ship', 951450).
confidence_modifier(ship, harbor, 1.25).
confidence_modifier(harbor, ship, 1.25).

confidence_modifier(helicopter, ship, 1.20).
confidence_modifier(ship, helicopter, 1.20).

%% fact('adjacent_to', 'large_vehicle', 'small_vehicle', 67014).
confidence_modifier(large_vehicle, small_vehicle, 1.15).
confidence_modifier(small_vehicle, large_vehicle, 1.15).

confidence_modifier(FacilityA, FacilityB, 1.10) :-
    sports_facility(FacilityA),
    sports_facility(FacilityB),
    FacilityA \= FacilityB.

%% fact('cooccurs', 'roundabout', 'small_vehicle', 17566).
confidence_modifier(small_vehicle, roundabout, 1.10).
confidence_modifier(roundabout, small_vehicle, 1.10).
```

Code snippet 5: Confidence boosting rules

```

%% =====
%% RULE SET 2: PLAUSIBILITY PENALTIES (Weight < 1.0)
%% =====

confidence_modifier(ship, 'Bridge', 0.1).
confidence_modifier('Bridge', ship, 0.1).
confidence_modifier(ship, roundabout, 0.1).
confidence_modifier(roundabout, ship, 0.1).

confidence_modifier(plane, harbor, 0.2).
confidence_modifier(harbor, plane, 0.2).
confidence_modifier(plane, 'Bridge', 0.2).
confidence_modifier('Bridge', plane, 0.2).

confidence_modifier(large_vehicle, roundabout, 0.75).
confidence_modifier(roundabout, large_vehicle, 0.75).

confidence_modifier(Infra, Facility, 0.7) :-
    infrastructure(Infra),
    sports_facility(Facility).
confidence_modifier(Facility, Infra, 0.7) :-
    sports_facility(Facility),
    infrastructure(Infra).

confidence_modifier(Vehicle, Facility, 0.5) :-
    vehicle(Vehicle),
    Vehicle \= helicopter, % A helicopter can land on a field
    sports_facility(Facility).

```

Code snippet 6: Plausible penalty rules

A.6 Extended Results

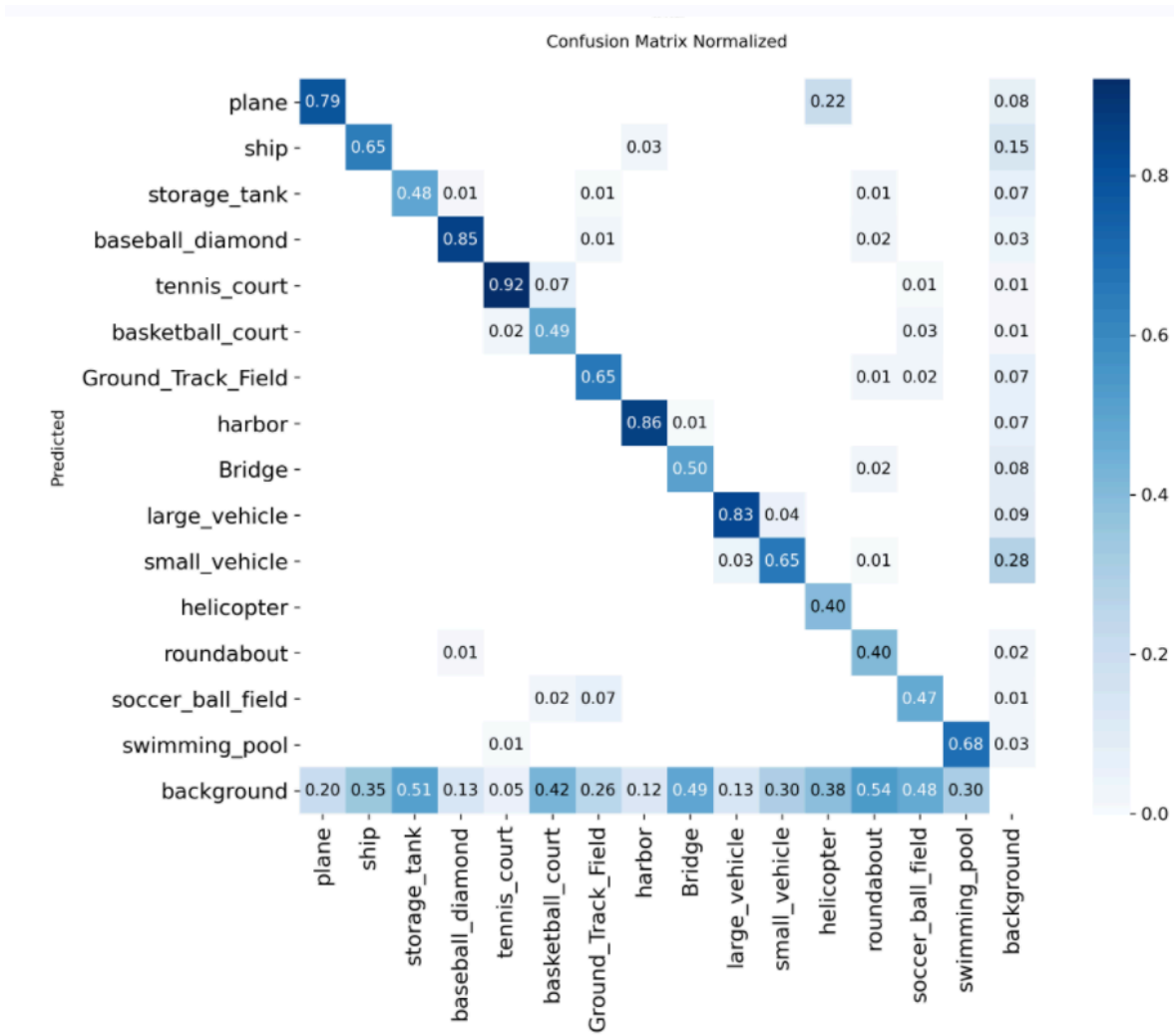


Fig A: Normalized confusion matrix



Fig B: Validation set labelling



Fig C: Validation set predictions from YOLO model

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AFFIDAVIT

Under the penalty of perjury, I, **Pradyumna Swarnalatha Ramanna**, hereby declare under oath, that this master thesis, titled

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has been independently authored by me, that I have used no other sources or aids other than those quoted, and that I have denoted all direct and indirect citations thereof within the text. This manuscript has never before been published or submitted to this or another academic institution for examination.

Berlin, Germany, 03/09/2025	
Location, Date	Signature of Author